Mansoura University

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Automated Skincare system based on deep learning (Pearl)

Graduation Project

by

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Eng / Kholoud Srour

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Abstract

"Pearl" is a cutting-edge skincare platform meticulously crafted to empower users in managing their skincare and overall well-being. The application leverages sophisticated AI algorithms, utilizing deep learning to process user-uploaded facial images and determine their unique skin type. In addition to this, the app intelligently analyzes responses to tailored questions about the user's skin, forming the foundation for a personalized routine.

The primary challenge we address is the pervasive lack of accessible and personalized skincare solutions. Our target Is addressing acne and its severity that can turn into serious diseases, Globally, acne affects an estimated 9.4% of the population. Its prevalence is particularly high among adolescents and young adults, with a notable decline in occurrence after the age of 30. In addition to that, we try to reduce some habits like exposure to ultraviolet rays that lead to severe diseases. Skin cancers are primarily caused by exposure to ultraviolet radiation (UVR), either from the sun or from artificial sources such as sunbeds1. In 2020, over 1.5 million cases of skin cancers were diagnosed globally, and over 120,000 skin cancer-associated deaths were reported. It's also worth noting that ultraviolet radiation from the sun causes more than 95% of all skin cancers. A study found that 91% of all melanomas were linked with ultraviolet (UV) radiation, mostly due to sun exposure.

Our decision to tackle this problem is rooted in the profound impact it has on the well-being and self-confidence of young individuals, especially to avoid depression and anxiety. One study found 96% of people with Acne have reported feeling depressed over their condition. This has led 46% to develop self-esteem issues over their complexions and has caused 31% to stay more at home, declining to participate in social activities.

Our commitment stems from recognizing that teenagers may lack easy access to personalized advice and routines, contributing to the persistence of these skin-related challenges.

Pearl goes beyond merely utilizing advanced AI to detect skin type, addressing acne and its severity, offering tailored routines, thereby promoting overall skin health, and bolstering the self-esteem of the younger demographic.

Keywords Skin; Deep Learning; Routine; Products; Doctor

Acknowledgement

We would like to express our heartfelt gratitude to our esteemed project supervisors, Dr. Fatma El Zahraa and Engineer Kholoud Srour. Their invaluable guidance, unwavering support, and wealth of knowledge significantly influenced the successful completion of this project, "Pearl." Their mentorship and expertise have been instrumental in shaping the project into what it is today.

We also extend our thanks to the dedicated team members who worked tirelessly to bring "Pearl" to life. Each team member's contribution, from software development to data analysis and user experience design, was essential to the project's success.

Furthermore, we want to acknowledge all individuals who contributed, directly or indirectly, to this project's progress. Your assistance in data collection, word processing, and data analysis was exceptional and played a pivotal role in the project's realization.

In the end, we owe our deepest gratitude to everyone who has been part of this journey. Your unwavering support and contributions have been the cornerstone of this project's success.

Thank you for believing in the vision of "Pearl" and for being part of this transformative experience.

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Chapter 1

Chapter 1: Introduction

Introduction

"Pearl" is an innovative skincare platform specifically designed for teenagers, combining advanced AI algorithms with personalized routines. The platform addresses the unique challenges of teen skincare by providing tailored solutions that encompass both skincare and lifestyle choices. Through seamless integration, "Pearl" aims to empower teens to adopt healthier habits for comprehensive well-being.

Problem Definition

The primary challenge we address is the pervasive lack of accessible and personalized skincare solutions, a critical issue, especially for teenagers. The prevalence of acne, a significant skin-related challenge among adolescents and young adults, underscores the urgency of our mission. Studies, such as the one conducted by Ghodsi and colleagues, reveal a staggering 93.3% prevalence of acne among 1,002 adolescents, with 14% experiencing moderate to severe acne. Moreover, Collier et al. report a lifetime acne prevalence of 73.3% among adults aged 20 years and older, with higher rates in women.

Our decision to tackle this problem is rooted in the profound impact it has on the well-being and self- confidence of young individuals. Acne, among other skin issues, can be emotionally distressing, affecting not only physical health but also mental and emotional aspects. Our commitment stems from recognizing that teenagers may lack easy access to personalized advice and routines, contributing to the persistence of these skin-related challenges.

It's crucial to note that acne is just one facet of the myriad skin issues our application addresses. The statistics highlighted underline the widespread nature of the problem, emphasizing the pressing need for comprehensive, user-centric solutions. "Pearl" goes beyond merely utilizing advanced AI to detect skin type, addressing acne and its severity, and offering tailored routines, thereby promoting overall skin health and bolstering the self-esteem of the younger demographic.

Project Objectives

User-Centric Objectives:

- ***Objective 1*:** Address Acne and Its Severity and Encourage Healthy Skincare Practices:
  - *Feature 1:* Detecting User's skin type.
  - *Feature 2:* Highlight acne and address its severity to guide users in their skincare journey.
  - *Feature 3*: Analyze and provide solutions for specific skin concerns through personalized routines.

- **Objective 2:** Enhance user engagement and motivation through personalized content.
  - *Feature 1: Implement a "skin shooter" for better photo quality.*
  - *Feature 2*: Collect medication information through photo and question answers for skincare analysis.
  - *Feature 3*: Customized Diets for Optimal Skincare.
  - *Feature 4*: Consider suggesting doctors.

- ***Objective 3**:* Maximize data advantages while respecting user privacy.
  - *Feature 1:* Clarify data usage, especially regarding photos.
  - *Feature 2:* Capture skin type information with app-driven advice.

User Acquisition Objectives:

- ***Objective**:* Attract users through compelling features and functionalities.
  - *Feature 1:* Offer a user-friendly profile page for seamless navigation.
  - *Feature 2:* Encourage users with the allure of comprehensive self-knowledge.

Doctor-Centric Objectives:

- ***Objective 1**:* Streamline patient history access for efficient consultations.
  - *Feature 1:* Centralize patient history with data and preview shots.
  - *Feature 2:* Enable in-app appointment scheduling.

Project Scope

"Pearl" skin care project aims to develop a user-centric platform for teenagers, providing personalized skincare routines, motivational features, and efficient doctor-patient interaction. The scope includes data privacy considerations, online appointment scheduling, and attractive user profiles. Technical components involve multi-platform development, AI integration, and secure data management. The project emphasizes continuous improvement, with user feedback shaping future updates for an evolving and adaptable skincare solution.

Contributions of This Study

1. Advancement in Skincare Technology:

"Pearl" introduces a novel approach to skincare that leverages advanced AI, machine learning, and data analysis. By offering personalized skincare routines, holistic well-being guidance, and efficient doctor-patient interactions, "Pearl" advances the state of skincare technology. It provides users with tailored recommendations to address their unique needs, fostering healthier habits and self-care.

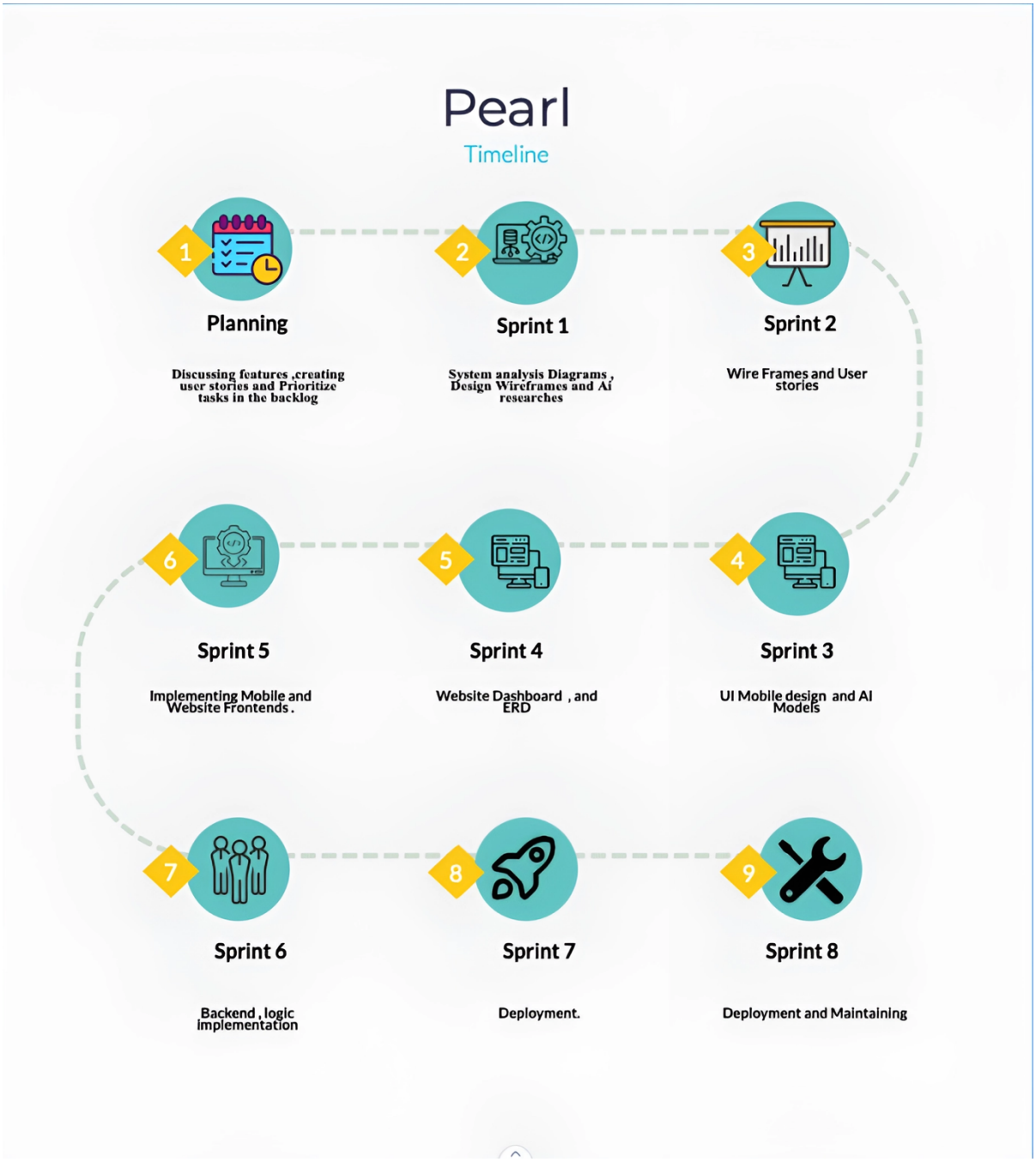
2. Enhanced Doctor-Patient Interaction:

Our project facilitates streamlined communication between patients and healthcare professionals. Doctors gain real-time access to patient data, optimizing consultation time and enhancing the overall healthcare experience. "Pearl" bridges the gap in teen skincare with efficient, personalized, and accessible solutions, setting a new standard for healthcare interactions.

3. Empowering Teens for Self-Care:

"Pearl" caters to the specific needs of teenagers, recognizing their unique skin concerns and offering a user-friendly platform to address these issues. The platform empowers teens to adopt healthier habits and provides them with easy access to quality medical advice and support. It fosters user engagement and motivation through personalized content and provides secure data sharing with healthcare professionals.

Project Timeline



Document Organization

This project consists of six chapters in addition to one appendix. These chapters are organized to reflect the scientific steps toward our main objective. A brief description about the contents of each chapter is given in the following paragraphs:

Chapter 1: introduces "Pearl," an innovative skincare platform designed for teenagers. It outlines the challenges faced by teenagers in skincare and sets the project objectives, focusing on user-centric and doctor-centric goals. The project scope covers personalized routines, motivational features, data privacy, and more.

Chapter 2

Chapter 2: Literature Review

In this chapter, we provide a background about the tools and techniques needed to build our system. We also review the previous related work to our system, and the common technologies that are used. In the end of the chapter, we provide a comparison between the system we are going to build and the related systems, and why our system is a treasure for

Introduction

The second chapter of this report embarks on a comprehensive exploration of the existing body of knowledge and research in the field of skincare, healthcare technology, and artificial intelligence. By delving into the extensive literature, this chapter lays the foundation for understanding the context and significance of "Pearl" in addressing the specific challenges faced by teenagers in their skincare routines and healthcare interactions. This review of relevant studies, technological advancements, and industry trends illuminates the path for the development and implementation of our innovative skincare platform.

Background

Our revolutionary platform, codenamed "Pearl," is a cutting-edge integration of advanced technologies aimed at redefining skincare routines and personalized dermatological care. This comprehensive system utilizes image processing, Artificial Intelligence (AI), and a mobile application to create an unparalleled skincare experience. The project encompasses various facets, including:

- Skin Health Analysis:
  - The core of our project revolves around in-depth skin health analysis. Leveraging image processing techniques, users can capture and submit images of their skin, allowing our platform to perform a detailed examination. This analysis forms the basis for personalized skincare recommendations.
- Image Processing:
  - Image processing serves as the backbone of our system, enabling the extraction of crucial data from user-submitted images. This includes the identification of skin types, detection acne and its severity, and the analysis of routine-related parameters.

- Image Segmentation:
  - To enhance the precision of our analysis, we employ image segmentation techniques. This enables the system to focus on distinct regions of the skin, ensuring a meticulous evaluation of various elements such as texture, tone, and potential abnormalities.

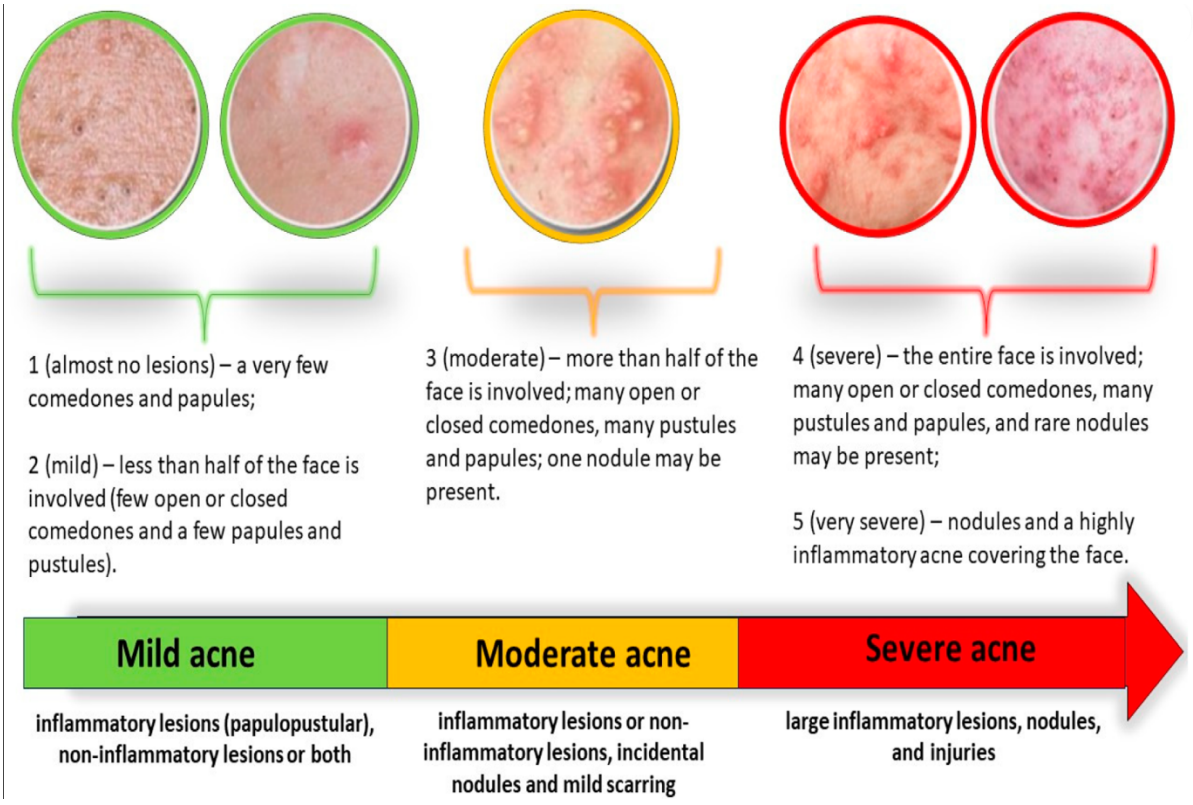
- Deep Learning Technology:
  - The integration of deep learning technology propels our platform to new heights of intelligence. Our AI models are trained to recognize patterns, trends, and anomalies within the submitted images. This level of sophistication allows for continuous learning and adaptation to evolving skincare needs.

Our system, “Pearl" is based on image processing, Artificial Intelligence technology, a mobile application, and a website for work. The following subsections provide brief background information on the tools and techniques needed to build our system, which will include the following:

- The Association Between Skin Types and Acne Vulgaris
- Image Processing
- Image Segmentation
- Deep learning Technology

4.1.1 **The Association Between Skin Types and Acne Vulgaris**

Acne vulgaris, a common chronic inflammatory condition of the pilosebaceous units, manifests through a variety of lesions including comedones, papules, pustules, nodules, and cysts. This condition can lead to significant scarring and pigmentation changes, affecting individuals' physical appearance and emotional well-being. The prevalence and severity of acne vulgaris are closely linked to skin type, a relationship that plays a crucial role in both its development and management. Skin types, broadly categorized as oily or dry based on sebum production, exhibit distinct characteristics that influence acne formation. Oily skin, with its higher sebum output and larger pores, tends to be more susceptible to clogged pores and bacterial growth, leading to more severe acne. On the other hand, dry skin, characterized by a weakened lipid barrier and increased transepidermal water loss, is vulnerable to irritation and bacterial invasion, which can also trigger acne. Understanding the association between skin types and acne vulgaris is essential for devising effective skincare strategies and treatment plans tailored to individual needs. This insight not only aids in better management of acne but also contributes to improved overall skin health.



4.1.2 Image processing

Image preprocessing is essential to ensure our images are in the best possible condition for training our models effectively. We started by cleaning up any noise in the images using color conversions, which also helped standardize how colors are represented. Resizing the images was crucial not only to fit them into our model but also to optimize memory usage without losing important details. We also focused on removing backgrounds that aren't relevant to our analysis and enhancing image quality overall. To make our model more robust, we added variations like flipping and rotating images, which also helps augment our dataset.Advanced techniques such as Contrast Limited Adaptive Histogram Equalization (CLAHE) were integrated to enhance image contrast locally, thereby improving feature visibility. Connectivity analysis ensured structural integrity during processing, crucial for maintaining image coherence across segmented regions. Shuffling of the dataset was implemented to achieve balanced training, thereby enhancing model robustness and generalization capabilities.

4.1.3 Image segmentation

Image segmentation is all about breaking down images into meaningful parts so our models can understand them better. In our approach, we used color codes to separate foreground elements (what we're interested in) from background elements (what we're not interested in). This process included techniques to remove background noise effectively, allowing us to focus only on the parts of the image that matter for our analysis. By identifying and isolating these foreground and background components, we created clearer, more focused images that our model can analyze more accurately. This segmentation step is crucial for extracting the right features from our images and improving how well our model can classify and understand them.

4.1.4 Feature extraction

In skin image analysis for skin type detection and disease discovery, several features are typically extracted:

1. **Color Features**: Skin diseases often present as changes in skin color. Therefore, we extract statistical color features such as mean, standard deviation, and skewness from color channels in various color spaces (e.g., RGB, HSV). These features highlight variations in skin tone and can indicate conditions like redness or pigmentation changes..
2. **Texture Features** we utilize **Contrast Limited Adaptive Histogram Equalization (CLAHE)** for enhancing texture details in skin images. This method

improves local contrast and facilitates better representation of fine textures associated with skin types

3. **Shape Features**: Certain skin conditions exhibit distinct shapes or structural abnormalities. Shape features are essential for capturing these manifestations, aiding in the identification and classification of specific dermatological issues.
4. **Deep Learning Features** : Leveraging **MobileNetV2**, a powerful Convolutional Neural Network (CNN) pretrained on ImageNet, we extract high-level features from skin images. MobileNetV2's architecture allows it to automatically learn and represent complex patterns and visual motifs associated with various skin conditions.

These features are then used as input for machine learning models to classify the skin type and detect Acne Severity.

Detecting facial skin type using AI involves analyzing various characteristics of the skin, such as oiliness, dryness, texture, and pigmentation. Here is a walkthrough the process.



3.1.4. Classification

In this system classification is the main part. Basically, it is a user part. It first loads the pre-trained image database that is generated earlier by using CNNs. It also takes an image from the user and does pre-processing to extract the features and classify it according to the dataset.

In the following figure, we will discuss some of the division techniques used by scientists

in previous contributions.

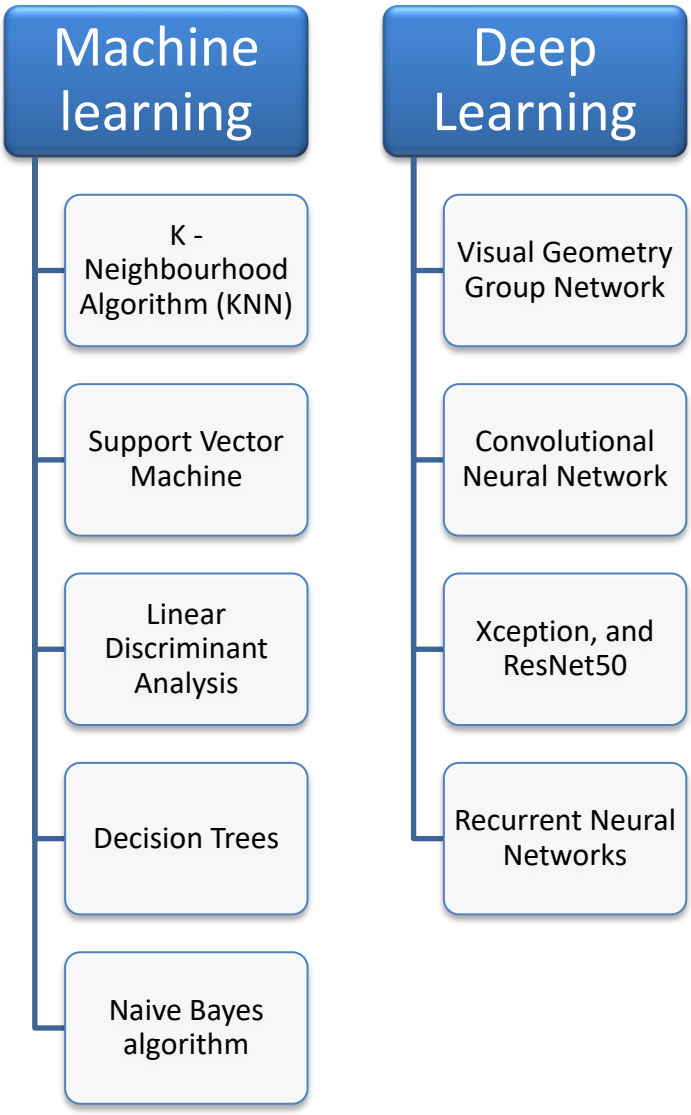


Table I Division Techniques

3.1.5 Deep learning

Deep learning is a machine learning technique that constructs artificial neural networks to mimic the structure and function of the human brain. In practice, deep learning, also known as deep structured learning or hierarchical learning, uses large numbers of hidden layers - typically more than 6 but often much higher - of nonlinear processing. It falls under the umbrella of machine learning and AI, eliminating some of machine learning's data preprocessing with algorithms. Deep learning drives many AI applications that improve the way systems and tools deliver services, such as voice-enabled technology .While machine learning uses data reprocessing driven by algorithms, deep learning strives to mimic the human brain by clustering data to produce accurate predictions.

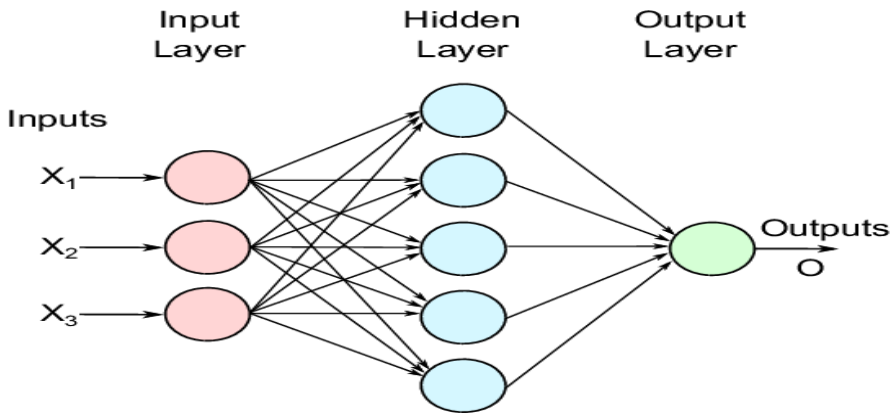


Figure I Neural Networks

3.1.6 Tensor Flow

TensorFlow is an open-source end-to-end platform for creating Machine Learning applications. It is a symbolic math library that uses dataflow and differentiable programming to perform various tasks focused on training and inference of deep neural networks. It allows

developers to create machine learning applications using various tools, libraries, and community resources.

Currently, the most famous deep learning library in the world is Google's TensorFlow. Google product uses machine learning in all its products to improve the search engine, translation, image captioning or recommendations.

Review of Relevant Work

Table II.1 . Review of Relevant Work

Reference, year	Methodology	dataset	Result
Ema Utami et al [1]	CNN with MobileNet architecture, CNN-SVM	The HAM10000	The results of the CNN classification show an accuracy of 93.47%, with high precision, recall, and F1-score, at 93.55%, 93.74%, and 93.62%, respectively. Meanwhile, the CNN-SVM model with "poly," "rbf," "linear," and "sigmoid" kernels exhibits varied performances. Overall, the CNN-SVM model performs

			lower than the CNN model.
Ting-Yu Lin et al [2]	DeepLab-v3+ model for pixel-level prediction , CV	Captured Images by Logitech's C310 camera , The database of this work contains 110 skincare products.	80% segmentation accuracy.
Sirawit Saiwaeo et al [3]	MobileNet-V2 EfficientNet-V2 Inception-V2 ResNet-V1	It's is the collection of a total of 60 subject's data.	89.52% 91.55% 92.22% 72.94% Accuracies respectively to the used methods
Patricia Angela Abu et al [4]	MATLAB with the use of the Image Processing ToolboxTM ,	Dermnet as well as images from the Health Sciences	an average F-measure of 88.67%

	Feed-Forward Neural Network	department in the Ateneo De Manila University	
Manerka et al. [19]	SVM	Clinical Images	Accuracy: 96–98%
Amarathunga et al. [15]	MLP	Clinical Images	Accuracy: 90%
Chatterjee et al. [24]	SVM	Dermoscopic Images	Sensitivity: 99.01% Specificity: 95.35% Accuracy: 98.79%
Chakraborty et al. [16]	NN- NSGA-II	Dermoscopic Images	Accuracy:90.56 %, F-measure:90.87% Precision:88.26 %, Recall:93.64%,

Review on related mobile applications

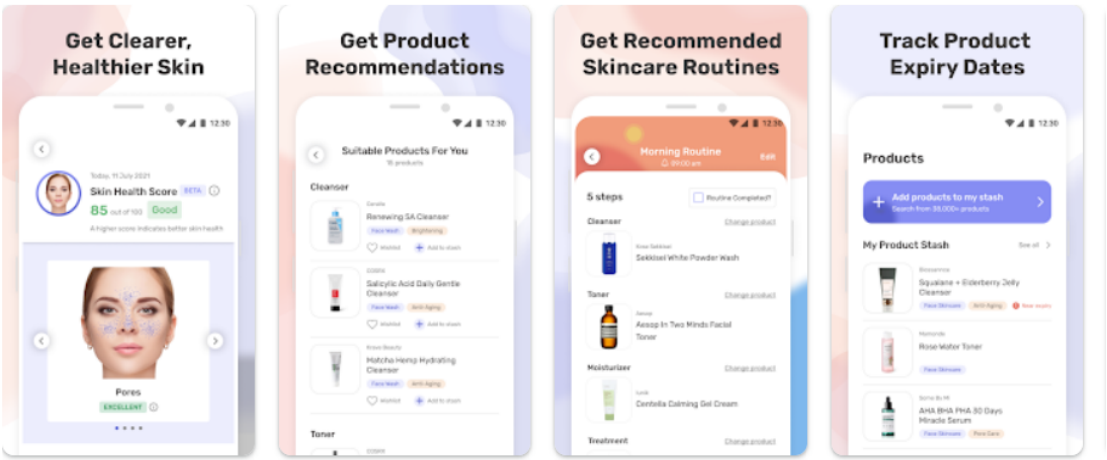
1. **TroveSkin**

- Pros :

- Products Recommendations
- Skincare Routines Recommendations
- Track Product Expiry dates
- Connecting with people interested to SkinCare.

- Cons :

- Features are not free , subscribe needed
- Ui isn't that friendly
- The output is not accurate
- The application glitches



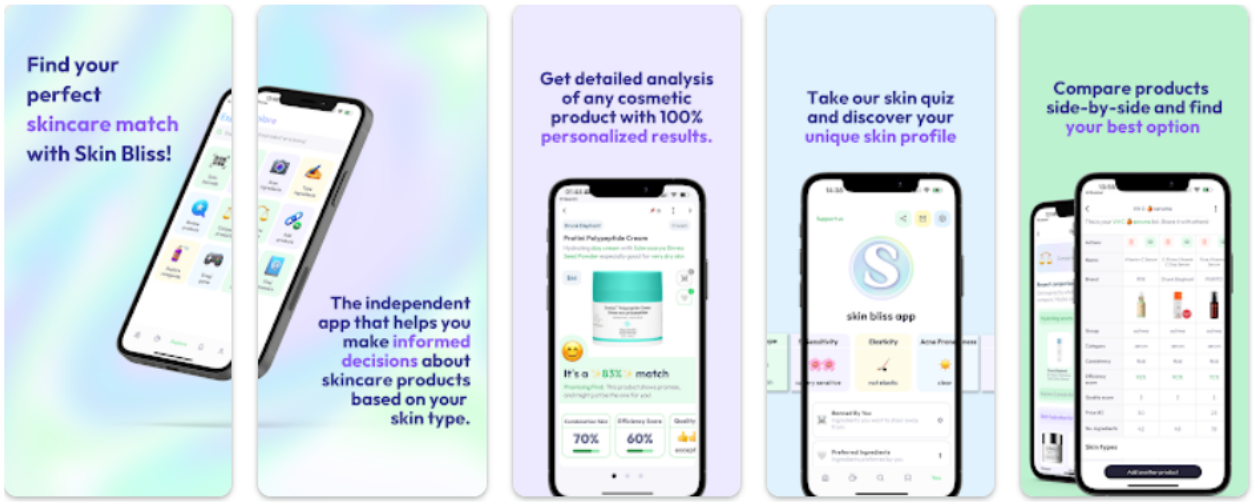
2. Skin Bliss

- Pros :

- Getting details of cosmetic products
- Skin quiz to discover your skin
- Easy tracking
- Free to download

- Cons :

- Ui is confusing
- Not user friendly
- Wifi needed all the time
- Long time to check the product ingredients



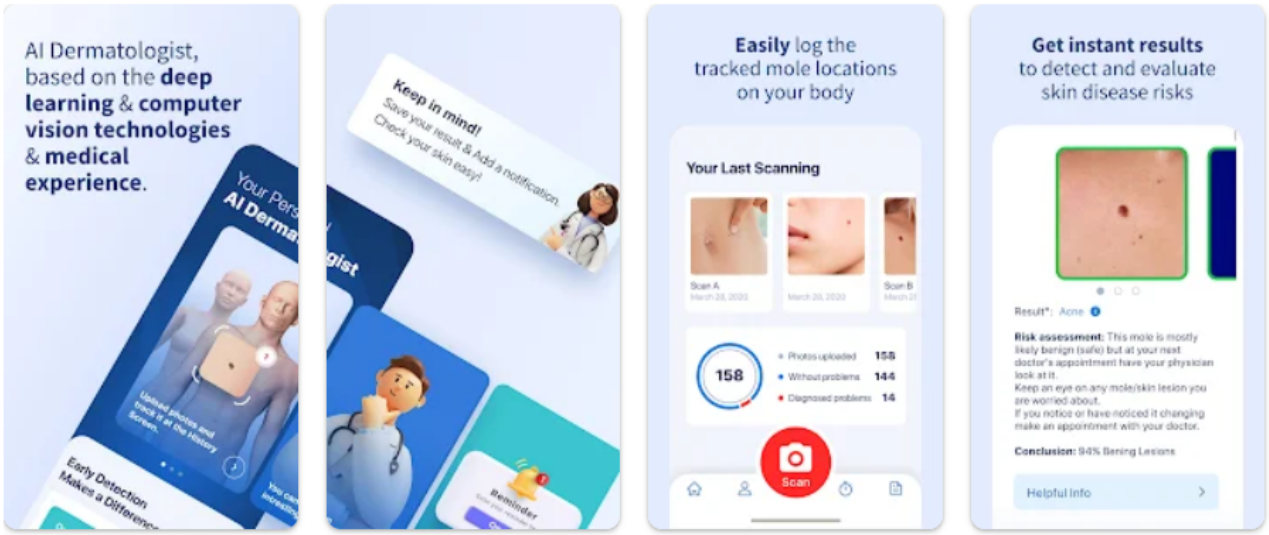
3. Ai Dermatogolist

- Pros :

- Easy to track more locations
- Getting instant results
- Result savings
- Free to download

- Cons :

- Only one scan is free
- Accuracy is low
- Ui Isn’t friendly
- Not Reliable



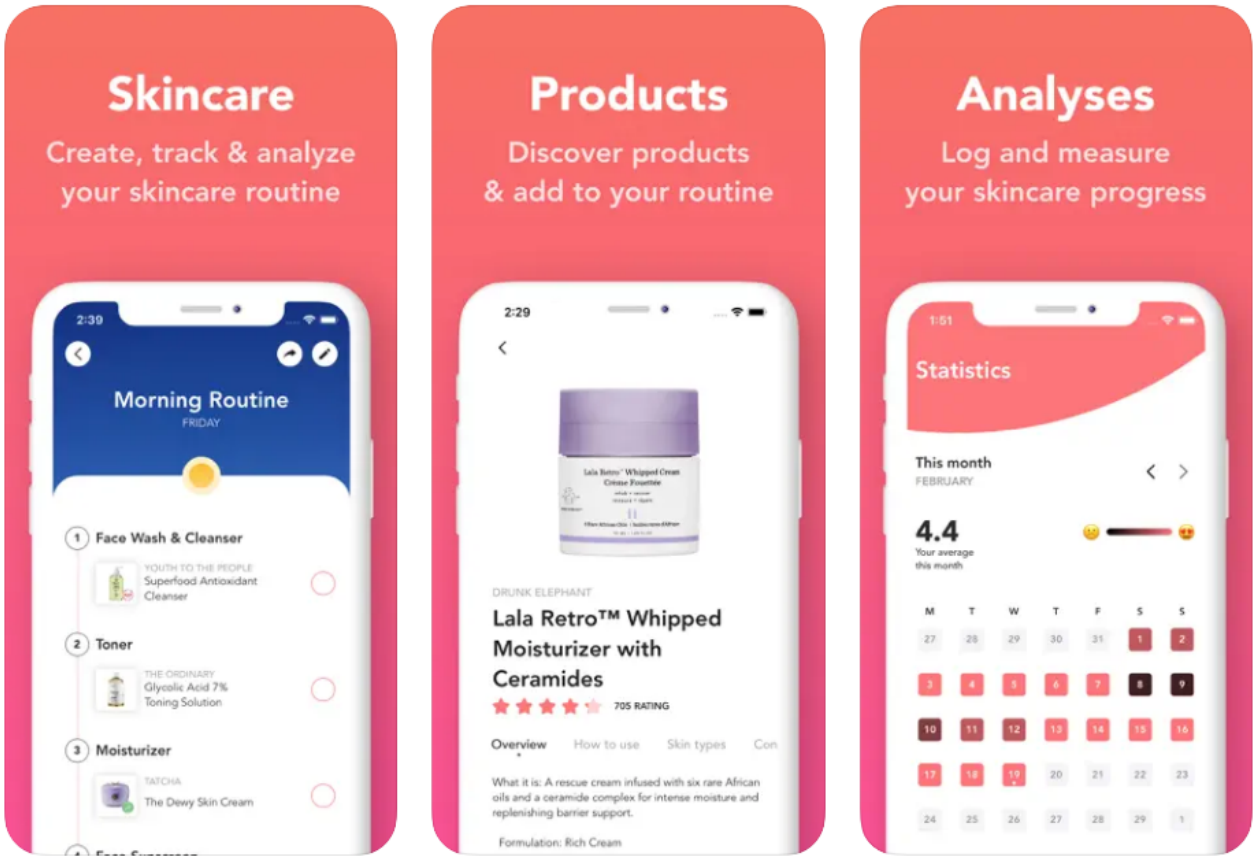
4. Charm

- Pros :

- Ui Friendly
- Creating skincare routine
- Tracking routine
- Analyzing your routine

- Cons :

- Crashes a lot
- Can’t add too much products
- Tracking isn’t great
- Premium needed for full functionality



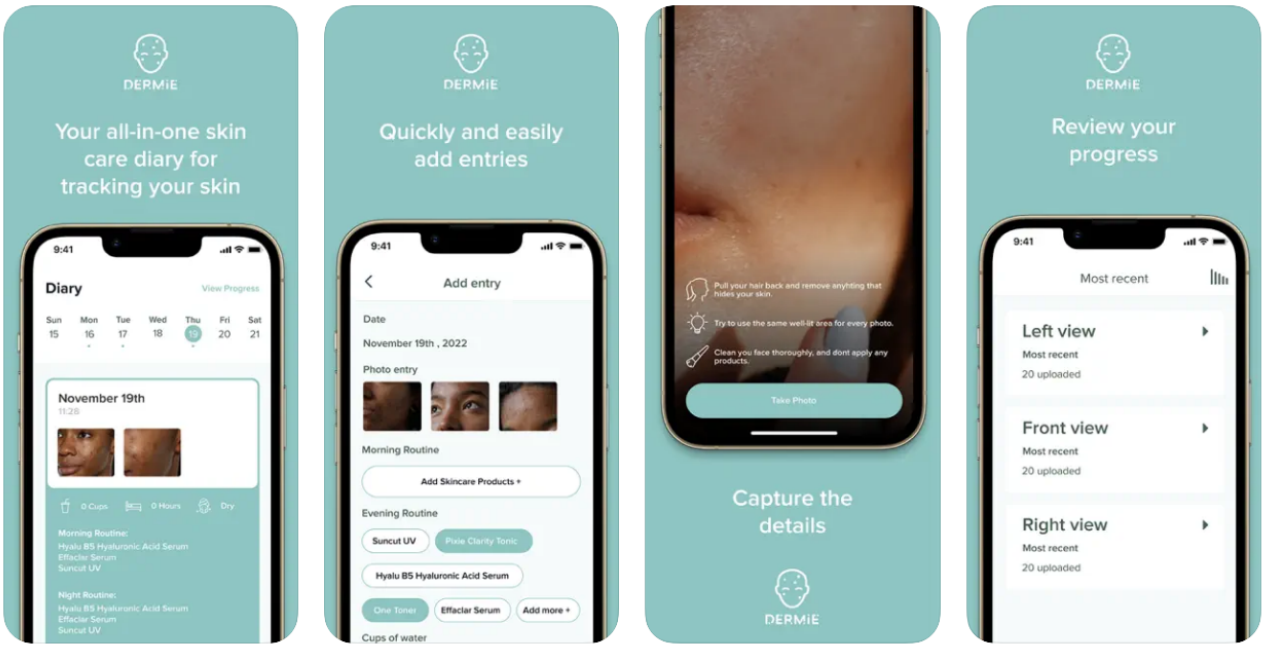
5. Derma Ai

- Pros :

- Ui Friendly
- Easy scanning for your face
- Reviewing your progress
- Details are important

- Cons :

- Lack of features
- Limited UI for the users
- Tracking isn’t great
- Premium needed for full functionality



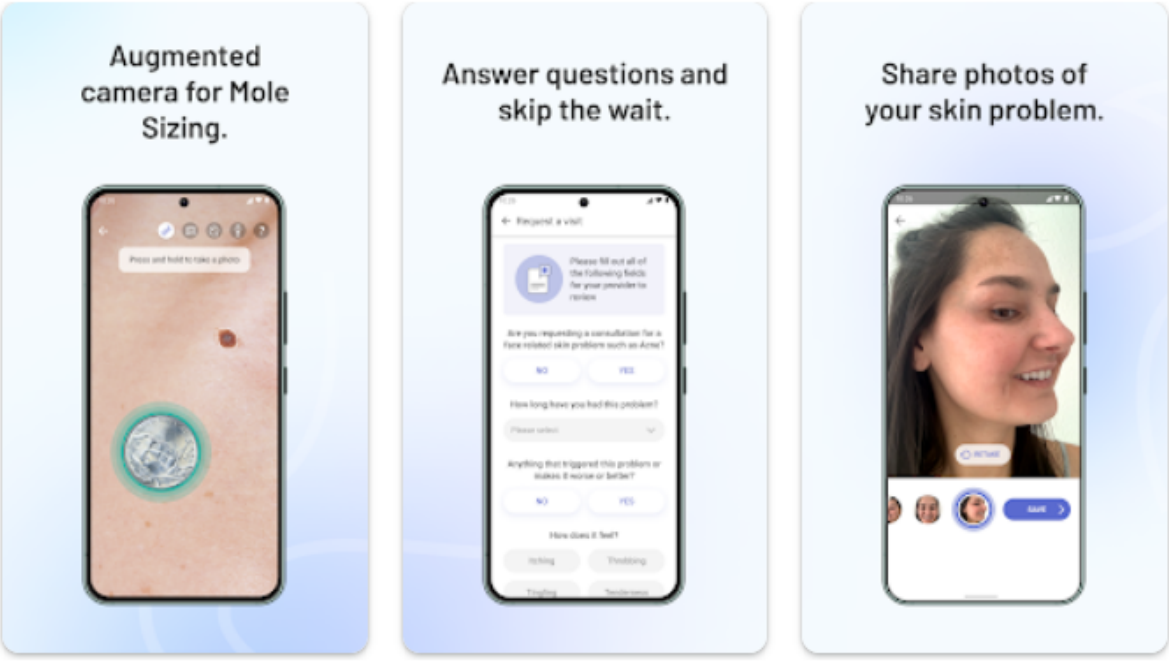
6. MiiSkin

- Pros :

- Ui Friendly
- Mole Mapping that allow users to monitor changes in moles
- Reviewing your progress by timelapse videos of skin images.
- Educational Content related to skin health
- Skin cancer awareness

- Cons :

- Self monitoring tool
- No treatment recommendations
- Limited diagnoses
- Premium needed for full functionality



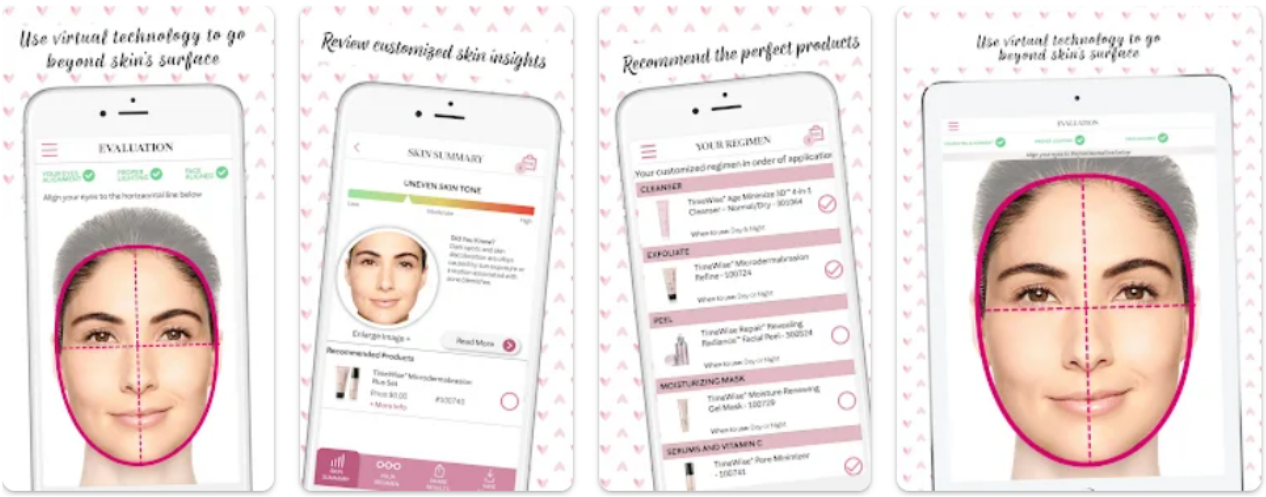
7. Mary Kay Skin Analyze

- Pros :

- Ui Friendly
- Personalized skincare recommendations
- Progress Tracking
- Skin analysis includes skin types and problems

- Cons :

- Accuracy Limitations
- Recommends Only the Owner’s Products
- Doesn’t Provide Comprehensive healthcare advice



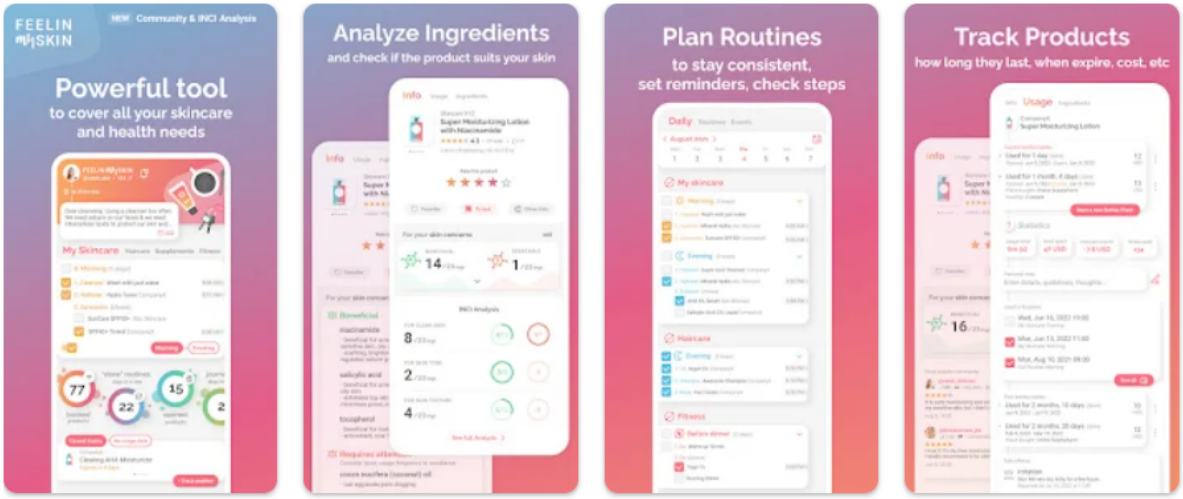
8. **Feeling My Skin**

- **Pros :**

  - Analyzing product's ingredients and checks if it suits you
  - Tracks product usage and expiry date
  - Reminders for routines

- **Cons :**

  - Bad UI/UX
  - No Product recommendations
  - Manual Skin tracking changes



Relationship between the Relevant Work and Our Own Work

The relationship between our work and the relevant existing solutions is characterized by a strategic amalgamation of key features, design enhancements, and technological advancements. Our project serves as a synthesis of the best elements extracted from various relevant works in the skincare domain. By meticulously studying and optimizing existing solutions, we have curated a unique approach that combines user-centric design, cutting-edge AI algorithms, and deep learning methodologies.

Our work goes beyond a mere aggregation of features; it represents a significant evolution in the realm of skincare applications. The incorporation of advanced AI and deep learning not only refines the process of skin analysis but also enables the generation of highly personalized and effective skincare routines. The design enhancements contribute to a more intuitive and engaging user experience, fostering increased user adherence to skincare regimens.

In essence, our project stands at the forefront of innovation, building upon the foundations laid by previous works to deliver a comprehensive, efficient, and user-friendly skincare platform. The strategic integration of the most effective features from relevant works ensures that our project is not only informed by the successes of its predecessors but also poised to set new standards in the skincare application landscape.

Summary

The Literature Review chapter provides a comprehensive overview of existing skincare applications, analyzing their strengths and weaknesses. It navigates through a diverse range of solutions, identifying key features and methodologies employed in the skincare domain. Our exploration of relevant works reveals common practices, such as routine recommendations and skin analysis, along with variations in design and functionality.

Moreover, the chapter sheds light on the limitations of existing solutions, paving the way for our project to address these gaps through innovative approaches. Our emphasis on artificial intelligence (AI) and deep learning emerges as a transformative element, revolutionizing the accuracy and personalization of skincare routines. By leveraging these technologies, our project introduces a dynamic dimension to user interaction, fostering a more tailored and effective skincare experience.

In summary, the Literature Review serves as the foundation for our project's conceptualization, drawing insights from existing works while strategically incorporating advanced technologies. The exploration of pros and cons informs our decision-making process, guiding the development of a skincare platform that not only builds upon the successes of its predecessors but also pioneers new standards through the integration of AI and deep learning methodologies.

Chapter 3

Chapter 3: System Analysis

In this chapter, we delve into the critical aspects of system analysis, laying the foundation for the "Pearl" skincare platform. Our primary objective is to define the system requirements, encompassing both functional and non-functional elements. These requirements are instrumental in shaping the platform's capabilities, ensuring it meets the needs and expectations of both users and healthcare professionals.

Furthermore, we explore the specific user requirements that drive the development of "Pearl." By understanding the expectations and preferences of our target audience, we aim to create a user-centric platform that offers personalized skincare solutions and seamless doctor-patient interactions.

Throughout this chapter, we will present various diagrams and visual representations, offering a comprehensive view of the system's architecture, data flow, and functionality. These visual aids serve as valuable tools for understanding the inner workings of "Pearl" and how it fulfills its mission of enhancing teen skincare.

3.1 Analysis of Existing Systems

3.1.1 Purpose

The analysis of existing systems plays a pivotal role in the development of our project. The objective is to gain a comprehensive understanding of the existing landscape of skincare applications and services. This process helps us discern why our project is essential and how it can provide unique advantages to users.

3.1.2 Methods

We employed a variety of methods to gather insights into the existing systems:

Document Review: We conducted extensive document reviews, delving into reports, articles, and literature related to skincare applications and solutions. This process allowed us to understand the historical context and user experiences.

User Interviews: To capture the user perspective, we conducted interviews with potential users. These interviews provided invaluable insights into their needs and expectations.

Surveys: A Google survey was administered to a diverse group of individuals. Their responses contributed to a broader understanding of users' skincare habits, preferences, and challenges.

Data Collection: We collected data related to skin issues, user preferences, and the effectiveness of different skincare routines. This data informed our project's design and functionality.

Observation: Observing existing systems in action helped us understand user interactions and the limitations of current solutions.

3.1.3 Findings

The analysis revealed several key findings:

- Existing systems often lack comprehensive user engagement and personalized features, leaving users dissatisfied with their skincare experiences.
- Users face challenges in obtaining personalized advice and skincare routines tailored to their unique skin types and conditions.
- There is a need for streamlined doctor-patient interaction within skincare applications.
- Many existing systems lack effective user data privacy mechanisms, raising concerns about data security.

3.1.4 Informing our project

The insights gained from the analysis of existing systems have significantly influenced the goals and design of our project. By addressing the shortcomings of existing solutions, we

aim to create a comprehensive skincare platform that prioritizes user-centricity, personalization, data security, and effective doctor-patient interaction.

3.1.5 Addressing Limitations

Our project is committed to addressing the limitations and gaps identified in existing systems. By combining personalized skincare routines, motivational features, and efficient doctor-patient interaction, we strive to provide a unique and holistic solution for teen skincare.

3.1.6 Project Impact

We believe our project will have a substantial impact on the skincare landscape. By offering personalized solutions, enhancing user engagement, and prioritizing data security, we aim to empower users to make informed decisions about their skincare. Our project will not only benefit teens but also anyone seeking effective and user-friendly skincare solutions.

3.2 System Requirements

This section outlines the detailed system requirements for the "Pearl" skincare platform. It encompasses functional and non-functional requirements, performance constraints, and resource considerations. Additionally, it may include design objectives, focusing on improving the user experience. User requirements are presented to ensure that the system meets the specific needs and expectations of its intended users. This section serves as a foundational guide for the development and design of the platform.

3.2.1 Functional Requirements

This subsection outlines the essential functions and features that our skincare platform must possess to fulfill the needs of users and healthcare professionals. It defines the specific capabilities the system should offer. We, also, provide a description for each function Table VI.1 shows the functional requirements for our system.

Table III.1 Functional Requirements

3.2.2 Non Functional Requirements

#	Functional Requirement	Requirement Description
1	User Registration and Profile Management	Users can create accounts, set up profiles, and update personal information.
2	Skin Image analysis	Users can take pictures for skin analysis, and the system processes these images to assess skin conditions.
3	Routine Generation	The system collects skin-related data from users and processes it to generate personalized skincare routines.
4	Questionnaire Editing	Users can modify their answers to the initial skincare questionnaire, influencing their skincare routines.
5	Online Dermatologist Consultation	Users can request appointments with dermatologists, and doctors can review patient profiles before consultations.
6	User Engagement Features	Motivational messages, streaks, and rewards to encourage consistent app usage.
7	User Friendly Profile	A visually appealing and user-friendly profile page for easy navigation.

Non-functional requirements cover all the remaining requirements which are not covered by the functional requirements. They specify criteria that judge the operation of a system, rather specific behaviors. Table 3.2 shows the nonfunctional requirements for our system.

Table IV.2 Non Functional Requirements

#	Non-Functional Requirement	Requirement Description
1	Response Time	The system should provide quick responses to user inputs, with a maximum response time of 2 seconds for common operations.
2	Scalability	The platform should be scalable to accommodate a growing user base without significant performance degradation.
3	Security	Robust security measures must be in place to protect user data, including encryption and secure authentication.
4	Reliability	The system should be available 24/7 with a minimum uptime of 99.9%.
5	Compliance	The system should comply with relevant healthcare and data protection regulations.
6	Accuracy	The system should have a high level of accuracy in analyzing users' skin conditions and providing personalized skincare recommendations.
7	Performance	Response times for common operations should be within milliseconds, even under heavy usage.

3.2.3 Design Objectives

Utilizing OKRs (Objectives and Key Results) in the 'Design Objectives' section ensures that our design goals are well-defined and measurable. It enables clear communication, team alignment, and a focus on user-centered design. OKRs also help us track progress and drive continuous improvement in our design work.

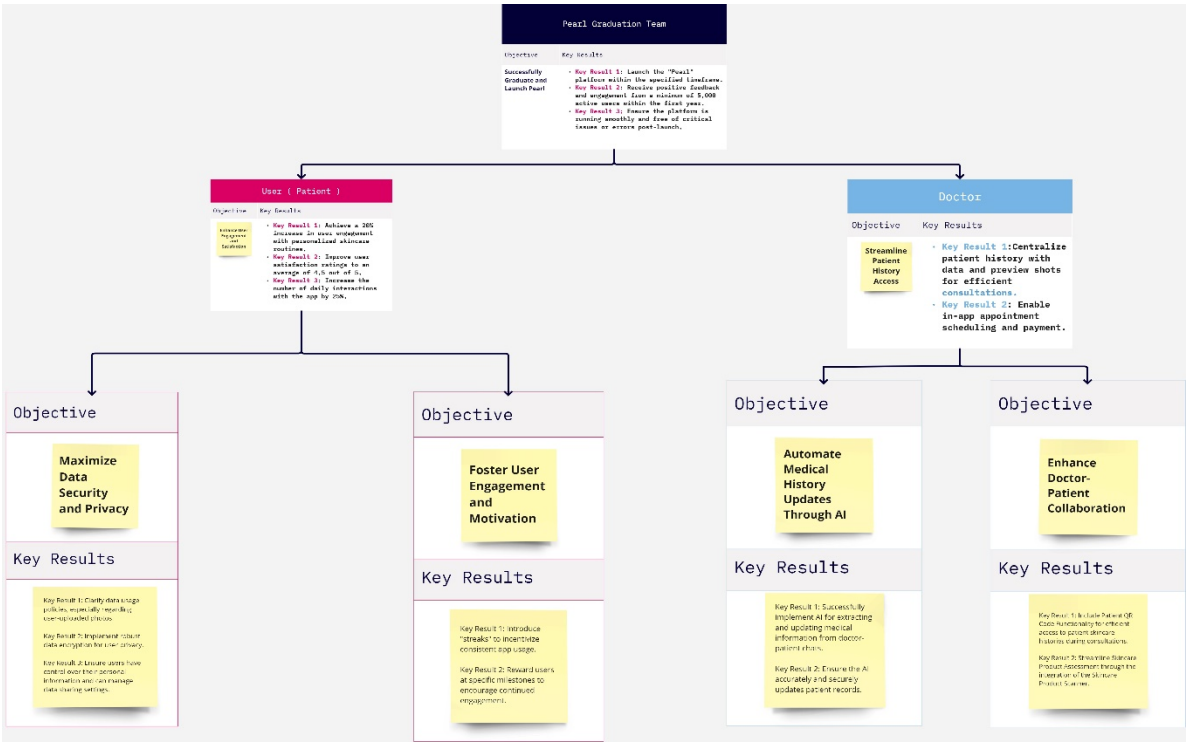


Figure III OKR Diagram

3.2.4 User Requirements

As a Teenage User:

- I want to have a personalized skincare routine.
- So that I can improve my skin condition and maintain its health.

As a Teenage User:

- I want to be able to edit my personal information and preferences.
- So that I can update my skincare routine to reflect any changes in my lifestyle or skin condition.

As a Teenage User:

- I want to take a picture of my face for skin analysis.
- So that I can receive personalized skincare recommendations based on the analysis.

As a Teenage User:

- I want to have access to online dermatologist appointments.
- So that I can consult with a professional for personalized advice and treatment.

As a Teenage User:

- I want to have a user-friendly and visually appealing interface.
- So that I can easily navigate the platform and find the information I need.

As a Dermatologist:

- I want to access patient profiles before appointments.
- So that I can provide more efficient and personalized consultations.

3.3 System Architecture

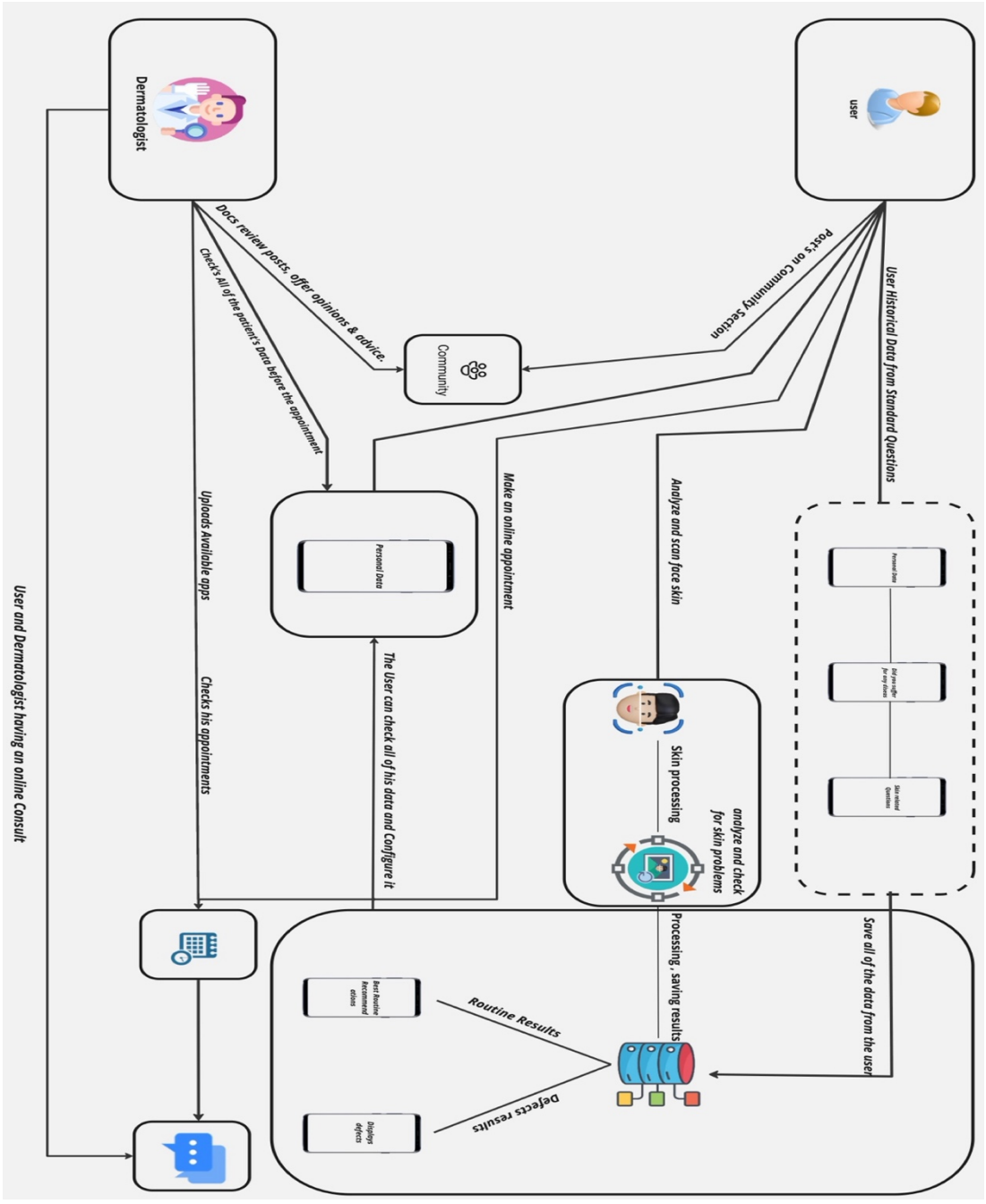


Figure IV System Architecture

Architecture diagram Description:

The architecture of the "Pearl" skincare platform is designed to provide a seamless and user-centric experience. Here's a detailed description of the system:

User Profile and Data Collection:

Users start by creating a profile, answering personal and skin-related questions.

The responses are stored in a secure database for later use in determining skin type and creating personalized routines.

Skin Analysis:

Users have the option to take a picture for a detailed skin analysis.

The image undergoes a processing stage, and the results are saved in the database for reference.

Routine Customization:

Users can edit their responses to the initial questions at any time, allowing for adjustments to their skincare routine.

Online Dermatologist Consultation:

Users can schedule appointments with dermatologists through the platform.

Dermatologists have access to the patient's profile and skin analysis results, streamlining the consultation process.

Continuous Data Management:

The system manages a comprehensive database of user information, skin analysis results, and appointment details.

This data is used to offer tailored skincare recommendations, and it can be accessed by both users and dermatologists.

Data Security:

Strong data security measures are in place to protect user information and ensure privacy.

3.4 Development Methodology

The development methodology chosen for this project is UML, which stands for Unified Modeling Language. UML is a widely accepted and standardized visual modeling

language used in software engineering. It provides a set of diagrams and notations to represent the system's structure, behavior, and interactions. In the context of this project, UML is utilized for modeling the system's architecture, use cases, sequence diagrams, and other visual representations to ensure a clear and consistent understanding of the project's design and functionality. UML aids in the efficient communication of ideas and concepts among the project team members and stakeholders, facilitating a smoother development process.

3.4.1 Use Case Diagrams

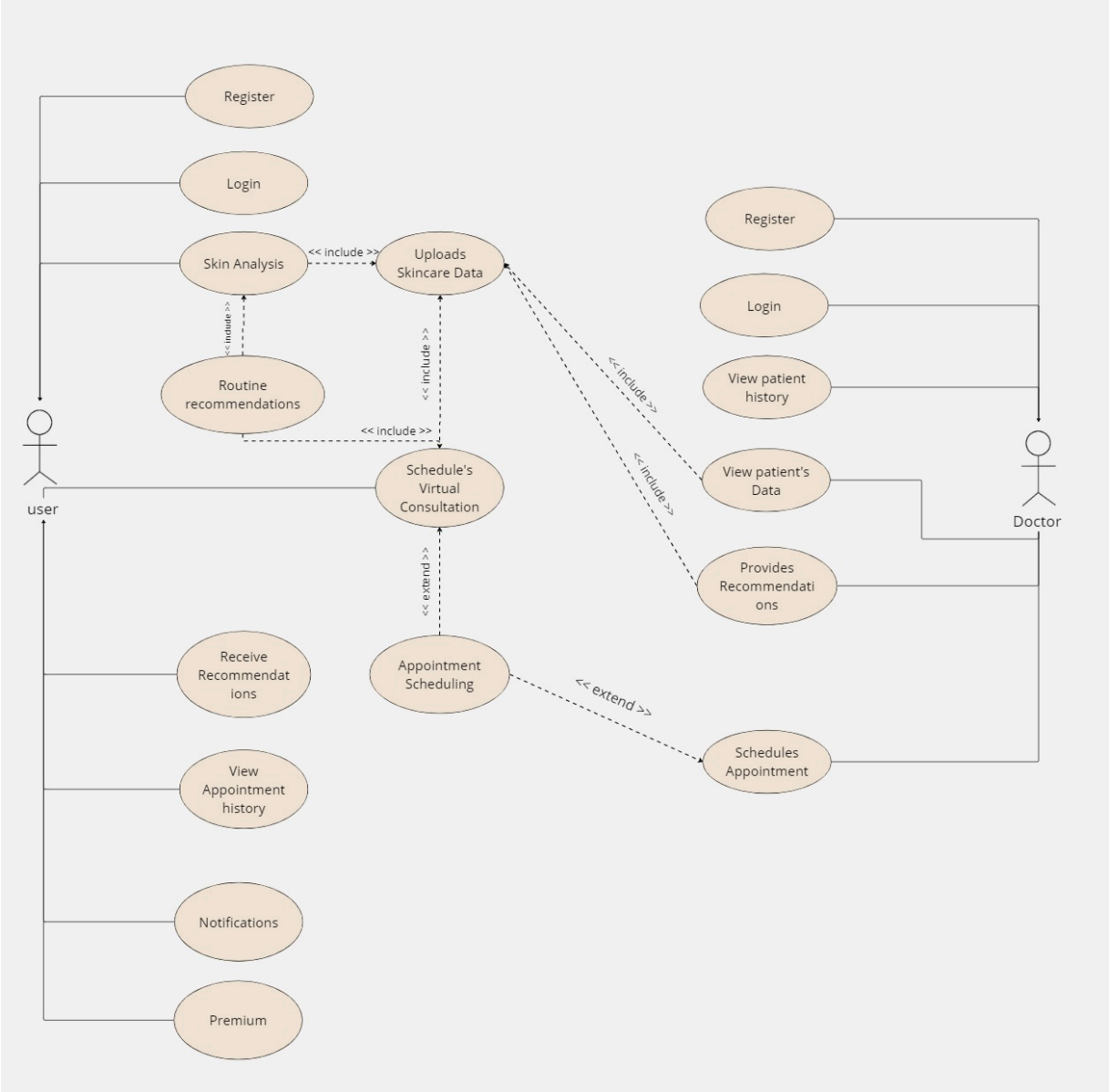


Figure V Use Case Diagram

3.4.2 Sequence Diagram

The way we explained how it works, have been detailed in Sequence diagrams, in the next couple of diagrams .

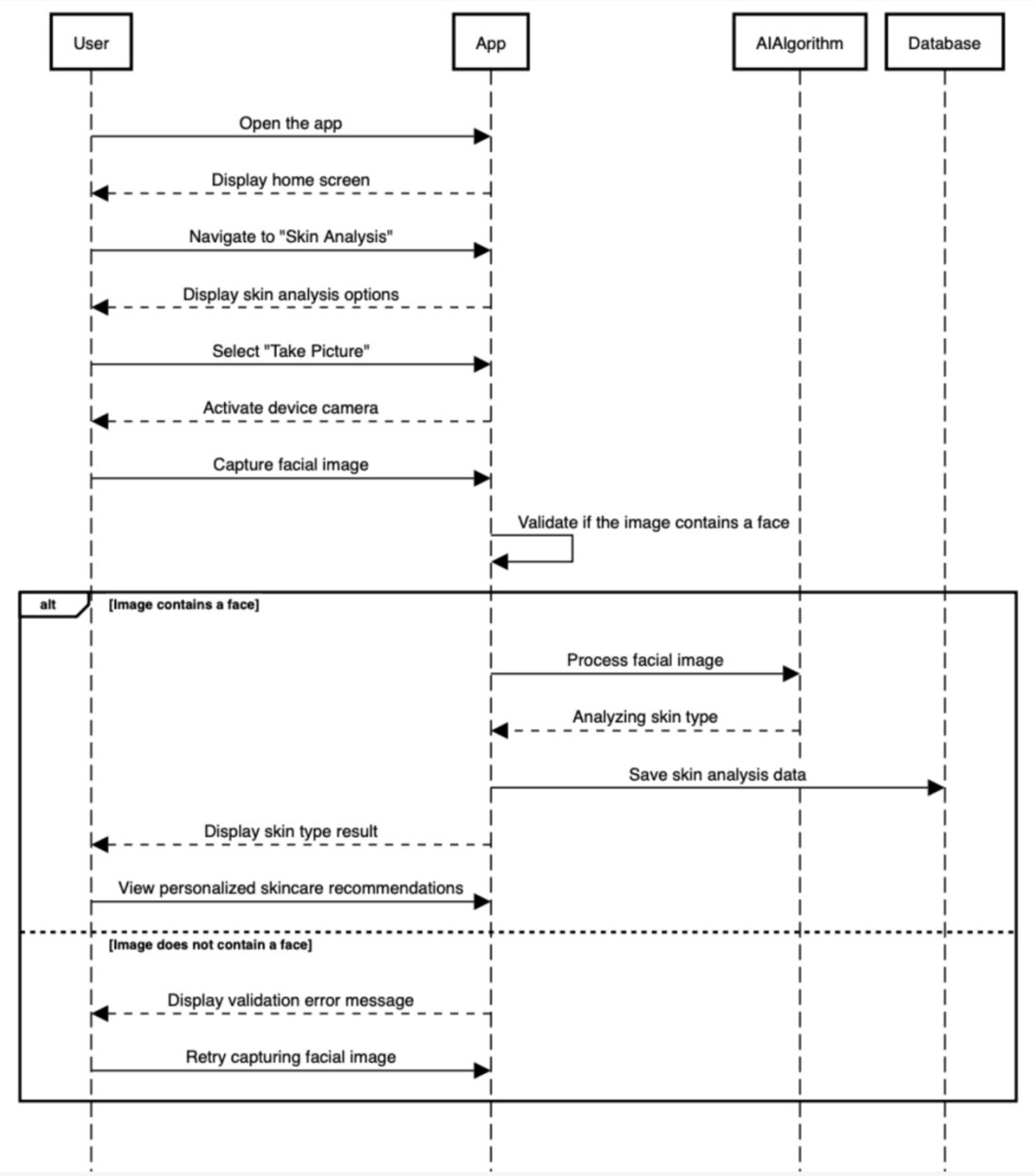


Figure VI Sequence Diagram " Skin Analysis "

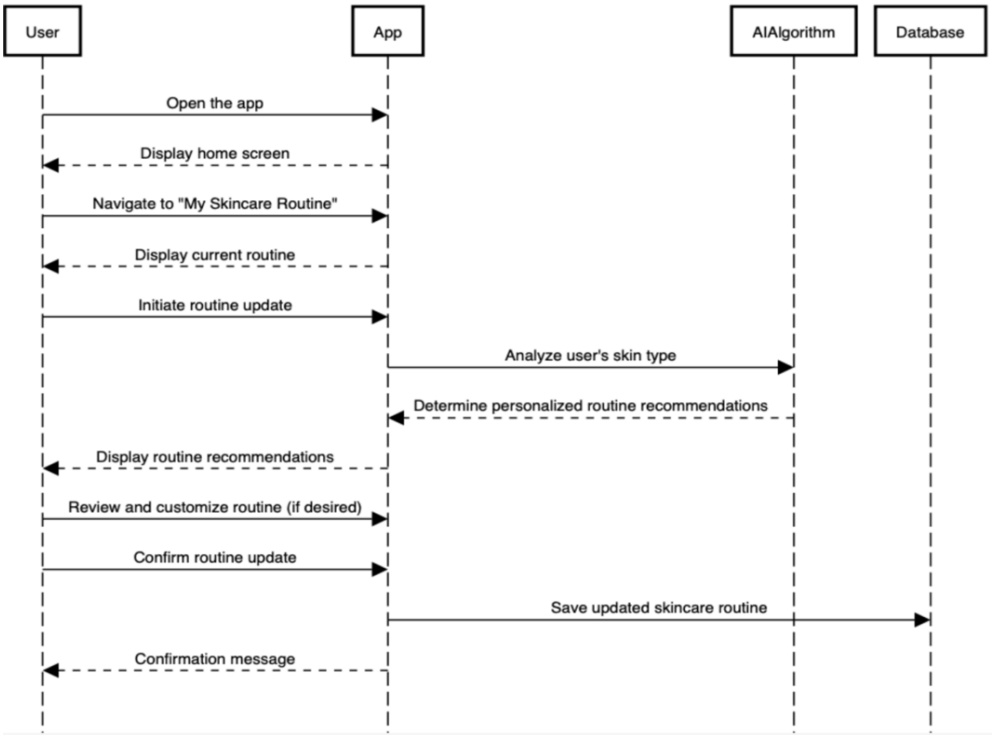
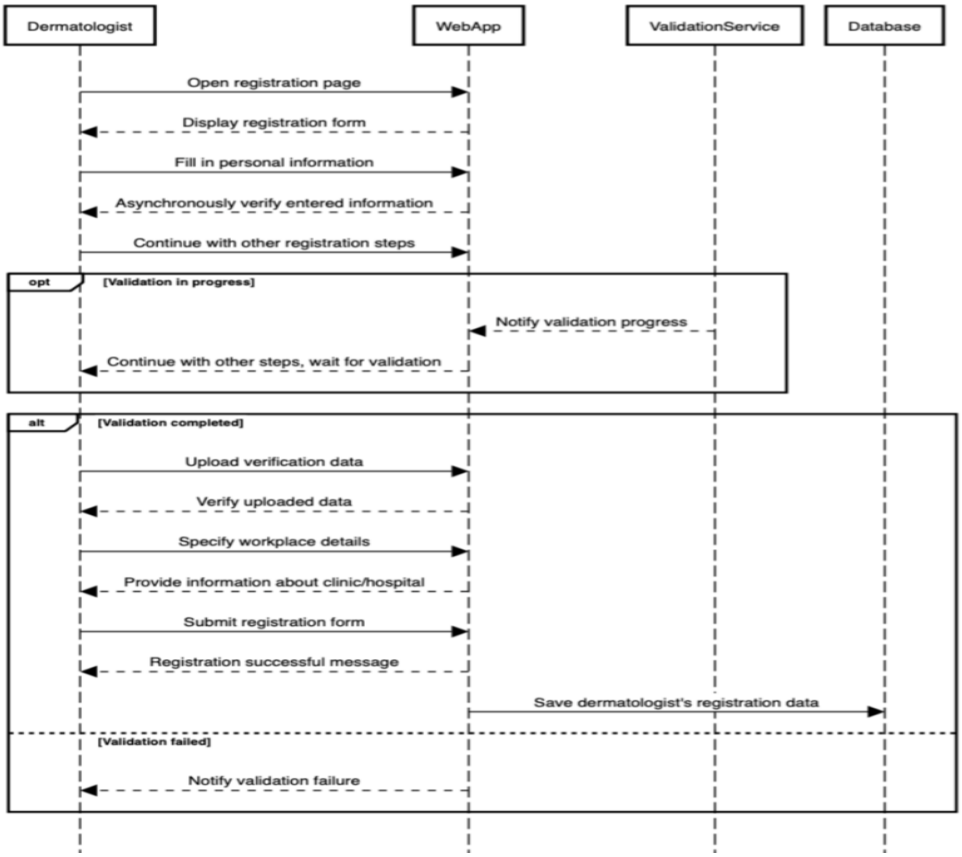


Figure VII Sequence Diagram " Routine Recommendation "



3.5 Tools and Languages

UI / UX

Figma

Figma is a powerful cloud-based design and prototyping tool used for creating user interfaces, interactive designs, and user experiences. It's known for its collaborative features, allowing team members to work together in real-time on design projects. Figma is widely used for creating wireframes, mockups, and prototypes, making it an ideal tool for UI/UX designers to streamline their design workflows and efficiently communicate design ideas.

Photoshop

Adobe Photoshop is a renowned raster graphics editing software that enables designers to manipulate, edit, and enhance digital images. It is frequently used for tasks such as image editing, photo retouching, and graphic design. Photoshop offers a wide range of tools for creating visual content, making it a fundamental tool for graphic designers, digital artists, and photographers.

Illustrator

Adobe Illustrator is a vector graphics editor that allows designers to create and edit vector-based illustrations, logos, and graphics. Unlike raster graphics, vector graphics are infinitely scalable without loss of quality, making Illustrator an essential tool for creating graphics and logos that need to maintain crispness at any size. It's commonly used in graphic design, logo design, and illustrations for both print and digital media.

Business analysis

Slack

Slack is a team collaboration platform designed for communication and file sharing. It provides channels for teams to organize their work and discussions. Slack enables real-time chat, file sharing, and integration with various tools and services, making it an effective hub for team communication.

Jira

Jira is a widely-used project management and issue tracking tool developed by Atlassian. It is designed to help teams plan, track, and manage their work. Jira allows for the creation and assignment of tasks, issue tracking, and agile project management. It is often used for software development and other project-related tasks.

Microsoft Teams

Microsoft Teams is a unified communication and collaboration platform that combines chat, video meetings, file storage, and app integration. It's part of the Microsoft 365 suite and enables teams to communicate and collaborate in a single workspace, making it particularly useful for remote or distributed teams.

Confluence

Confluence, developed by Atlassian, is a collaboration and documentation tool. It allows teams to create, share, and collaborate on projects and documentation. It supports the creation of wikis, knowledge bases, and other shared content repositories. Confluence is often used for creating and maintaining project documentation and knowledge sharing.

Miro

Miro is an online collaborative whiteboard platform that facilitates real-time visual collaboration. Teams can use Miro to brainstorm, plan, and work on various visual projects. It's particularly useful for diagramming, flowcharts, wireframing, and mind mapping.

Why these apps specially?

These tools are integrated to create a cohesive ecosystem for team collaboration, project management, and documentation. Slack serves as the communication hub where team members can discuss tasks and projects. Jira connects to Slack to provide task and project updates. Microsoft Teams allows video meetings and screen sharing to enhance team communication.

Confluence integrates with Jira to provide a space for detailed project documentation and knowledge sharing. Miro enhances visual collaboration and diagramming, making it a valuable addition to the suite.

This integration ensures that team members have seamless communication, easy access to project updates, robust documentation, and efficient visualization tools, enhancing the team's productivity and collaboration.

Development

Mobile

Kotlin

Kotlin is a modern, statically-typed programming language designed to be fully interoperable with Java, making it an excellent choice for Android app development. Developed by JetBrains, Kotlin offers concise syntax, expressive language features, and seamless integration with existing Java codebases. It is officially supported by Google for Android development.

Android

Android development refers to the process of creating applications for devices running the Android operating system. Android is a mobile operating system developed by Google, and it is widely used for smartphones, tablets, smartwatches, and other devices. Android applications are typically written in Java or Kotlin programming languages.

Frontend

HTML and CSS

HTML (Hypertext Markup Language) and CSS (Cascading Style Sheets) are fundamental web development technologies. HTML is used to structure web content, while CSS is used for styling and layout. These technologies are essential for creating web pages and user interfaces.

JavaScript

It is a high-level programming language used for adding interactivity to web pages. It allows manipulation of the DOM, handling user interactions, and communication with web servers.

Bootstrap

Bootstrap is a popular open-source front-end framework for web development. It provides a collection of pre-designed HTML and CSS components, making it easier to create responsive and visually appealing web applications. Bootstrap simplifies the process of designing and styling web interfaces.

Visual Studio Code

Visual Studio Code (VS Code) is a free, open-source code editor developed by Microsoft. It is widely used by developers for writing and editing code across various programming languages. VS Code is known for its flexibility and extensibility, offering numerous extensions and features that enhance the development workflow.

Backend

PHP

PHP is a server-side scripting language commonly used for web development. It is known for its simplicity and versatility in creating dynamic web pages and web applications. PHP is often used to interact with databases, handle form data, and perform server-side processing.

REST API

REST is an architectural style for designing networked applications. REST APIs enable communication between different software systems over the internet. They follow principles such as statelessness, uniform interfaces, and resource-based interactions. RESTful APIs are widely used for web services, providing data and functionality to clients.

Laravel

Laravel is a popular open-source PHP web application framework. It offers a rich set of tools and features for building web applications, including robust routing, authentication, and ORM (Object-Relational Mapping). Laravel simplifies common web development tasks and promotes clean, maintainable code.

Visual Studio Code

Visual Studio Code (VS Code) is a free, open-source code editor developed by Microsoft. It is widely used by developers for writing and editing code across various programming languages, including PHP. VS Code offers a range of extensions and features that enhance the development workflow when working with PHP and related technologies.

AI , Machine Learning , Deep Learning

Jupyter Notebook

Jupyter Notebook is an open-source web application that allows users to create and share documents that contain live code, equations, visualizations, and narrative text. It is commonly used for data science and machine learning tasks. With Jupyter Notebook,

developers and data scientists can interactively work with code, data, and visualizations in a collaborative and user-friendly environment.

Google Colab

Google Colab, short for Google Colaboratory, is a free cloud-based service provided by Google that offers a Jupyter Notebook environment. It allows users to write and execute Python code, create interactive documents, and leverage the power of Google's cloud infrastructure for machine learning and data analysis tasks. Google Colab provides access to GPUs and TPUs, making it suitable for training machine learning models.

Visual Studio Code

Visual Studio Code (VS Code) is a versatile and widely used code editor for various programming languages, including Python. It offers an extensive range of extensions and plugins that make it a favorite among developers for Python development. VS Code provides features for code editing, debugging, and version control integration, enhancing productivity in Python development.

Python

Python is a high-level, general-purpose programming language known for its simplicity and readability. It is widely used in data science, machine learning, web development, and many other domains. Python's extensive libraries and frameworks, such as TensorFlow, make it a go-to choice for developing machine learning models and data-driven applications.

TensorFlow Libaries

TensorFlow is an open-source machine learning framework developed by Google. It offers a comprehensive set of libraries and tools for building and training machine learning models. TensorFlow is known for its flexibility, scalability, and support for deep learning. It is extensively used for tasks like image recognition, natural language processing, and neural network development.

Why these apps specially?

These tools and technologies are instrumental in data science and machine learning, enabling the development, training, and deployment of advanced machine learning models and data analysis. Python, with its rich ecosystem of libraries and frameworks, serves as the primary programming language for these tasks. Visual Studio Code is an excellent code editor for Python development, while Jupyter Notebook and Google Colab provide interactive environments for data exploration and analysis. TensorFlow libraries are essential for building and training machine learning model

3.6 Summary

In the development of "Pearl," we have leveraged a range of cutting-edge technologies and tools to create a sophisticated and highly personalized skincare platform. Our core technologies include AI and machine learning, cloud computing, mobile application development frameworks, and data analysis tools. These choices were made to ensure we can deliver tailored skincare recommendations, maintain scalability, provide a user-friendly mobile interface, and analyze data effectively. Complemented by the agile Scrum methodology, these technologies and tools collectively empower "Pearl" to offer a unique and personalized skincare experience while remaining adaptable to changing user needs and ensuring high-quality services.

Chapter 4

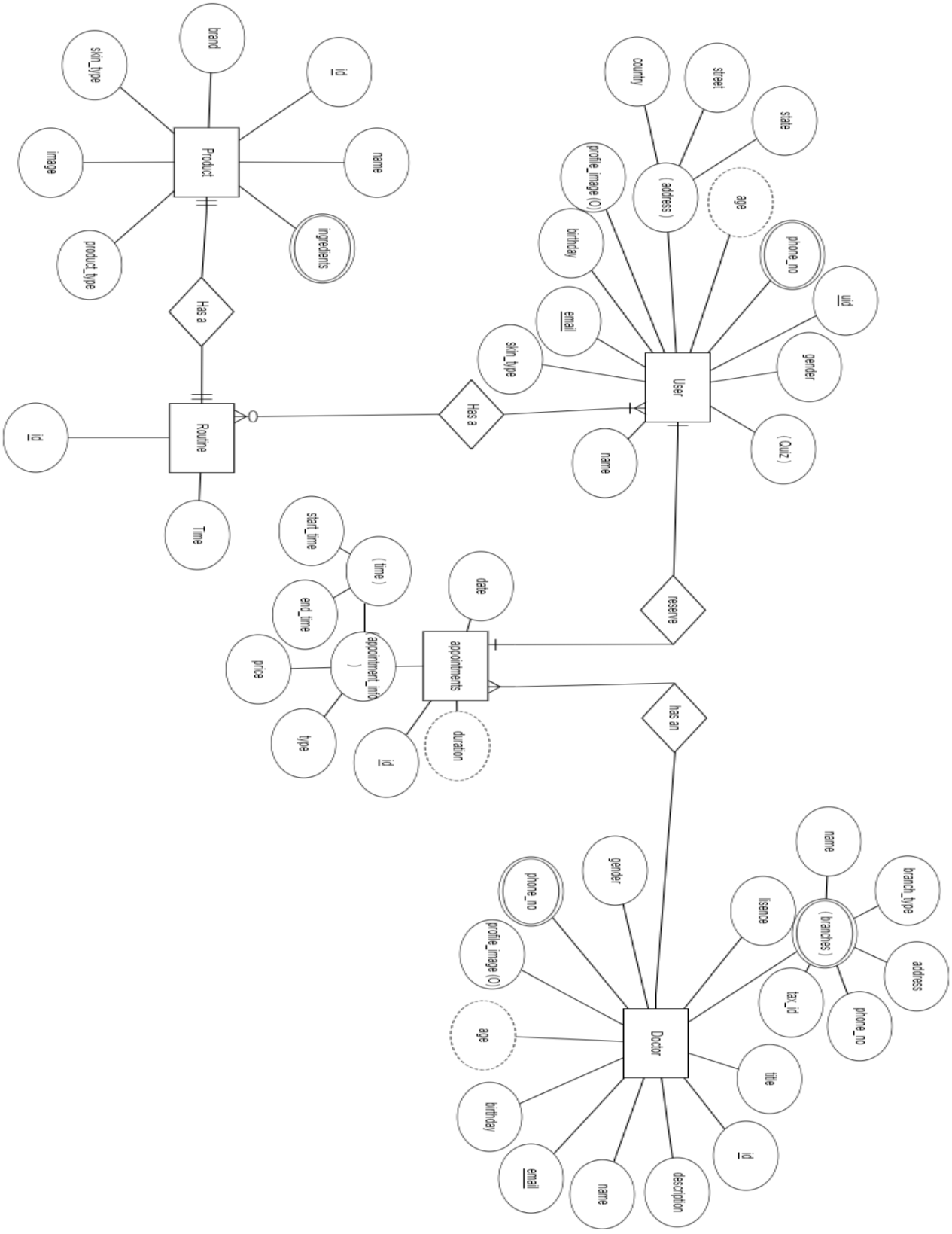
Chapter 4: System Design

The design of the investigation explains how the study was formulated to investigate each question or hypothesis and if appropriate, it identifies all variables and how they are manipulated.

4.1 Introduction

The System Design chapter provides a detailed exploration of key components essential to our project. Dive into the intricacies of Entity-Relationship Diagrams (ERD) and database design, understanding the nuances of structuring data effectively. Unpack the principles of interface design, emphasizing user-friendly experiences. Extend your knowledge to mobile and web design, ensuring a cohesive and optimized system across platforms. This chapter serves as a concise guide for developers, designers, and stakeholders, offering insights into the foundational aspects of our system design.

4.2 ER Diagram



4.3 Interface Design

Our UI design revolves around simplicity, clarity, and user-friendliness. We're committed to minimizing your time and effort, ensuring a seamless and enjoyable experience. We choose smooth and calming colors to be gentle on the eyes, these hues create an inviting environment, encouraging daily use. Embrace a soothing and seamless experience every day. That making your interaction with the app not only efficient but also visually refreshing. The following subsections show the designed views of the mobile application and description of each view

4.3.1 Choose Language Screen

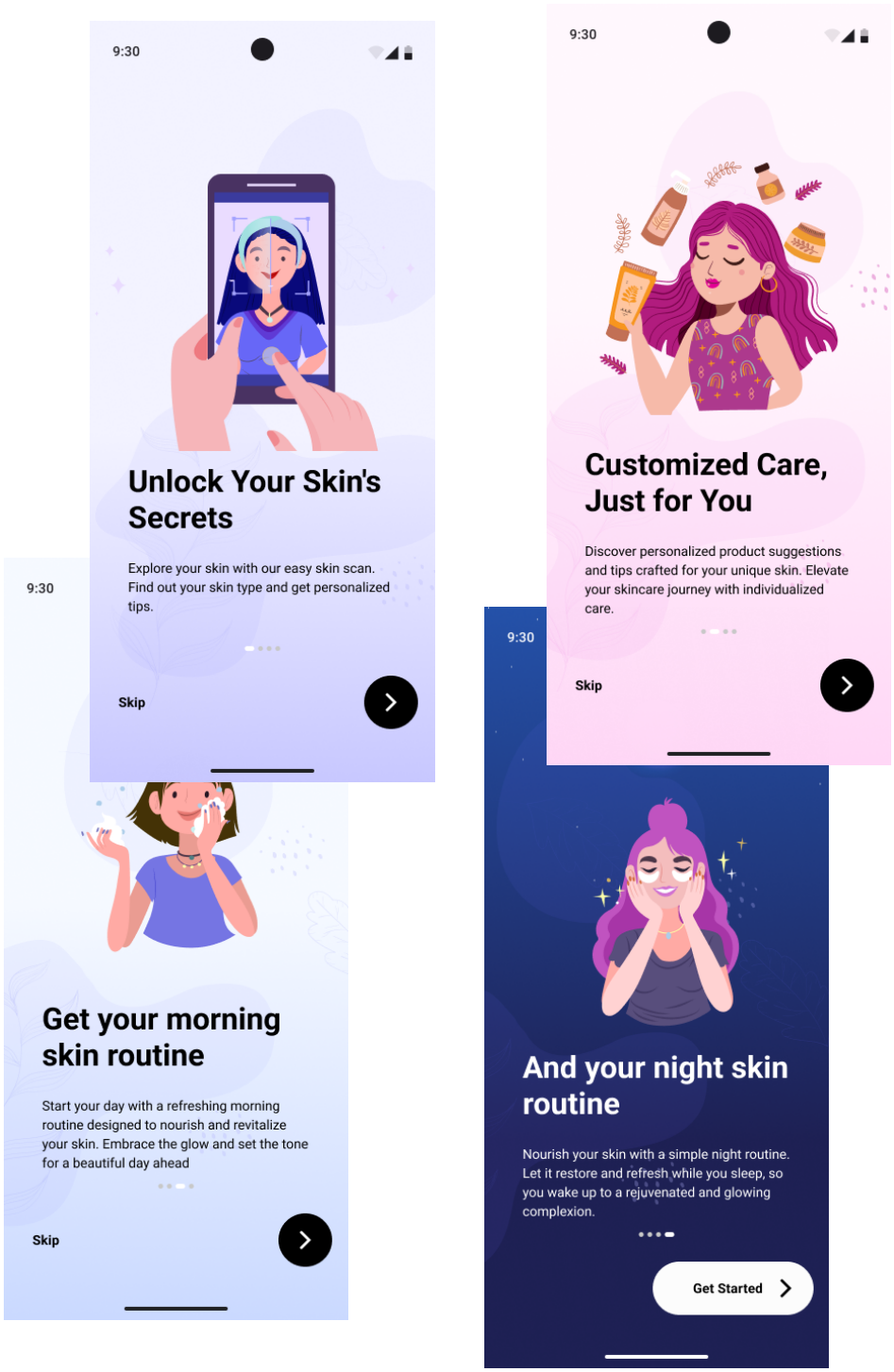
Allows users to choose their preferred language, to establish a personalized and comfortable experience.
Options: Arabic and English, providing diverse lan guage choices.





4.3.2 Onboarding Screens

As users enter the app for the first time, a series of welcome pages unfold, presenting core functions. These concise pages outline key features, culminating in a final page featuring a prominent "Get Started" button, beckoning users to initiate their interaction with the application.



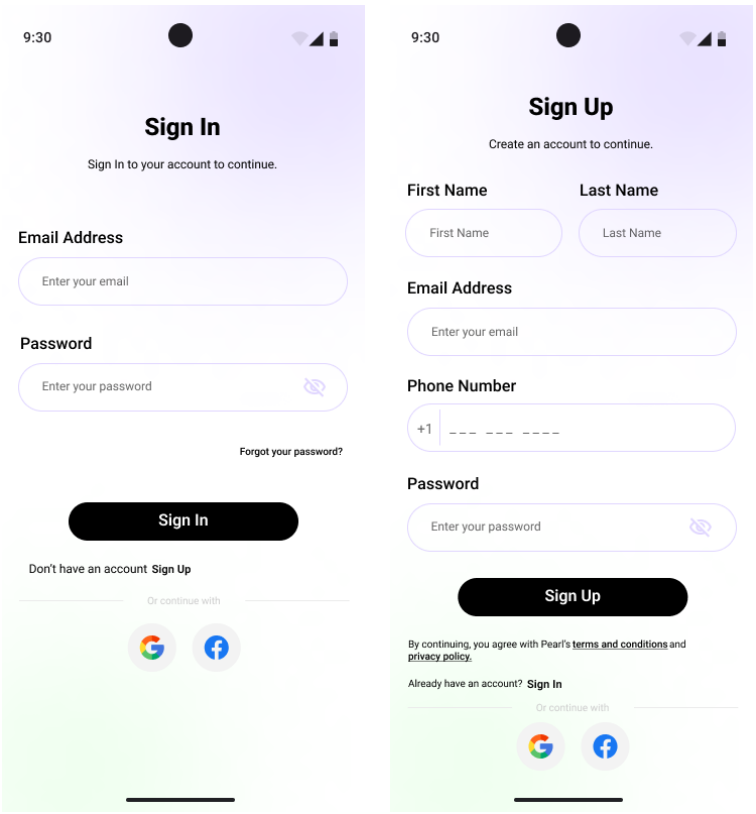
4.3.3 Registration

Sign in page:

1. Username Field: Enter your username.
2. Password Field: Enter your password.
3. Login Button: Confirm access to your account.
4. Facebook Button: Login with your Facebook account.
5. Google Plus Button: Login with your Google Plus account.

Sign up page:

1. Name Fields: To enter your first and last name.
2. Email Field: To enter your email as a primary contact and login credential.
3. Phone Number Field: To enter your phone number as an additional contact information.
4. Password Field: To enter your password to secure your account
5. Sign Up Button: Confirm registration process.
6. Gmail Button: To Signup with your Google account.
7. Facebook Button: To Signup with your Facebook account.

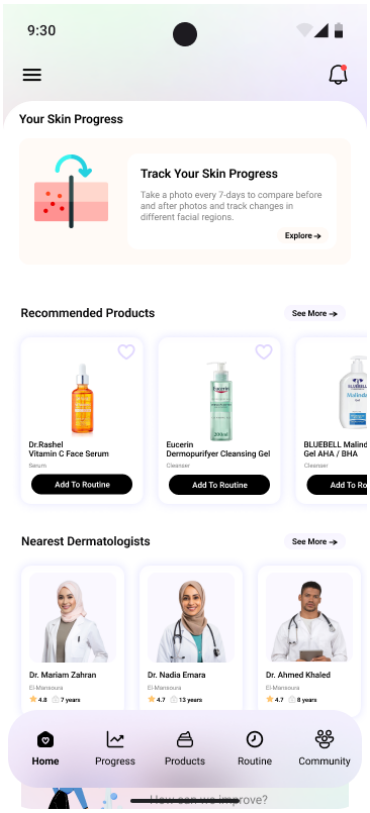


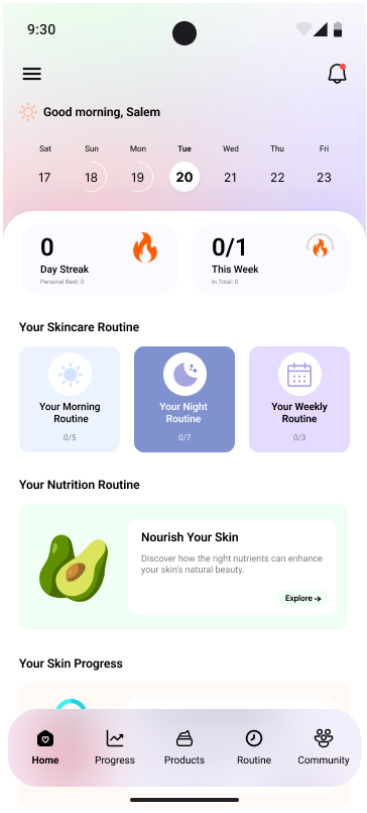
4.3.4 Home Page

The user can select the function wants from the home page. home page contain a lot of function as explained below:

1. Streaks Section: Monitors and displays daily and weekly usage streaks within the app and encourages regular and consistent engagement.
2. Routines: Houses your personalized morning, night, and weekly skincare routines.
3. Nutrition Routine: Provides dietary guidance tailored to skin type, issues, and allergies, and it includes suggestions for foods to avoid, stick to, and recommended recipes.
4. Skin Progress: Allows users to track and monitor their skin's journey and improvements over time.
5. Recommended Products: Presents a curated selection of skincare products tailored to your skin case. And provides detailed information on each recommended product.

6. Social Community Button: Connects users with a community of specialized experts and other users to facilitate engagement and knowledge-sharing.
7. Suggested Dermatologists: Enables users to discover, communicate with nearby dermatologists based on their location, and promotes accessibility to professional skincare advice.
8. Your Profile Button: Centralizes user data for easy access and modification, and empowers users to manage and update their personal information.
9. Notification Button: Allows users to view and manage notifications, and to Keep users informed about updates, progress, and community interactions.

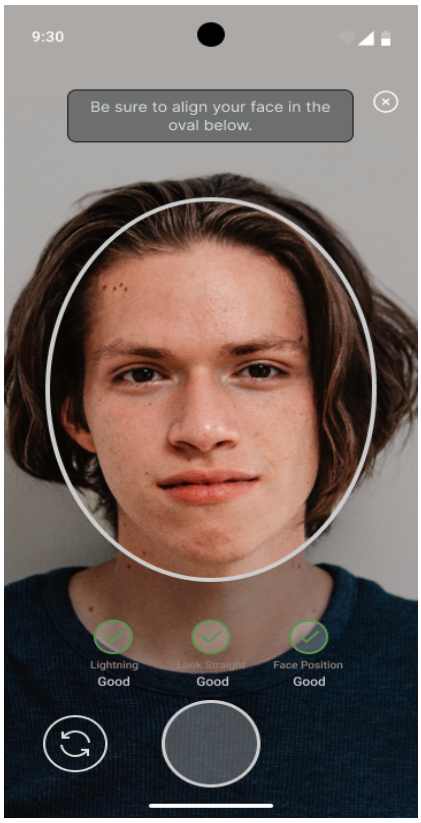




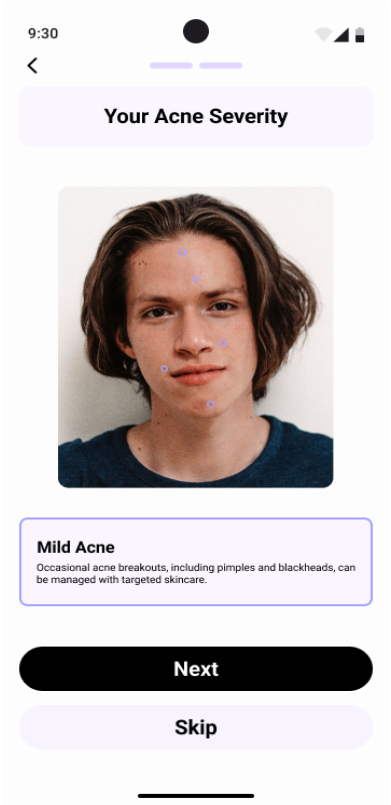
4.3.5 Scan and Diagnosis Pages

The Scan and Diagnosis page serves as the cornerstone feature of the app, dedicated to detecting both skin type and any associated skin concerns. Prior to initiating the scan, users are guided to ensure optimal conditions such as adequate lighting and proper facial positioning.

Once these prerequisites are met, the scanning process commences, leveraging sophisticated algorithms to analyze the user's skin comprehensively. The results encompass an accurate assessment of the user's skin type along with the identification of any prevalent skin concerns, including acne, dryness, or sensitivity. By offering precise insights into skin health, this feature empowers users to devise personalized skincare regimens tailored to their unique needs, thereby fostering a proactive approach to skincare management.


9:30
Be sure to align your face in the oval below.
Lightning
Good
Look Straight
Good
Face Position
Good

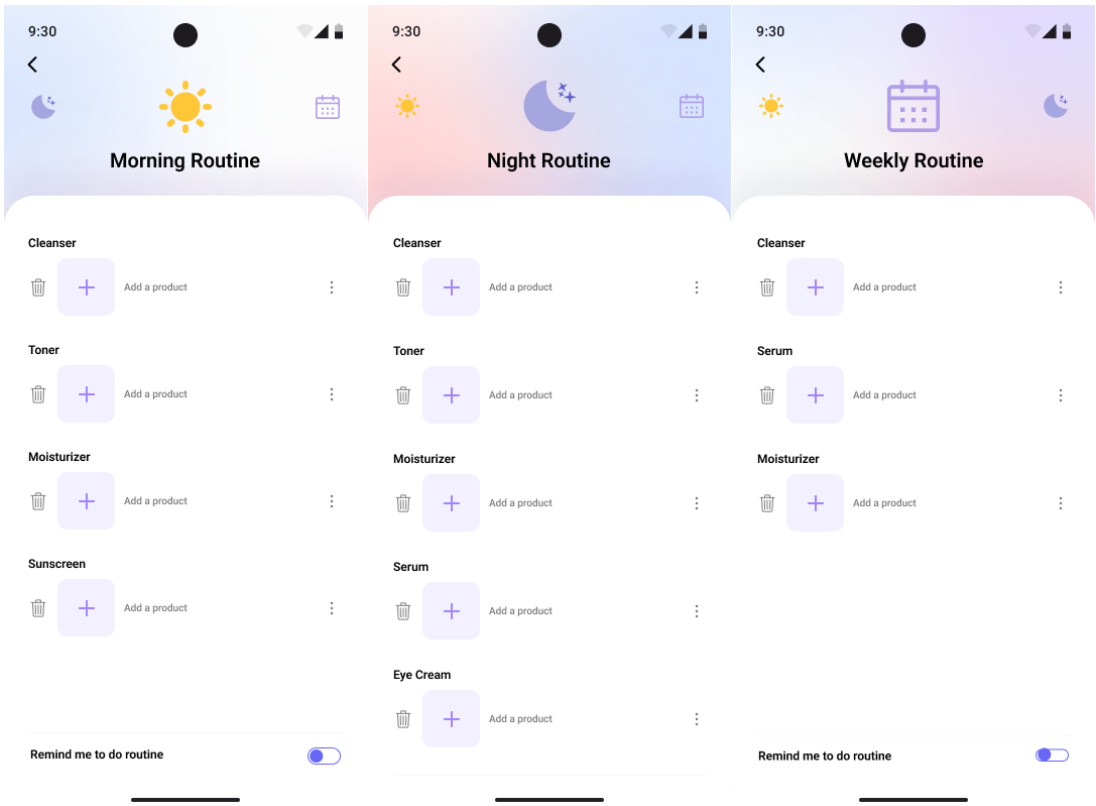

9:30
Your Skin Type
Dry
Next
Skip

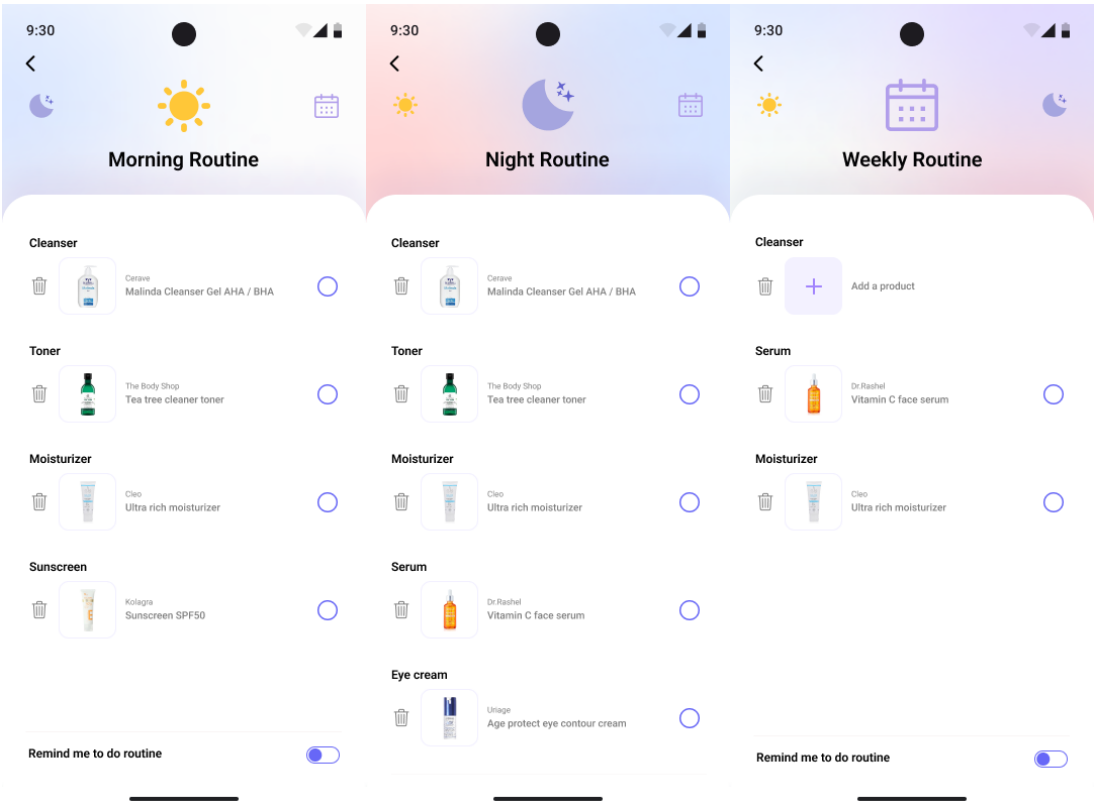

9:30
Your Acne Severity
Mild Acne
Occasional acne breakouts, including pimples and blackheads, can be managed with targeted skincare.
Next
Skip

4.3.6 Skincare Routine Pages

The Routine Pages (Morning, night and weekly routine) are designed to empower users with personalized and adaptable skincare routines, promoting user engagement and effective skincare practices

1. Add Product: Allows users to add new skincare products to their routine, and enables customization and adaptation of routines based on changing skincare needs.
2. Remove Product: Enable users to remove or replace products from their routine, and to ensure flexibility and adaptability in response to changing preferences or skin conditions.
3. View Product Details: Offers in-depth information about each skincare product. And empowers users to make informed choices and understand the impact of each product on their skin.
4. Check Used Products: Provides a quick overview of products used in the current routine, and facilitates tracking and consistency in daily skincare practices.
5. Activate Reminder: Enables users to set reminders for their skincare routines.


9:30
Morning Routine
Cleanser
Add a product
Toner
Add a product
Moisturizer
Add a product
Sunscreen
Add a product
Remind me to do routine
9:30
Night Routine
Cleanser
Add a product
Toner
Add a product
Moisturizer
Add a product
Serum
Add a product
Eye Cream
Add a product
9:30
Weekly Routine
Cleanser
Add a product
Serum
Add a product
Moisturizer
Add a product
Remind me to do routine



4.3.7 Nutrition Routine Pages

1. Water Intake Tracking: Allows users to monitor and track their daily water intake, and promotes hydration for healthier skin, and aids in achieving daily water intake goals.
2. Food Recommendations: Provides guidance on foods to “avoid” and “stick to” based on skin type, issues, and allergies, and tailors nutrition advice to support skin health and addresses individual dietary restrictions.
3. Recipe Suggestions: Offers a selection of suggested recipes tailored to skincare needs.

9:30

Water Intake

0 / 3000 ml

1 Cup

Nutrition Routine



Stick to or Avoid

Optimize your skin health by making mindful choices. Avoid skin-stressing foods and stick to nourishing options for a radiant complexion.

Explore →



Recipes for you

Indulge in a culinary journey with recipes customized for your skin. Discover delicious and nutritious options designed to enhance your skin's natural radiance.

Explore →

9:30

Nutrition Routine

Stick to



Fruits and Vegetables

Colorful fruits and vegetables are rich in antioxidants, which can help combat free radicals.



Omega-3 Fatty Acids

Salmon, chia seeds, and flaxseeds, may help reduce inflammation and contribute to healthy skin.



Whole Grains

Brown rice, quinoa, and oats have a lower glycemic index, which may help control skin oil production



Vitamin A-rich Foods

Carrots, spinach, and leafy greens support the shedding of dead skin cells, preventing them from pores.

Avoid



Dairy

Dairy products can trigger acne or worsen oily skin.

9:30

Recipes for you



Grilled Salmon & Quinoa

It provides omega-3 fatty acids, balanced nutrition, and essential vitamins, potentially regulating oil production.

374 249 42 128



Citrus Green Juice

This Juice provides hydration, antioxidants, vitamins, and potential anti-inflammatory benefits.

238 114 27 53



Lentil & Vegetable Soup

It provides a wholesome blend of plant-based protein, fiber, vitamins, and minerals, contributing to skin health.

84 72 15 24



Avocado & Chickpea Salad

It delivers a nourishing mix of healthy fats, fiber, antioxidants, and vitamins, supporting skin health.

127 97 32 41



Mediterranean Pizza

It delivers a nourishing mix of healthy fats, fiber, antioxidants, and vitamins, supporting skin health.

238 114 27 53

9:30

Grilled Salmon & Quinoa



Ingredients:

- Salmon fillets
- Quinoa
- Mixed vegetables (broccoli, carrots, bell peppers)
- Olive oil
- Lemon juice
- Garlic
- Salt and pepper to taste

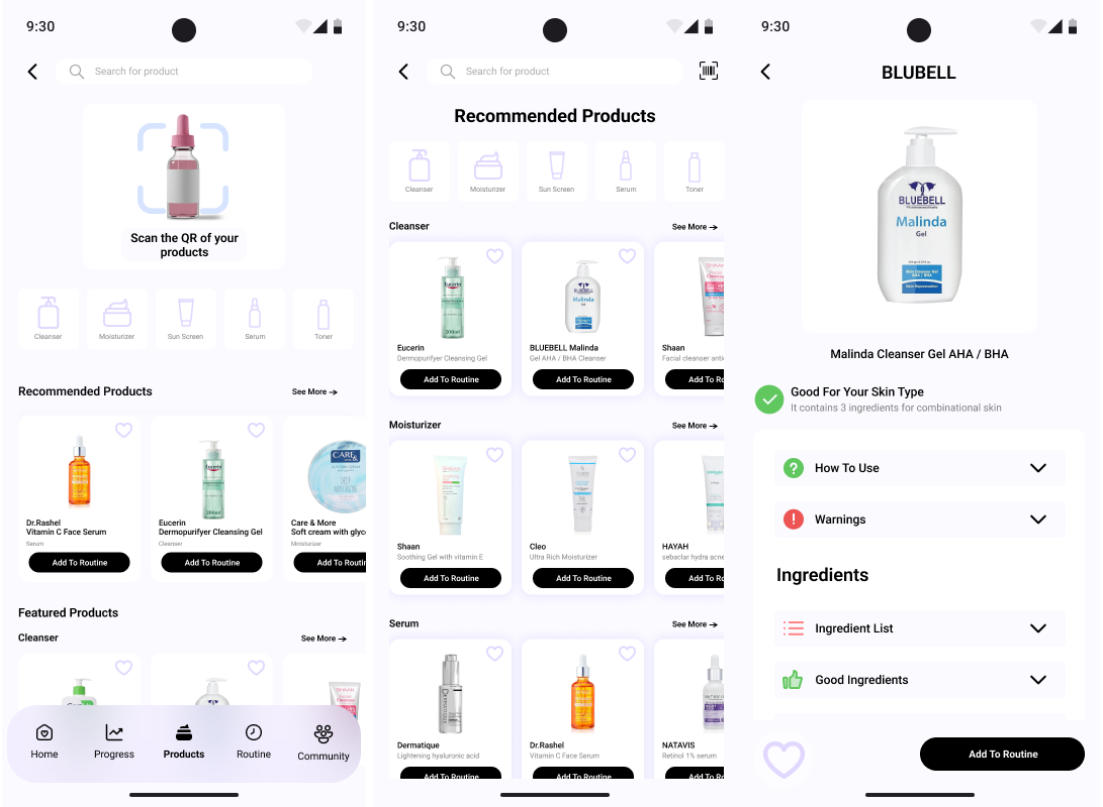
Instructions:

1. Grill the salmon fillets with a drizzle of olive oil, lemon juice, minced garlic, salt, and pepper.
2. Cook quinoa according to package instructions.
3. Steam a mix of colorful vegetables.
4. Serve the grilled salmon over a bed of quinoa and steamed vegetables.



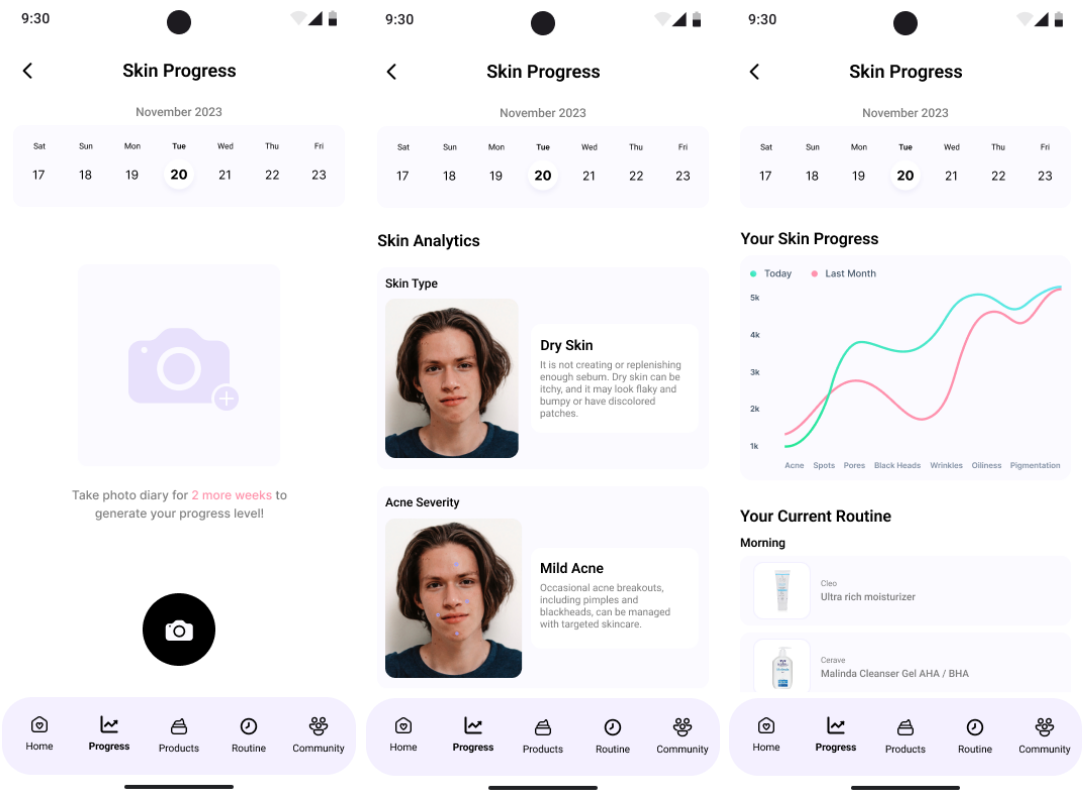
4.3.8 Products Pages:

1. Product Categories: Organizes products into different and facilitates easy navigation and browsing of products based on user preferences.
2. Recommended Products Section: Displays carefully selected product recommendations customized to match the user's specific skin type and skincare needs. it helps users discover new products aligned with their skincare needs, enhancing their overall skincare routine.
3. Usage Instructions: Provides clear and concise guidance on how to use the product effectively, and ensures users understand the proper application and usage frequency for optimal results.
4. Warnings and Precautions: Alerts users to any potential side effects or precautions associated with the product, it helps users use the product safely and avoid adverse reactions.
5. Ingredient List: Offers a detailed list of ingredients used in the product formulation. And enables users to identify specific ingredients and potential allergens.
6. Good and Bad Ingredients for Skin: Highlights beneficial and harmful ingredients present in the product, it empowers users to make informed decisions about product suitability based on ingredient analysis.
7. Add to Routine Button: Allowing users to easily incorporate the product into their skincare regimen with a simple click



4.3.9 Skin Progress

The Skin Progress feature enables users to track their skincare journey by taking a photo of their skin each week. This photo serves as a reference point for assessing progress over time. The app compares the new photo with previous ones, highlighting any changes in the user's skin condition. Depending on the user's specific skin concerns, such as acne, wrinkles, or hyperpigmentation, the app provides insights into the progress made. Additionally, users receive information about the products they used during the week, allowing them to correlate skincare routines with observed changes in their skin. This feature empowers users to monitor their skincare progress effectively and make informed decisions about their skincare routines.



4.3.10 Dermatologists Pages

By utilizing the user's location, the app recommends nearby dermatologists, ensuring accessibility and convenience. Each dermatologist's profile presents comprehensive information, including their experience, qualifications, ratings, and clinic location. Users can easily schedule appointments with dermatologists, choosing between online consultations or in-person visits, and paying for appointments using various payment methods. Additionally, user reviews offer valuable insights into the dermatologist's practice and patient satisfaction, aiding users in making informed decisions about their skincare journey. Overall, these features empower users to connect with trusted skincare professionals seamlessly and receive personalized care tailored to their needs.

9:30

Nearest Dermatologists

Location

El Mansoura, Egypt

Search for dermatologist

Dermatologists | Favorites | My Bookings



Professional Doctor

Dr. Mariam Zahran

El Mansoura

7 years

4.8 | 143 Reviews

Make Appointment



Professional Doctor

Dr. Nadia Emara

El Mansoura

13 years

4.7 | 128 Reviews

Make Appointment



Professional Doctor

Dr. Ahmed Khaled

El Mansoura

8 years

4.7 | 142 Reviews

Make Appointment

9:30

Dermatologist Details



Dr. Mariam Zahran

El Mansoura, El Gaish St

7+ Years Exp

4.8+ Rating

143 Reviews

About

Dr. Mariam, an accomplished dermatologist in El Mansoura with seven years of specialized experience, excels in diagnosing Read more

Working Hours

Saturday	12 pm : 10 pm
Tuesday	12 pm : 10 pm
Thursday	12 pm : 10 pm

Address

24 El Gaish St, El Mansoura

Book Appointment

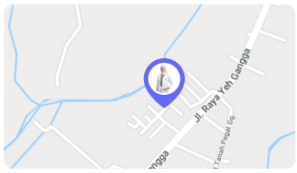
9:30

Dermatologist Details

Thursday	12 pm : 10 pm

Address

24 El Gaish St, El Mansoura



Reviews

Tap to Rate:

Add Review

Merna

Dr. Mariam's expertise and personalized care transformed my skin, grateful for her outstanding dermatological support!

Ahmed

Dr. Mariam's exceptional care and expertise made my skin journey remarkable, highly recommend!

Book Appointment

9:30

Book Appointment



Dr. Mariam Zahran

El Mansoura, El Gaish St

Select Package

Online Appointment — 150 LE

Chatting and videocall — /30 min

In Person — 200 LE

Visit the doctor — /30 min

Day

Today 20 OCT | Thu 22 OCT | Sat 24 OCT

Time

1:00 pm | 1:30 pm | 2:00 pm

Next

9:30

Payment Method

Credit and Debit Card

Add New Card

More Pyment Options

PayPal

Visa card

Master Card

Cash Payment

Cash Payment

Next

9:30

Review Summary



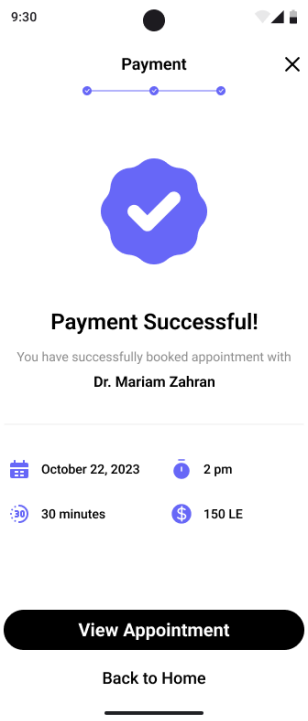
Dr. Mariam Zahran

El Mansoura, El Gaish St

Date	October 22, 2023
Package	Online Appointment
Time	2:00 - 2:30
Duration	30 minutes
Amount	150 LE
Duration (30 minutes)	1 x 150LE
Duration	30 minutes
Total	150 LE

Visa Card — Change

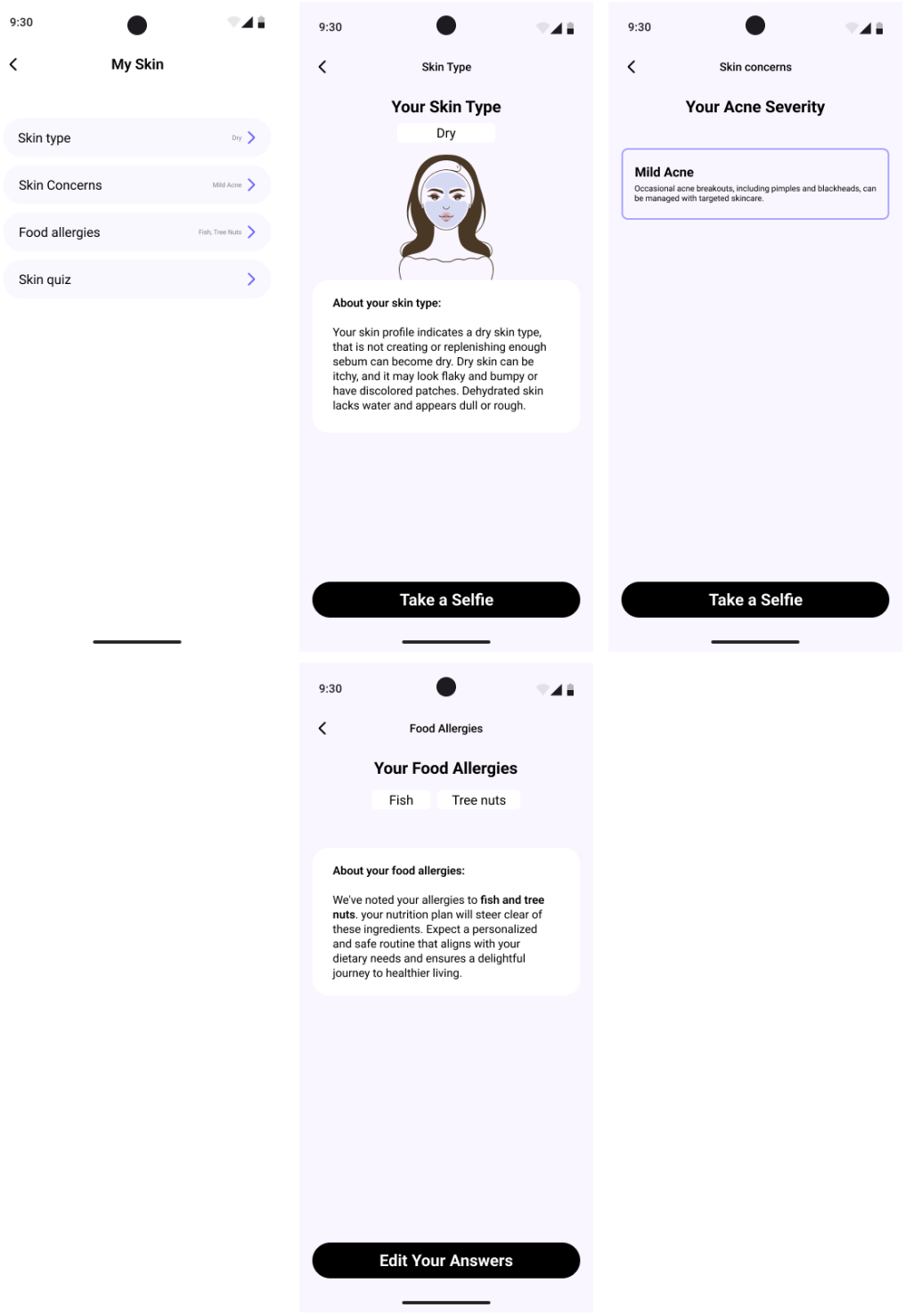
Pay Now



4.3.11 My Skin Pages

My Skin screens provides users with essential tools to understand and manage their skincare:

1. Skin Type: Displays information about the user's skin type based on the initial picture taken. Users can retake the picture if needed.
2. Skin Concerns: Offers insights into specific skin concerns identified from the initial picture. Users have the option to retake the picture for accurate results.
3. Food Allergies Button: Provides information based on the user's initial quiz responses regarding food allergies. Users can review and adjust their answers at any time to ensure accurate recommendations.
4. Skin Quiz Page: Allows users to revisit and update their answers to the initial skin quiz, ensuring ongoing accuracy and personalized skincare recommendations.



4.3.12 User Profile

User Profile screen allows users to conveniently review and edit their personal details.

9:30

Your Profile



Name

Salem Ahmed

Phone Number

+20 102 896 5872 Change

Email

salemahmed@gmail.com Change

Country

Egypt

Gender

Male

Date of Birth

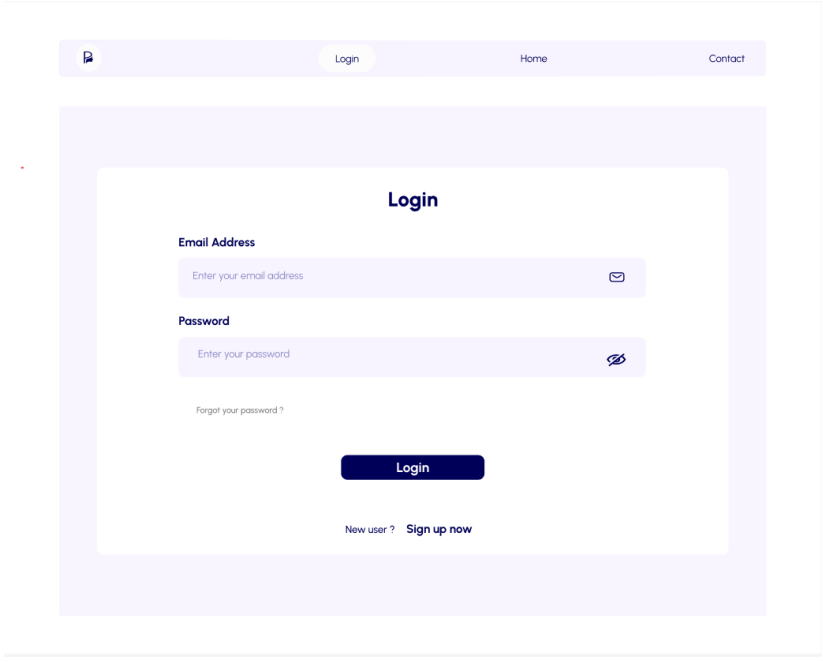
1 Sep 2002

Update Profile

4.4.13 website Our UI design revolves around simplicity, clarity, efficiency, and user-friendliness. We're committed to minimizing your time and effort, ensuring a seamless and effective experience. We choose smooth and clear colors to attract your eyes, these hues create an inviting environment, encouraging daily use. Embrace a soothing and seamless experience every day. That makes your interaction with the website not only efficient but also visually refreshing. The following subsections show the designed views of the website and description of each view.

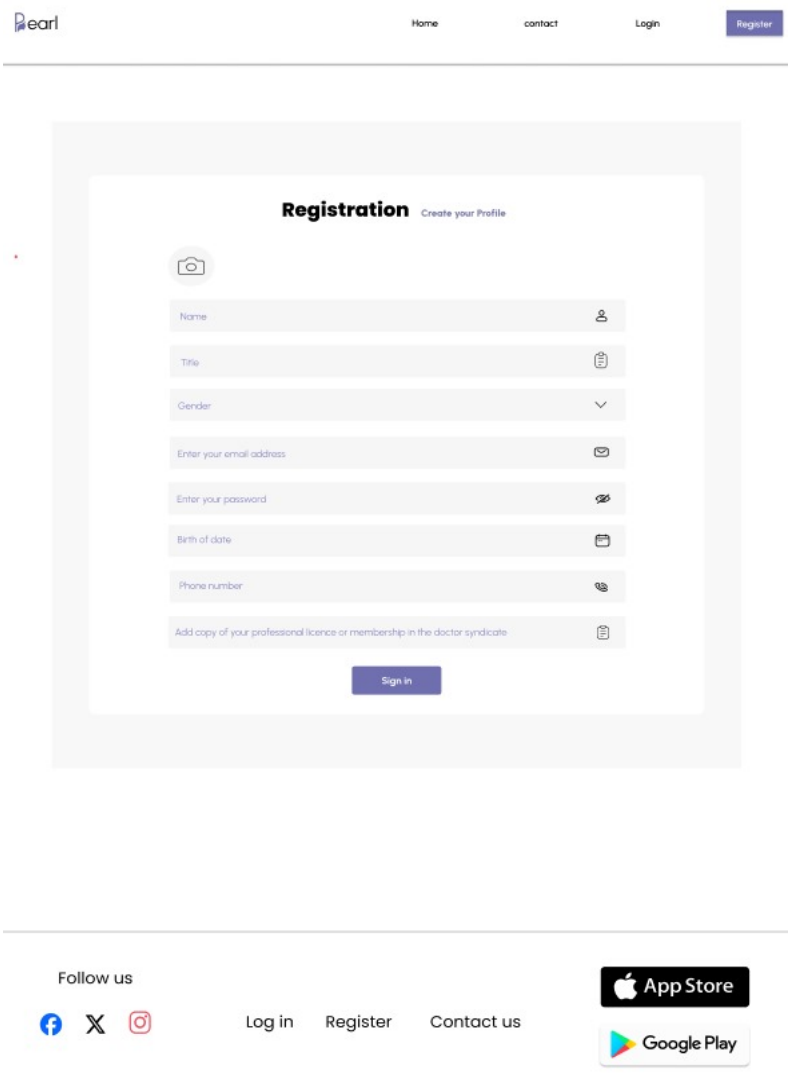
4.4.14 Login screen:

1. Email field: doctor enter his email.
2. Password: doctor enter his password.
3. Login button: enable doctor to access his account.
4. Signup button: enable the doctor to create his account.



4.4.15 Registration Screens

1. Name field: the doctor enters his first and last name.
2. Title field: the doctor enters his job title.
3. Email field: the doctor enters his email.
4. Password field: the doctor enters his password.
5. Birth of date field: the doctor enters his Birth date.
6. Phone number field: the doctor enters his phone number.
7. Verification field: the doctor uploads a copy of his professional license or membership in the doctor syndicate.
8. Profile Picture field: the doctor enters his profile picture.
9. Next button: takes the doctor to the next step of registration.



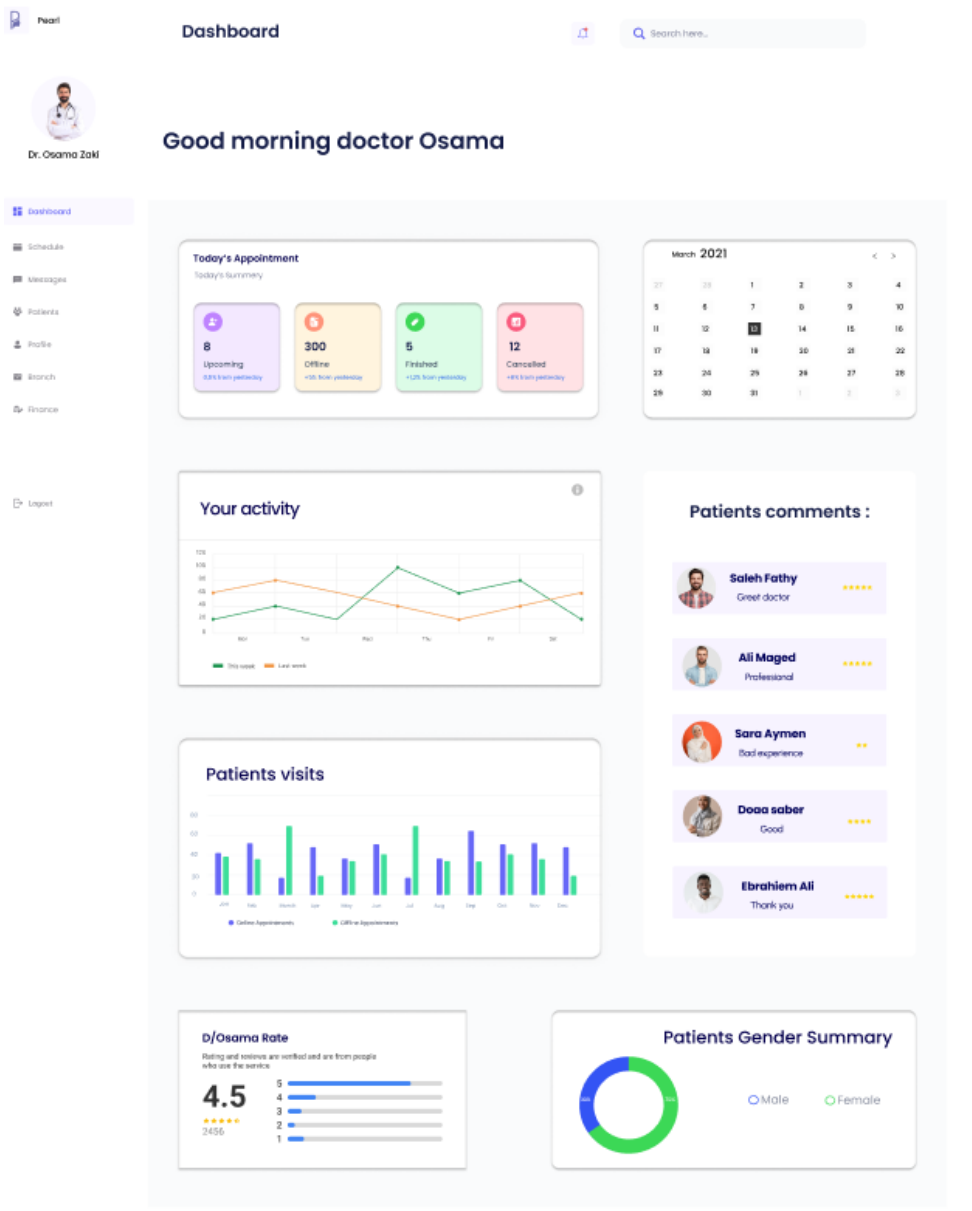
Now the doctor data is under verification, when pearl team make sure thar data is correct, the doctor will be able to login and access his dashboard.

4.4.16 Dashboard Page

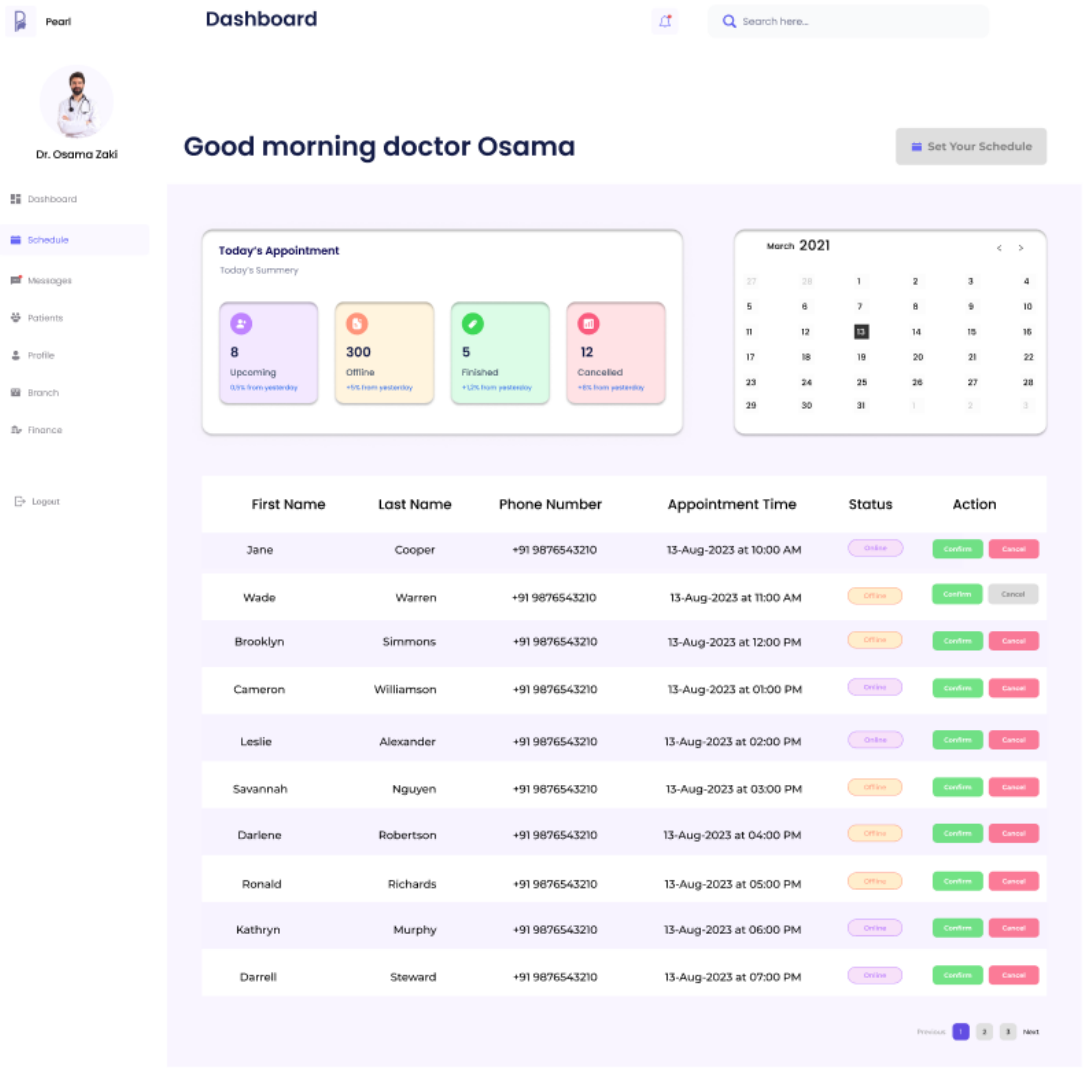
The doctor can select the function he wants to use from the side bar.

- ❖ Summery screen: display to the doctor many graphs, each one is responsible to measure feature.

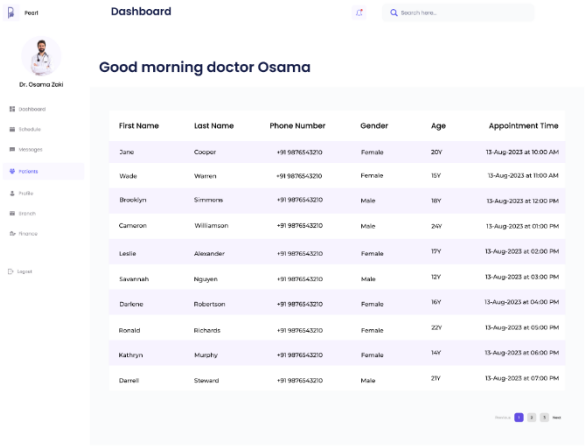
1. Today appointments card: display the upcoming, offline, finished, canceled appointments in the day.
2. Doctor activity graph: display the doctor activity on the website and compare it with his activity in the last week.
3. Patients visits graph: display the online and offline visits during the week.
4. Patients gender graph: display the female patient's percentage to the male.
5. Doctor rates graph: display the doctor rates.
6. Comments section: display patients comment about their experience with the doctor.


Pearl
Dashboard
Search here...
Dr. Osama Zaki
Good morning doctor Osama
Dashboard
Schedule
Messages
Patients
Profile
Branch
Finance
Logout
Today's Appointment
Today's Summary
8
Upcoming
300
Offline
5
Finished
12
Cancelled
March 2021
Your activity
Patients comments :
Saleh Fathy
Greet doctor
Ali Maged
Professional
Sara Aymen
Bad experience
Doaa saber
Good
Ebrahiem Ali
Thank you
Patients visits
D/Osama Rate
4.5
2456
Patients Gender Summary
Male
Female

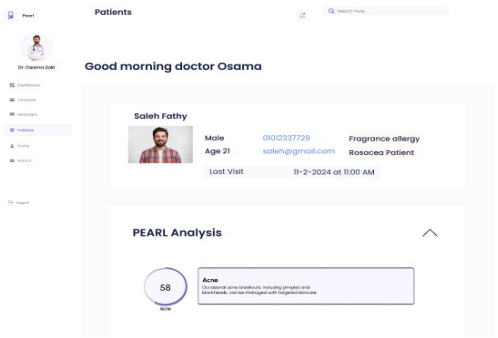
4.4.17 Appointments Page



4.4.18 Patients Page: it's about all patients' data such as first and last name, phone number, gender, age and last appointment date

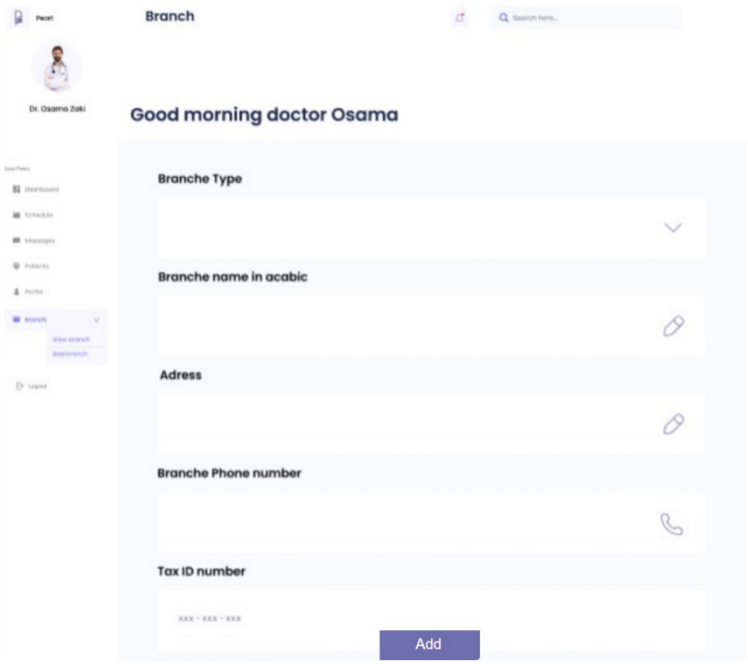


4.4.19 Patient Profile Page:



4.4.20 Doctor Branch Page: enable doctor to edit his branch registration data**.**

1. Branch type field: the doctor chooses his branch type between clinic or clinic center.
2. Address field: to add the branch address.
3. Branch name field: the doctor enters his branch name in Arabic.
4. Branch phone number field: the doctor enters his branch phone number.
5. Tax ID number field: the doctor enters his Tax ID number.
6. Add button: the new data edited successfully.



4.4.21 Doctor Personal Profile Page: enable doctor to edit his personal registration data.

1. Name field: the doctor enters his first and last name.
2. Title field: the doctor enters his job title**.**
3. Email field: the doctor enters his email.
4. Password field: the doctor enters his password.
5. Birth of date field: the doctor enters his Birth date.
6. Phone number field: the doctor enters his phone number.
7. Verification field: the doctor uploads a copy of his professional license or membership in the doctor syndicate.
8. Profile Picture field: the doctor enters his profile picture.
9. Save button: the new data edited successfully.

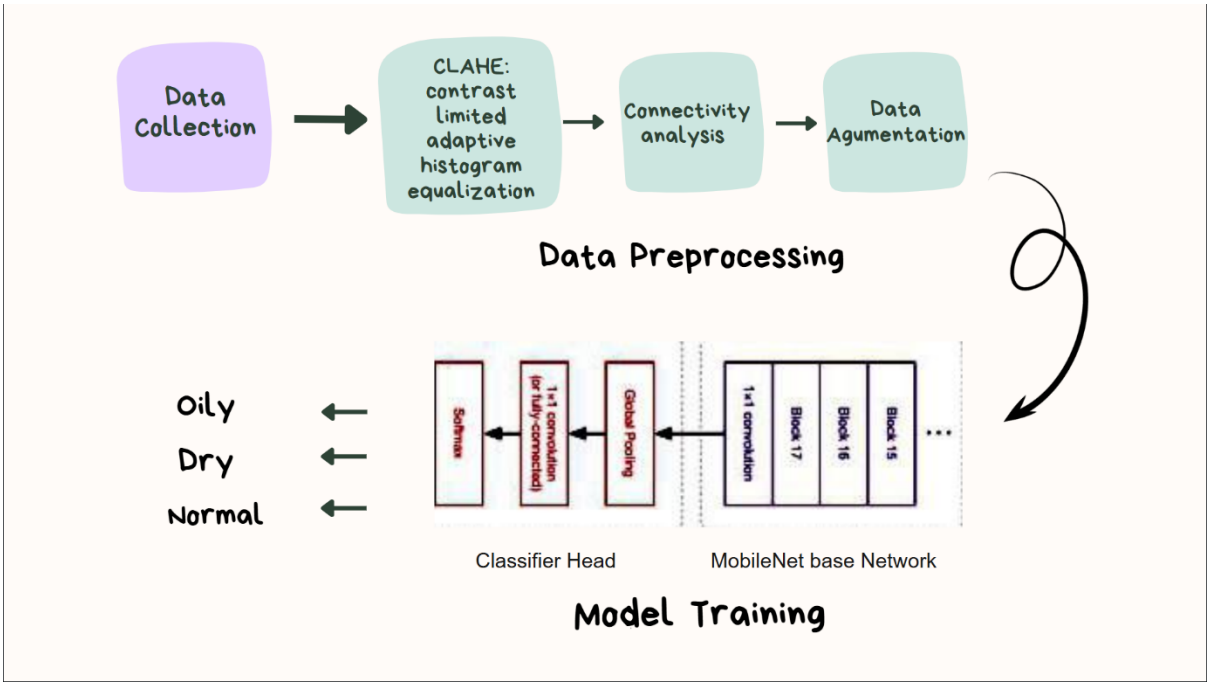
Chapter 5

Chapter 5: System Implementation

This chapter outlines the process of transforming the designed system from a theoretical concept into a fully operational reality. It details the activities involved in bringing the system to life, ensuring it functions as intended, and is smoothly integrated into the user environment.

5.1 AI Implementation

Skin Types Model :



1. Data Preparation:

- The code initializes paths for three skin type classes (Dry, Normal, Oily) and collects file paths with corresponding labels into a DataFrame (`skin_df`).

- It splits the data into training (`train_set`), validation (`val_set`), and testing (`test_images`) sets using train_test_split from sklearn.model_selection.

2. Data Augmentation:

- Using ImageDataGenerator from tensorflow.keras.preprocessing.image, it sets up image augmentation and preprocessing. Images are resized to (244, 244) pixels and undergo MobileNetV2's preprocess input function, preparing batches for training (`train`), validation (`val`), and testing (`test`).

4. Custom Preprocessing:

- CLAHE Function (`apply_clahe`): Enhances image contrast using Contrast Limited Adaptive Histogram Equalization (CLAHE) in LAB color space.

- Connectivity Analysis Function (`connectivity_analysis`): Performs connectivity analysis on binary images to count connected components using both 4-connectivity and 8-connectivity methods.

- Combined Preprocessing (`custom_preprocessing`): Integrates CLAHE enhancement and connectivity analysis into a preprocessing function suitable for MobileNetV2.

It enhances image contrast and extracts relevant features before applying MobileNetV2's standard preprocessing.

4. Model Building:

- MobileNetV2 is initialized as the base model, pre-trained on ImageNet, and excluding the top classification layer.

- A transfer learning model is built on top, adding a global average pooling layer, dense layers with dropout for regularization, and a softmax output layer for classifying into Dry, Normal, and Oily skin types.

5. Training and Evaluation:

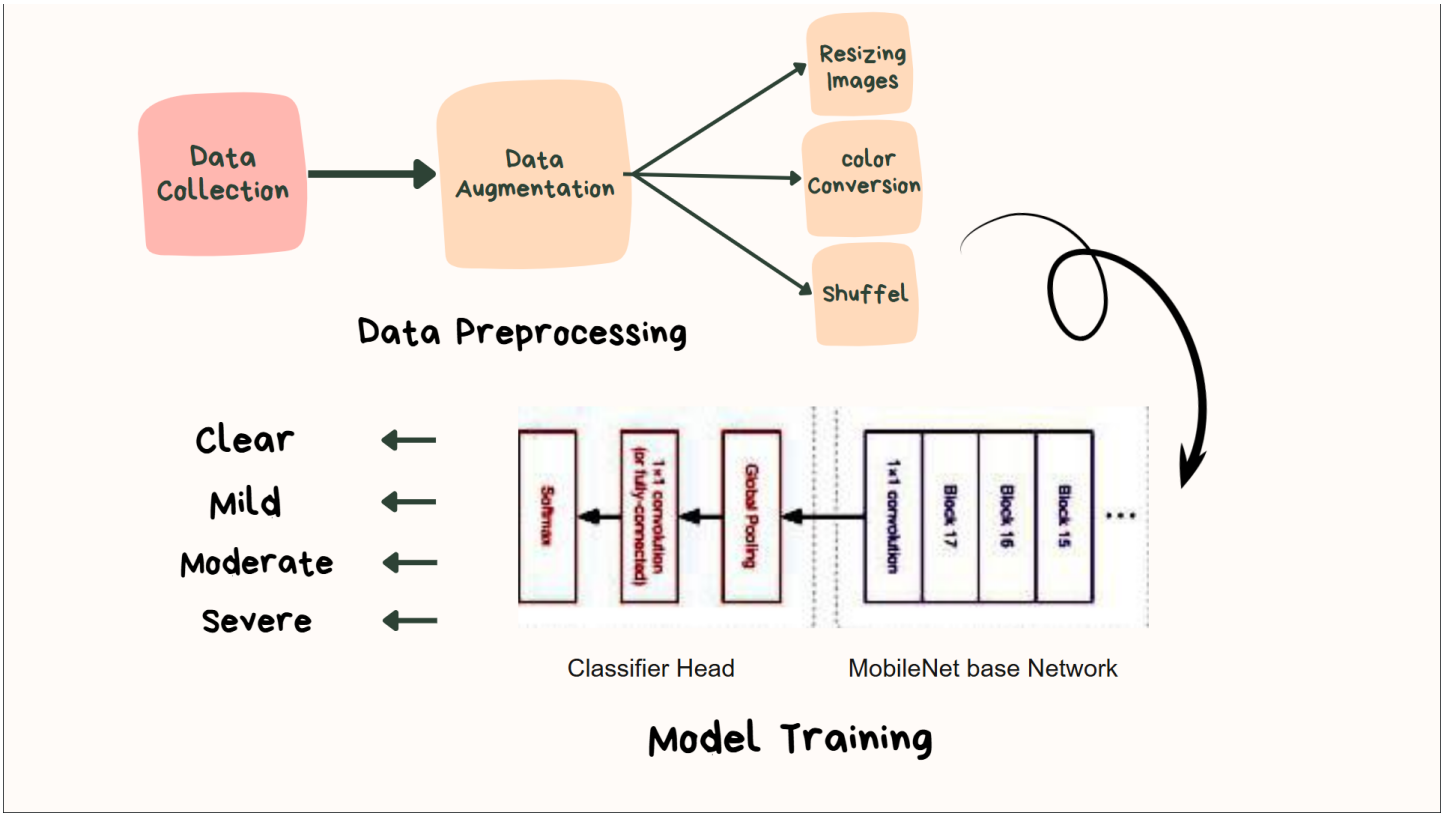
- The transfer learning model is trained with categorical cross-entropy loss and Adam optimizer (`learning_rate=0.001`), monitored with a ReduceLROnPlateau callback.

- After training, the model's performance is evaluated on the test dataset (`test`), calculating and printing test loss, accuracy, and generating a classification report with precision, recall, and F1-score.

- Training and validation accuracy over epochs are plotted to visualize learning progress.

6. Model Saving:

- The trained transfer learning model (`transfer_learning_model`) is saved in Keras format (`model.keras`) at a specified location.



1. Data Setup:

- Paths are defined for datasets representing different levels of acne severity: Mild Acne, Moderate Acne, Severe Acne, and Clear skin.

- File paths and corresponding labels are organized into a DataFrame (`Acne_df`), preparing the data for further processing.

2. Data Splitting:

- The dataset is split into training (`train_set`), validation (`val_set`), and testing (`test_images`) sets using a standard 70-20-10 split ratio. This ensures each model phase has sufficient data for effective training, validation, and evaluation.

3. Data Augmentation and Preprocessing:

- Images are augmented and preprocessed using TensorFlow's ImageDataGenerator. This includes resizing images to a uniform size of 244x244 pixels and applying the MobileNetV2-specific preprocessing function to enhance model input quality.

4. Model Construction:

- MobileNetV2, a pre-trained neural network model known for its efficiency and accuracy in image tasks, is employed as the base model. Its top layers, originally designed for ImageNet classification, are excluded.

- On top of MobileNetV2, a custom classifier is constructed using global average pooling, followed by dense layers with dropout regularization to prevent overfitting. The final layer employs a softmax activation function to classify images into one of the four acne severity categories.

5. Training and Optimization:

- The transfer learning model undergoes training over 15 epochs on the prepared training data (`train`), with ongoing evaluation against the validation set (`val`) to monitor and optimize its learning progress.

- To enhance learning efficiency, a learning rate reduction strategy (`ReduceLROnPlateau`) is implemented. This dynamically adjusts the learning rate based on validation loss performance, aiming to stabilize and improve model accuracy.

6. Performance Evaluation:

- Following training, the model's effectiveness is evaluated using the separate test dataset (`test`). Key metrics such as accuracy and loss are computed and reported to assess its predictive performance.

- Additionally, a detailed classification report provides insights into the model's precision, recall, and F1-score across all acne severity classes, offering a comprehensive view of its classification capabilities.

7. Model Persistence:

- The fully trained transfer learning model is saved in Keras format (`AcneClassification_MobileNetmodel.keras`). This allows for easy deployment and future use in various applications without the need for retraining, ensuring practicality and scalability.

	precision	recall	f1-score	support
Mild Acne	0.90	1.00	0.95	556
Moderate Acne	0.98	0.94	0.96	673
Severe Acne	0.88	0.90	0.89	634
Clear	0.97	0.87	0.92	505
accuracy			0.93	2368
macro avg	0.93	0.93	0.93	2368
weighted avg	0.93	0.93	0.93	2368

5.2 Website Implementation

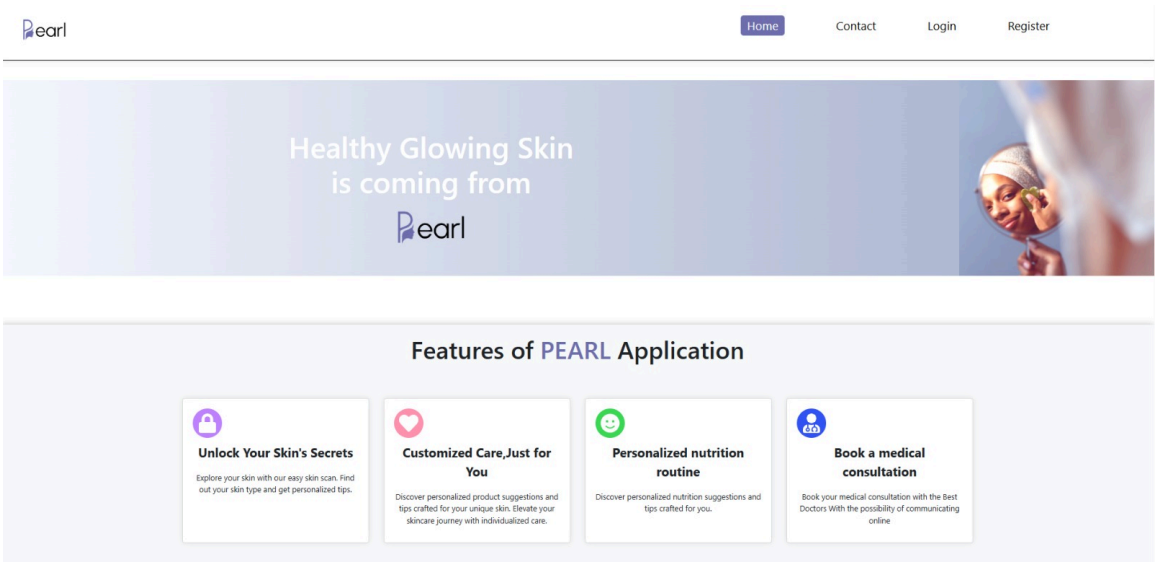
Used tools:

- Front-end: HTML, CSS, JavaScript, Bootstrap.
- Back-end: Laravel, Firebase.

Implementation:

1. Home Page:

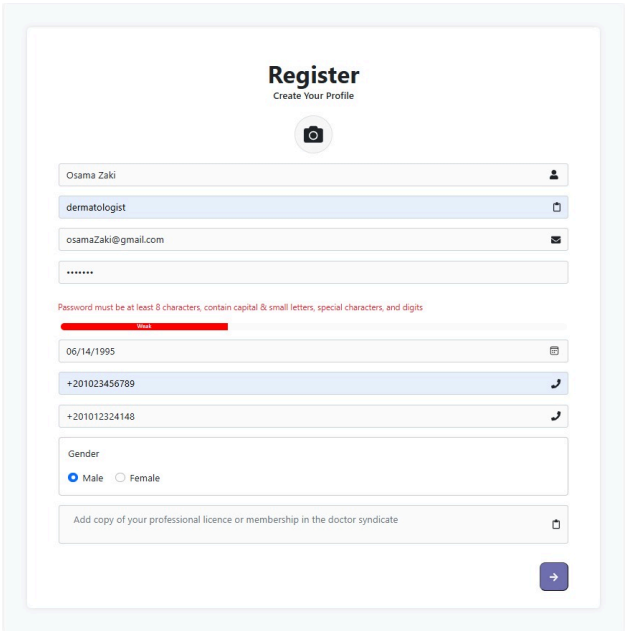
It’s the entry to our website, it’s just a description about “Pearl” showing its features.

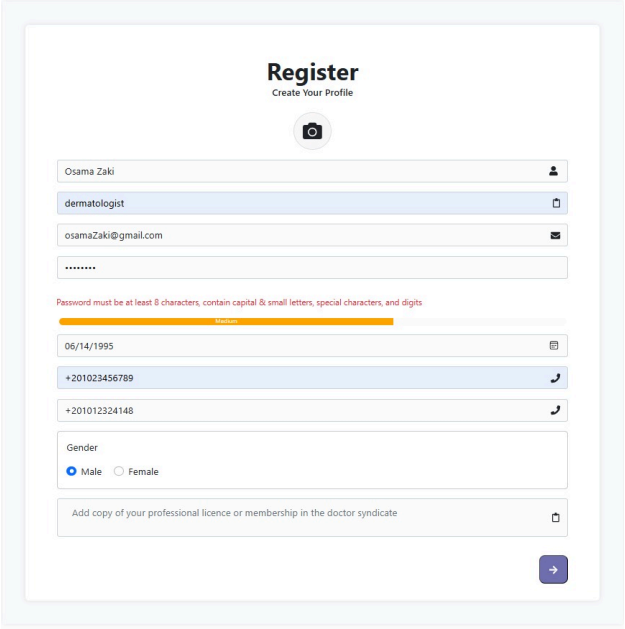


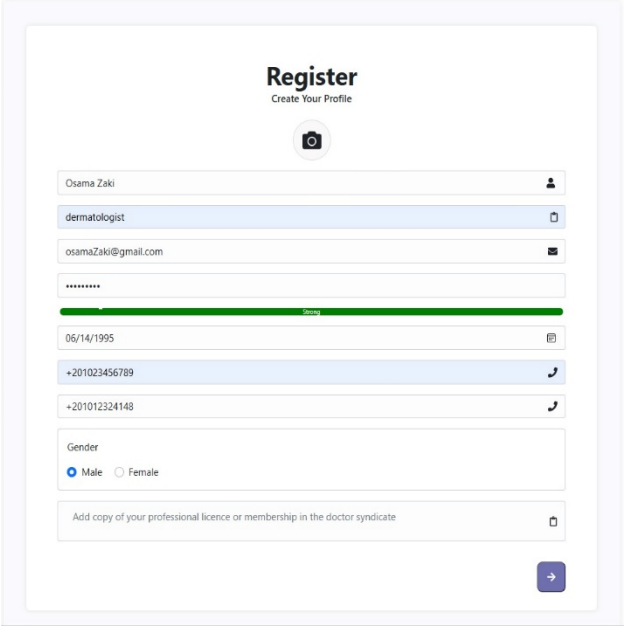
2. Registration:

- The doctor enters his data to register a new account into “Pearl”, one of the most important fields in registration is the doctor professional license, to make sure that the person trying to register is really a doctor.

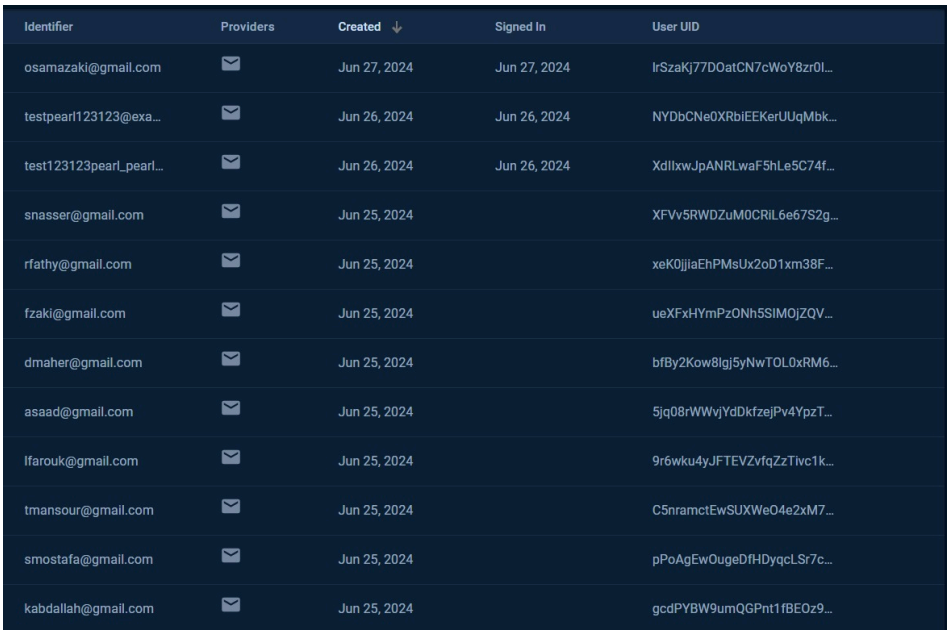
- As the password is sensitive data, we needed to apply a strong validation on it, to make sure the doctor has a strong password.







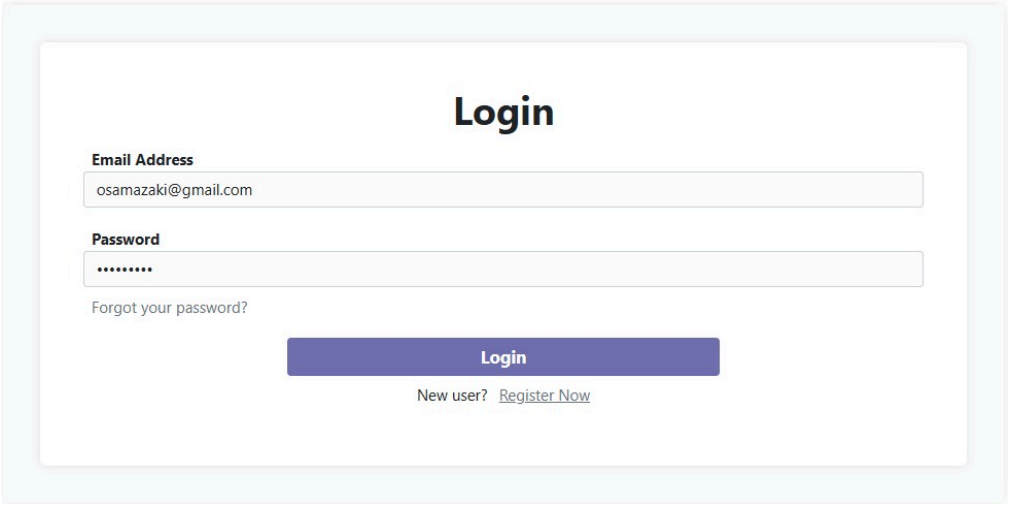
- When the doctor is done registering, his data is saved into our database.
- We are using firebase as it is a real time database.



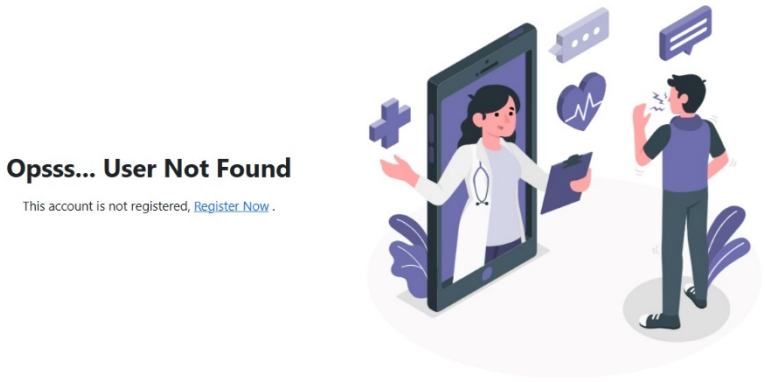
Identifier	Providers	Created ↓	Signed In	User UID
osamazaki@gmail.com	✉	Jun 27, 2024	Jun 27, 2024	IrSzaKj77DOatCN7cWoY8zr0l...
testpearl123123@exa...	✉	Jun 26, 2024	Jun 26, 2024	NYDbCNe0XRbiEEKerUUqMbk...
test123123pearl_pearl...	✉	Jun 26, 2024	Jun 26, 2024	XdIIxwJpANRLwaF5hLe5C74f...
snasser@gmail.com	✉	Jun 25, 2024		XFVv5RWDZuM0CRiL6e67S2g...
rfathy@gmail.com	✉	Jun 25, 2024		xeK0jjiaEhPMsUx2oD1xm38F...
fzaki@gmail.com	✉	Jun 25, 2024		ueXFxHYmPzONh5SIMOjZQV...
dmaher@gmail.com	✉	Jun 25, 2024		bfBy2Kow8lgj5yNwTOL0xRM6...
asaad@gmail.com	✉	Jun 25, 2024		5jq08rWWvjYdDkfzejPv4YpzT...
lfarouk@gmail.com	✉	Jun 25, 2024		9r6wku4yJFTEVZvfqZzTIvc1k...
tmansour@gmail.com	✉	Jun 25, 2024		C5nramctEwSUXWeO4e2xM7...
smostafa@gmail.com	✉	Jun 25, 2024		pPoAgEwOugeDfHDyqcLSr7c...
kabdallah@gmail.com	✉	Jun 25, 2024		gcdPYBW9umQGPnt1fBEOz9...

3. Login:

- The doctor enters his data to log into his account.

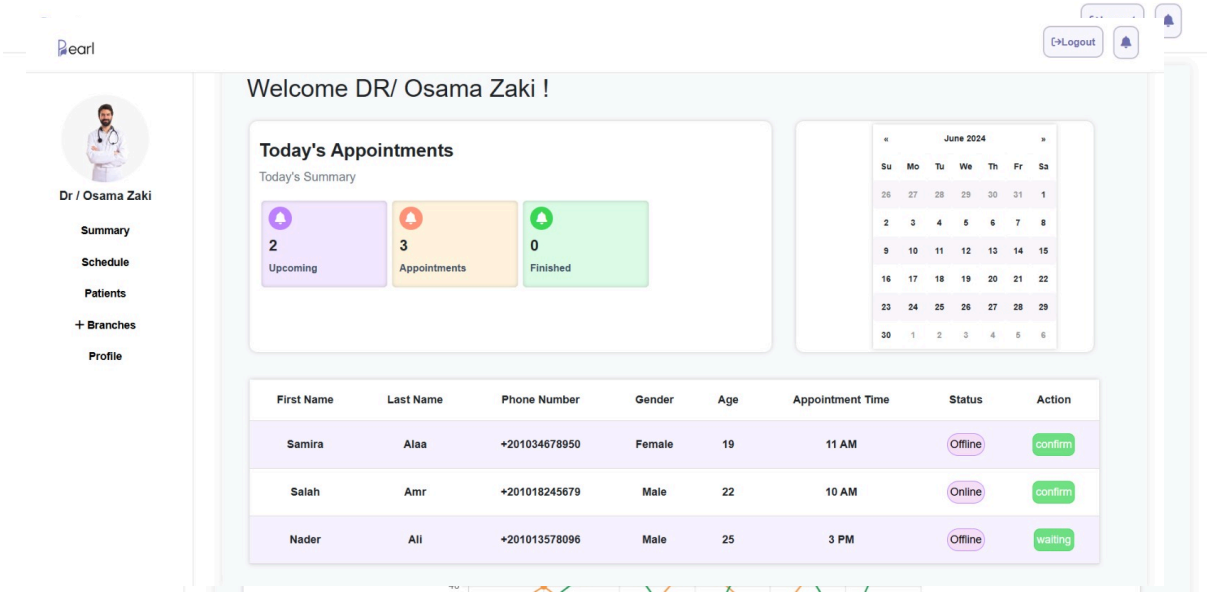


- If the doctor is not found in the database, then he may need to register.

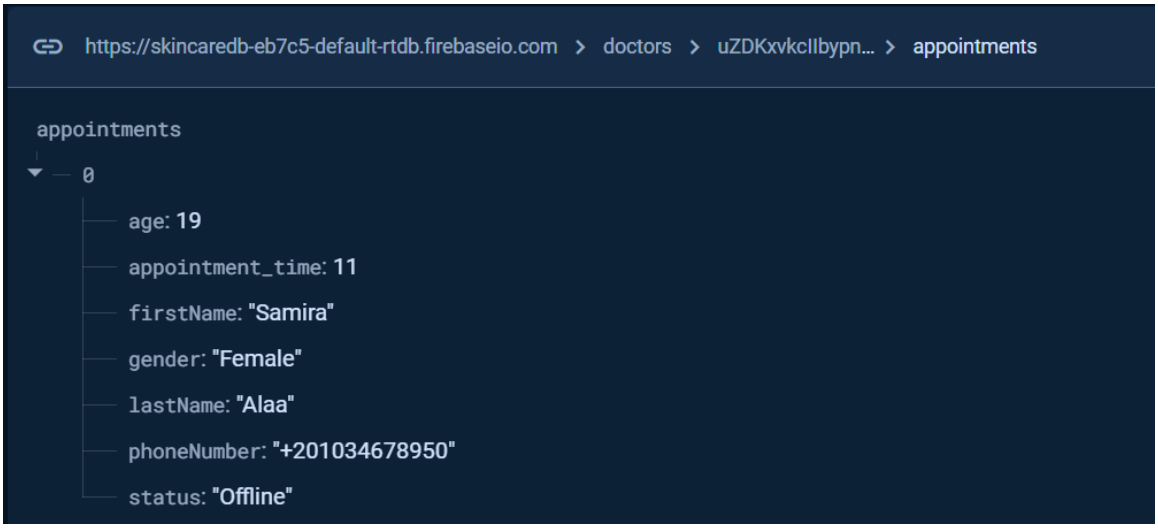


- If the doctor is found in the database, then he logged into “Pearl” successfully and redirected into his dashboard

4. Schedule:
   - It provides an overview of the doctor's appointments, including details of patients who have reserved slots. This page offers functionalities for confirming or canceling these appointments, ensuring an organized and efficient scheduling process.

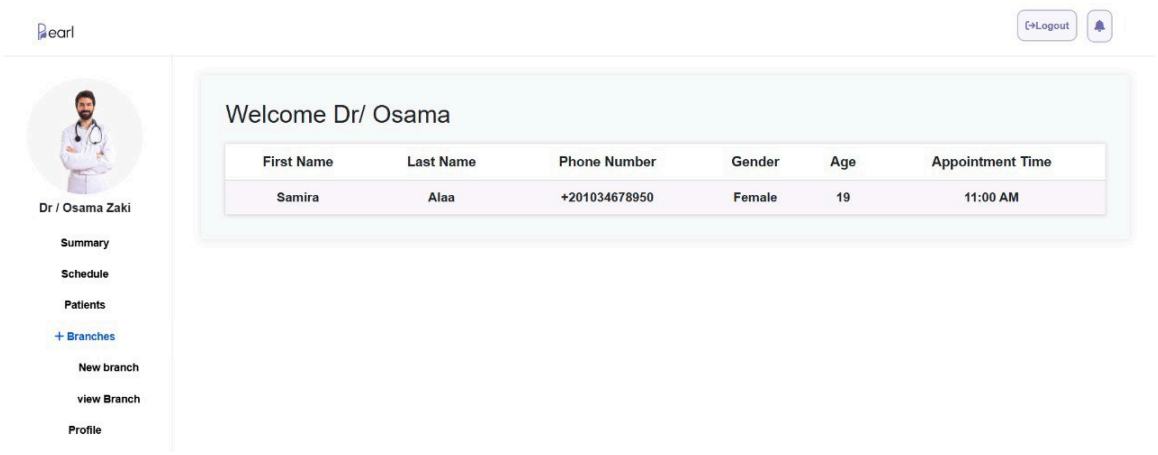


- Confirmed patients are automatically saved into the firebase.



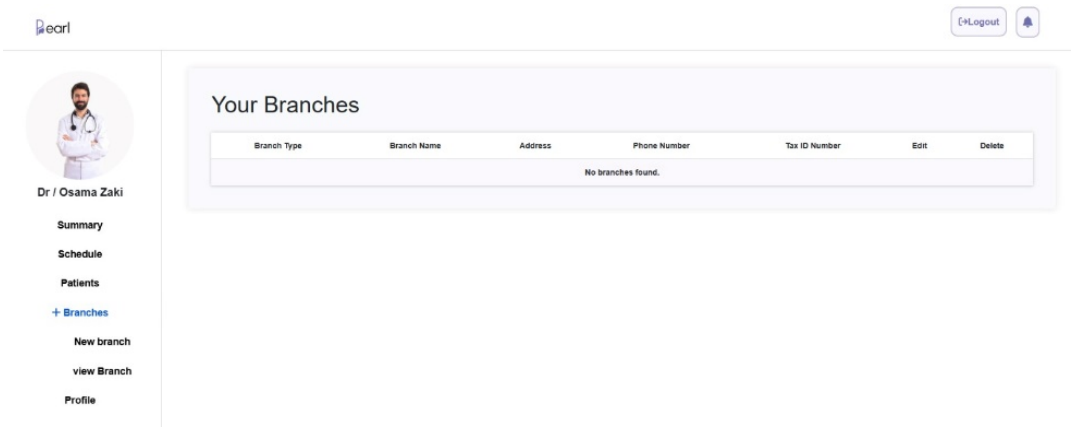
5. Patients:

The confirmed patients appear on this page.



6. Branches:

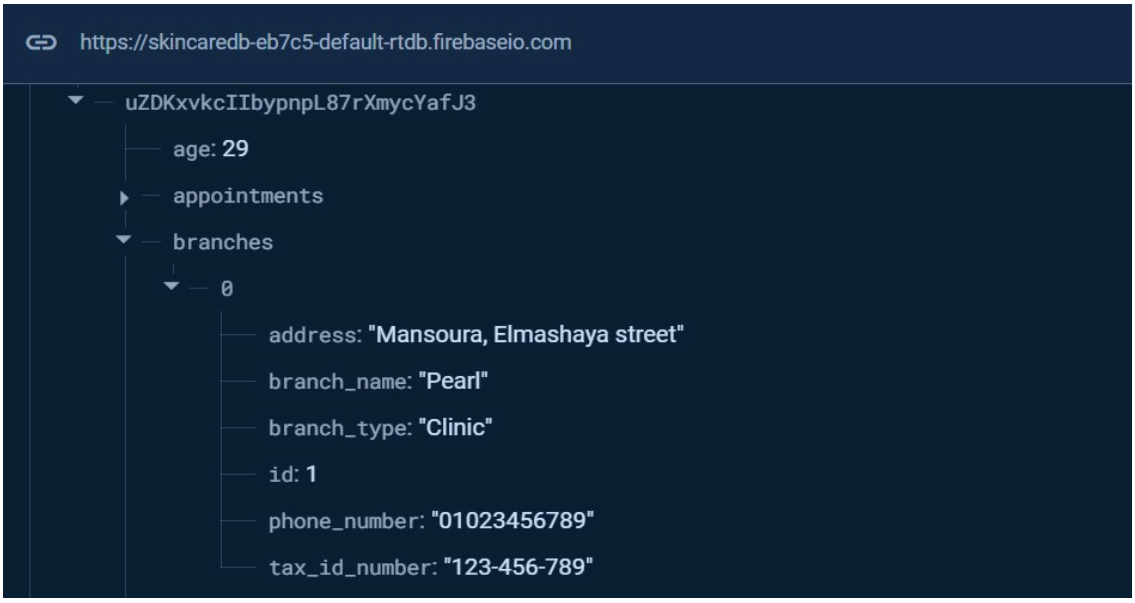
By default, the doctor doesn’t have any branch



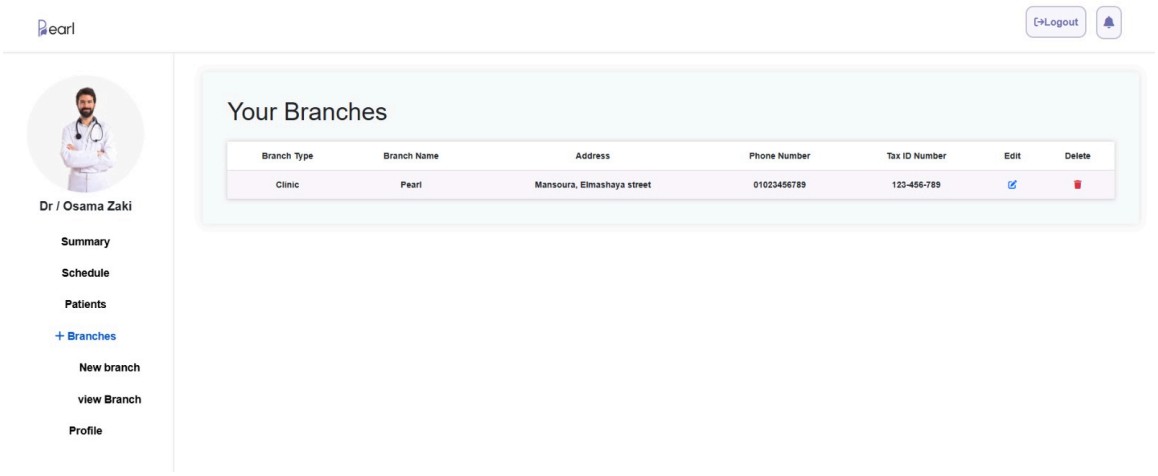
- The doctor can add a branch.



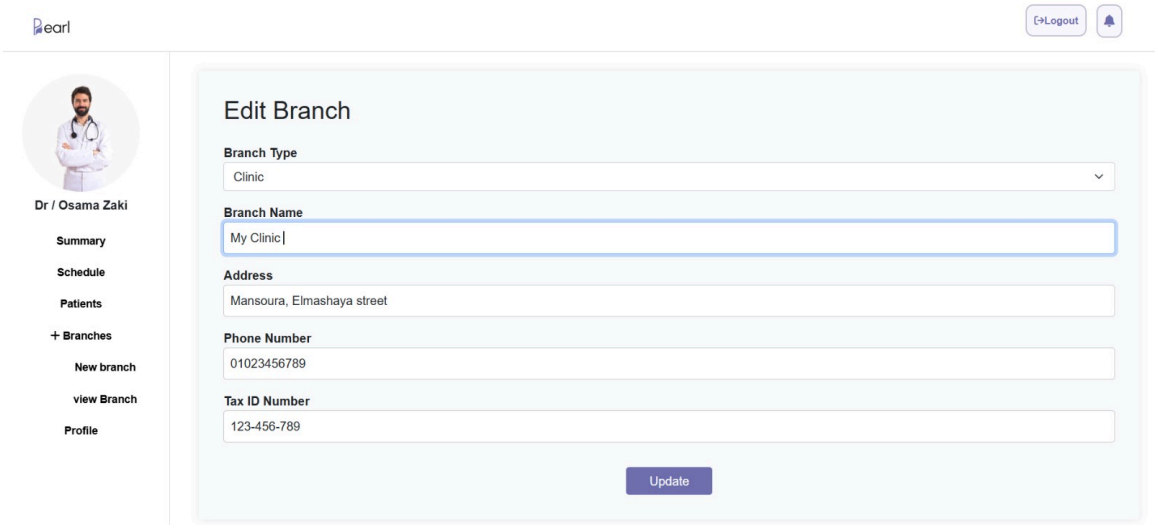
- Once the doctor submits the Adding form, the application processes the input data and updates the firebase. Specifically, the new clinic's details are added to doctor's document within the "doctors" collection. This update includes creating a new list within doctor's document to store information about all the clinics associated with him.



- After successfully adding the new clinic, the doctor can view and manage the clinic's information within the application, ensuring his profile is up-to-date and accessible for his patients. This seamless process allows for efficient management of doctor's multiple clinic locations.



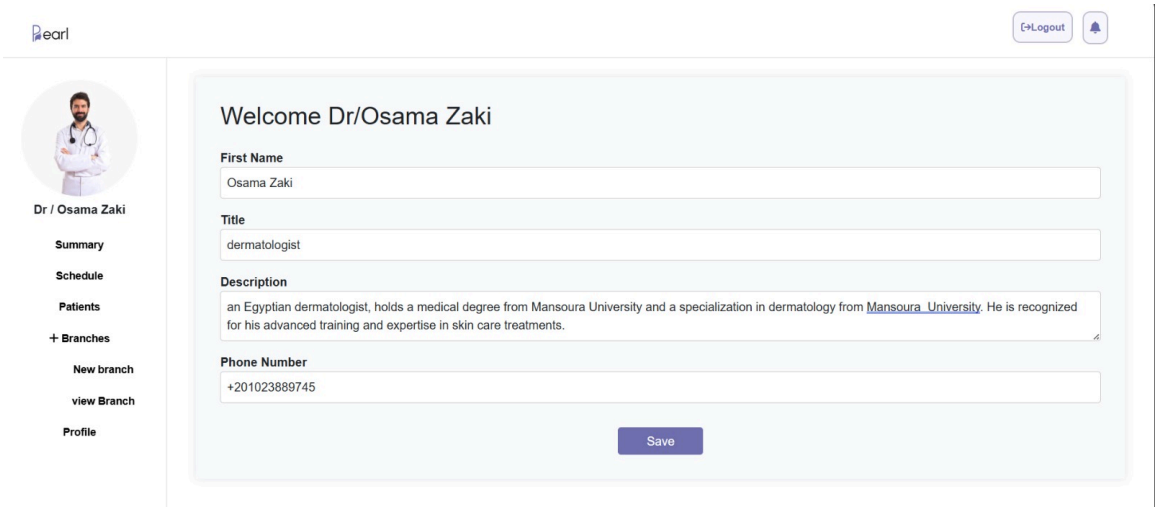
- The doctor can edit or delete any branch using the icons in the viewed branches.



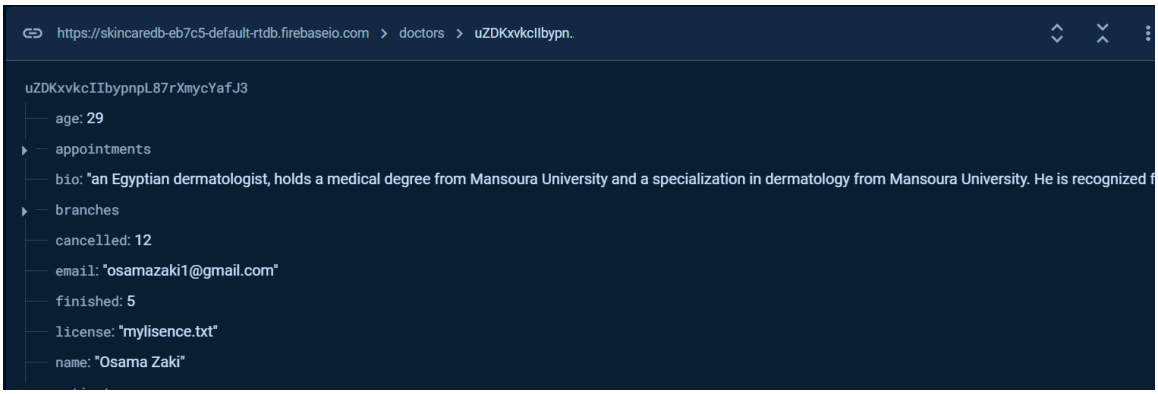
- The application processes the updated data and sends it to the firebase. The specific document corresponding to the clinic within doctor "branches" list in the "doctors" collection is updated with the new information. here, the doctor changed the name of his clinic from Pearl to My clinic.



7. Profile:
   - The doctor can edit his profile data and add his bio.



   - The data then is updated automatically in firebase.



5.3 Mobile Development

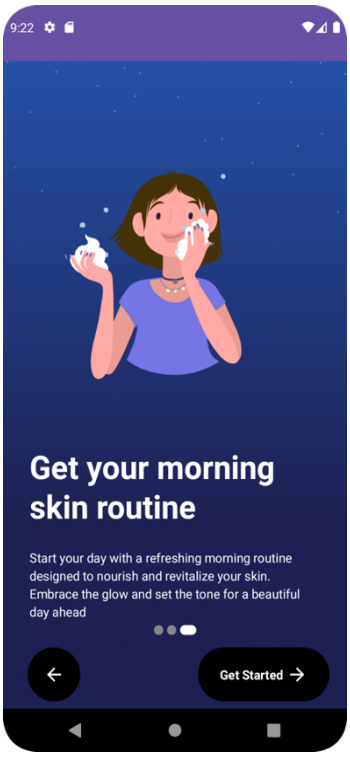
1. Introduction Screen

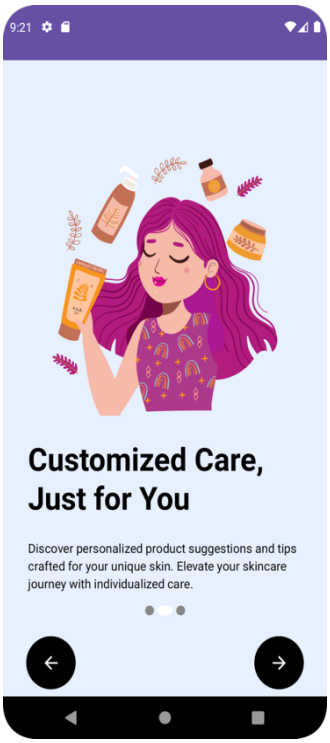
- It displays when it's the first time for the user to enter application.

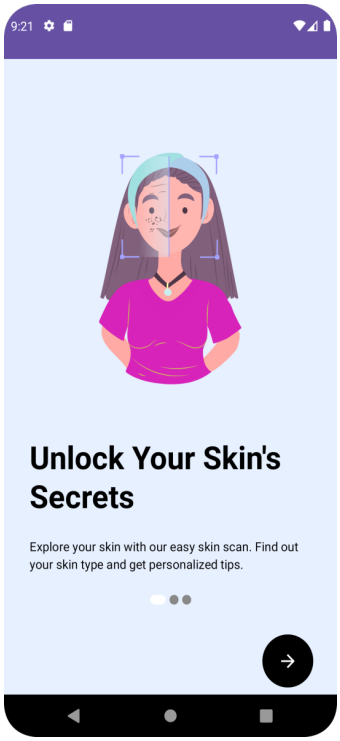


2. Onboarding Screen

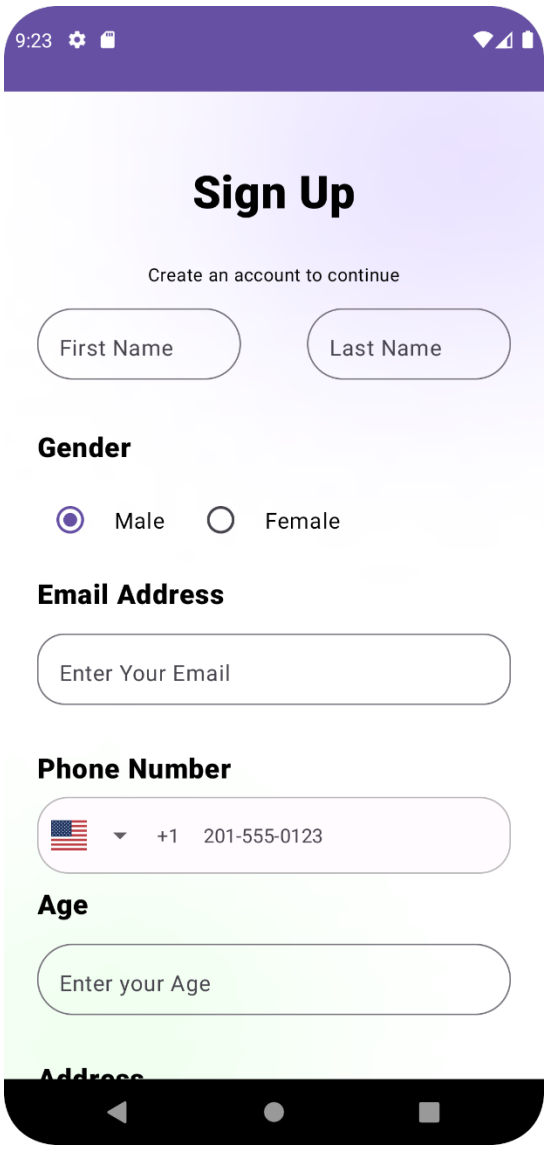
- It displays when it's the first time for the user to enter application.

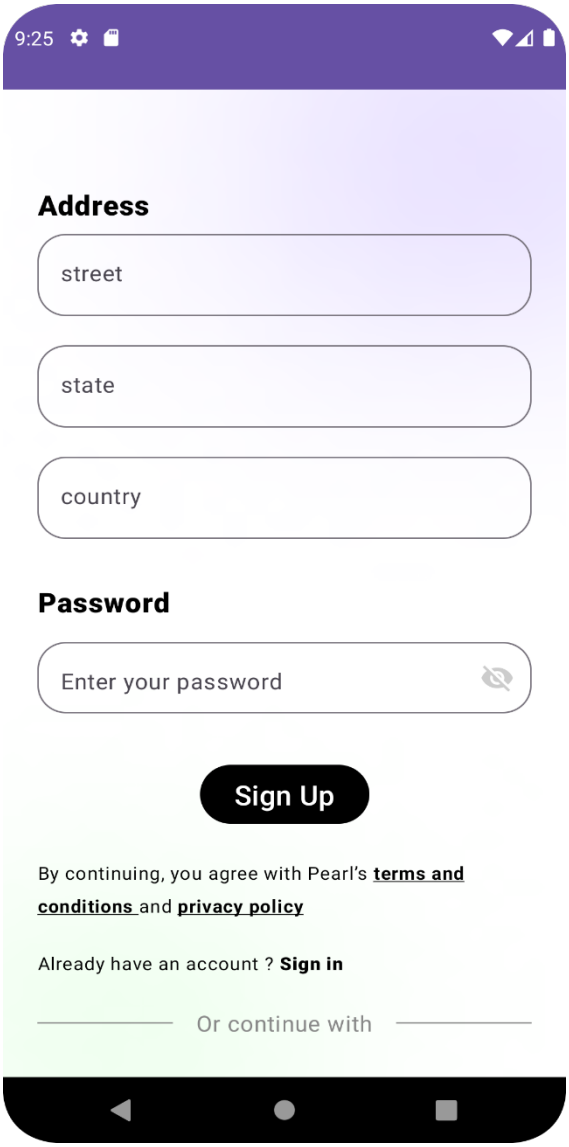




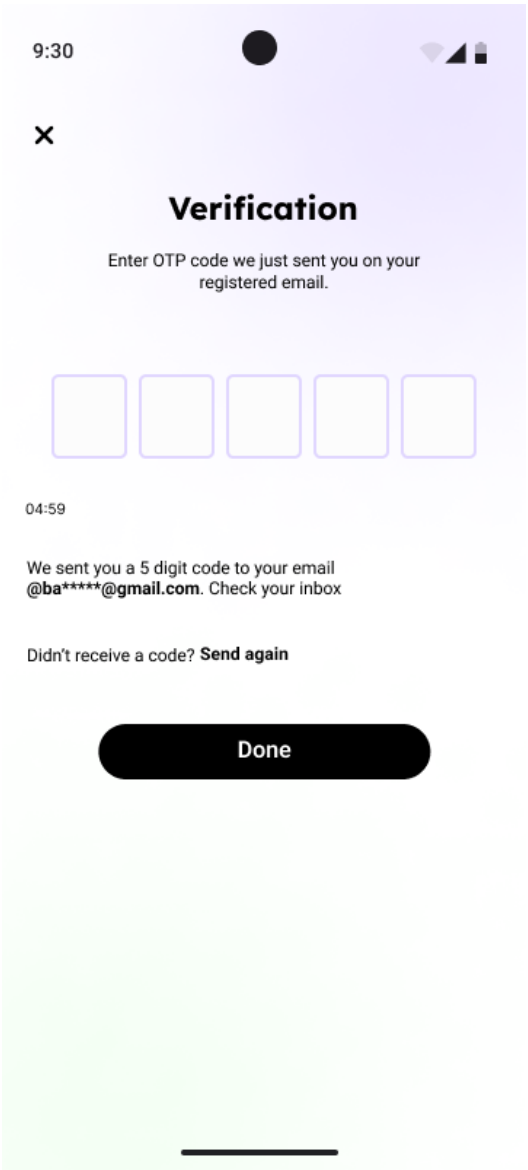


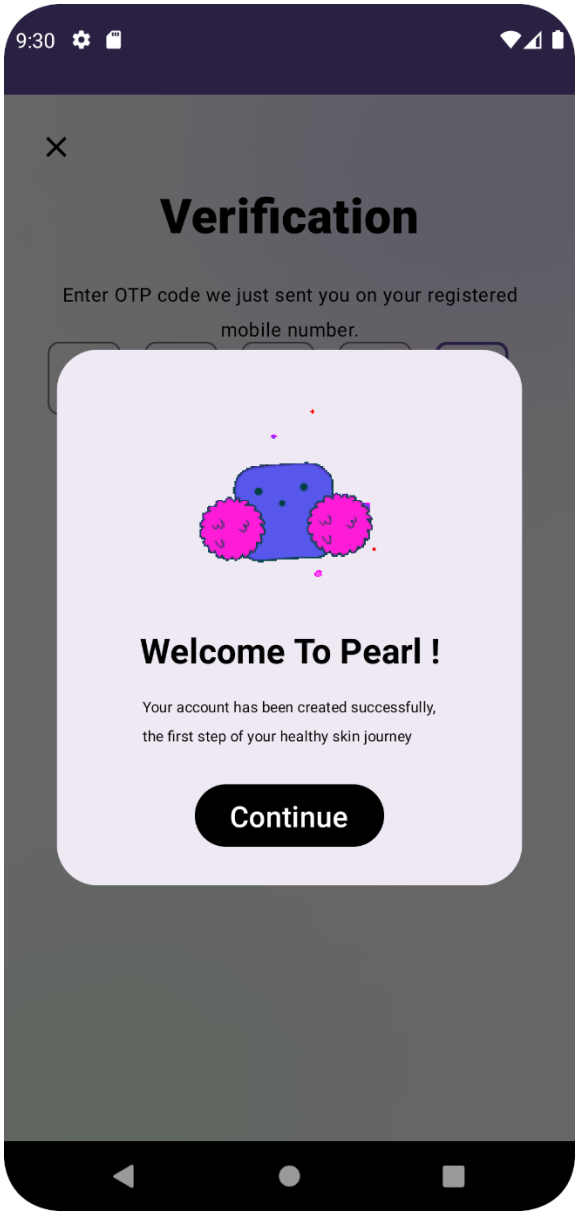
3. **Sign Up**


9:23
Sign Up
Create an account to continue
First Name
Last Name
Gender
Male
Female
Email Address
Enter Your Email
Phone Number
+1 201-555-0123
Age
Enter your Age

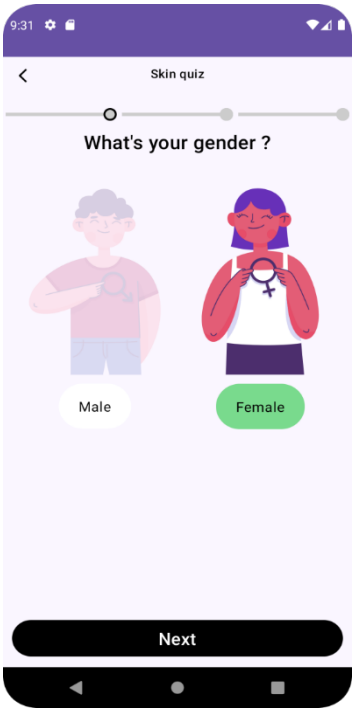

9:25
Address
street
state
country
Password
Enter your password
Sign Up
By continuing, you agree with Pearl's terms and conditions and privacy policy
Already have an account ? Sign in
Or continue with

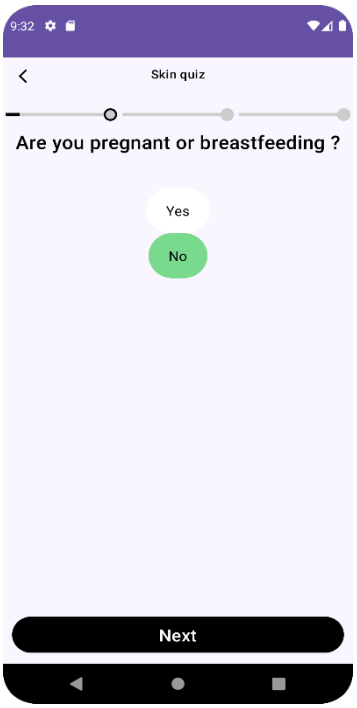
- Send OTP code to verify email/phone

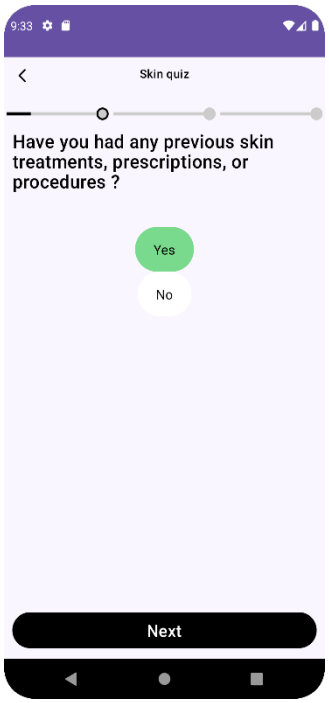


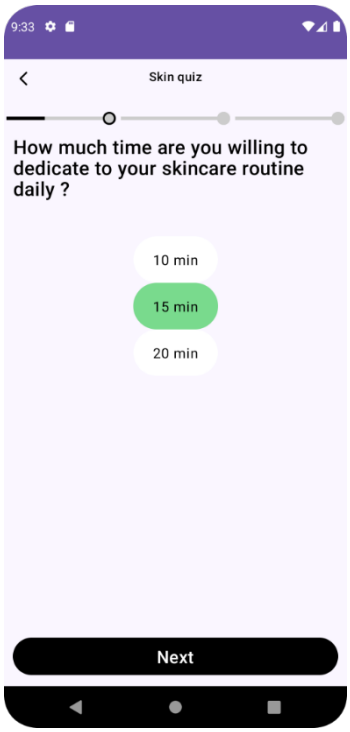


4. Survey Screens

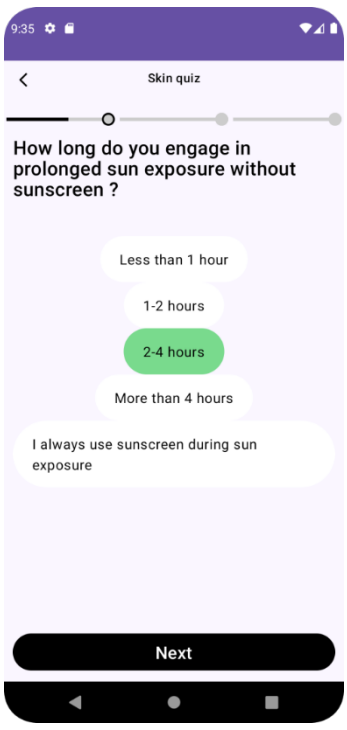




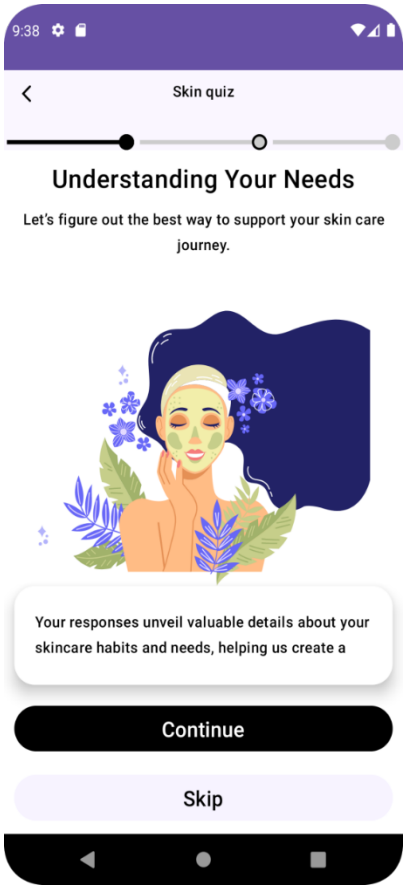


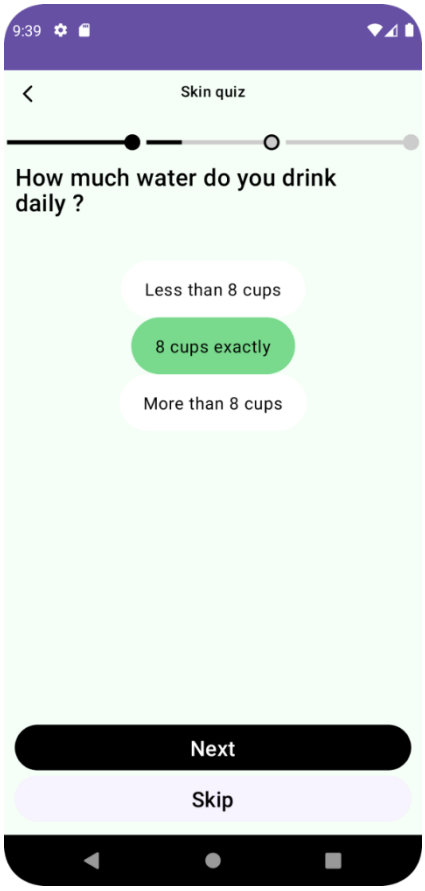


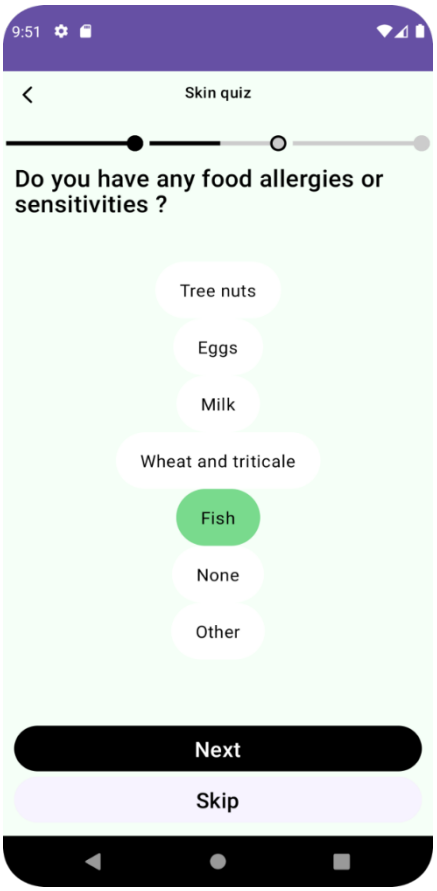




5. Nutrition Survey Screens

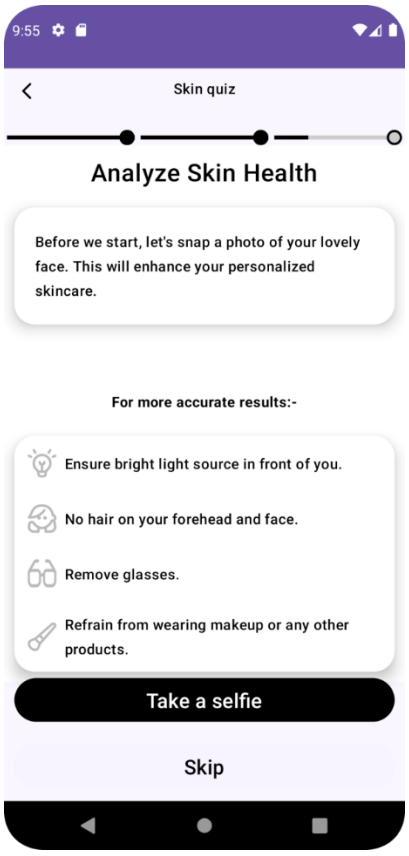




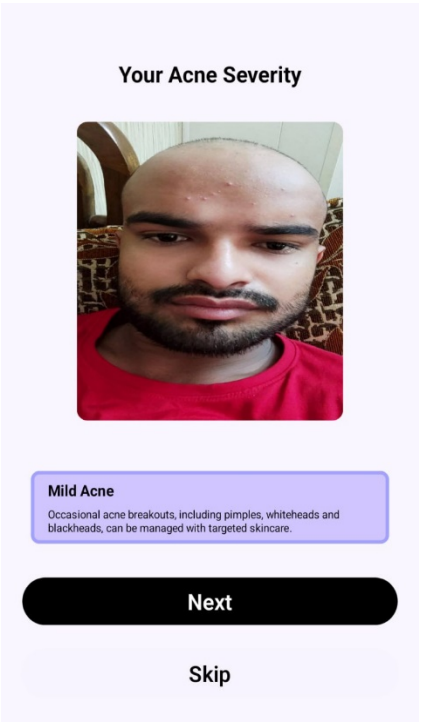


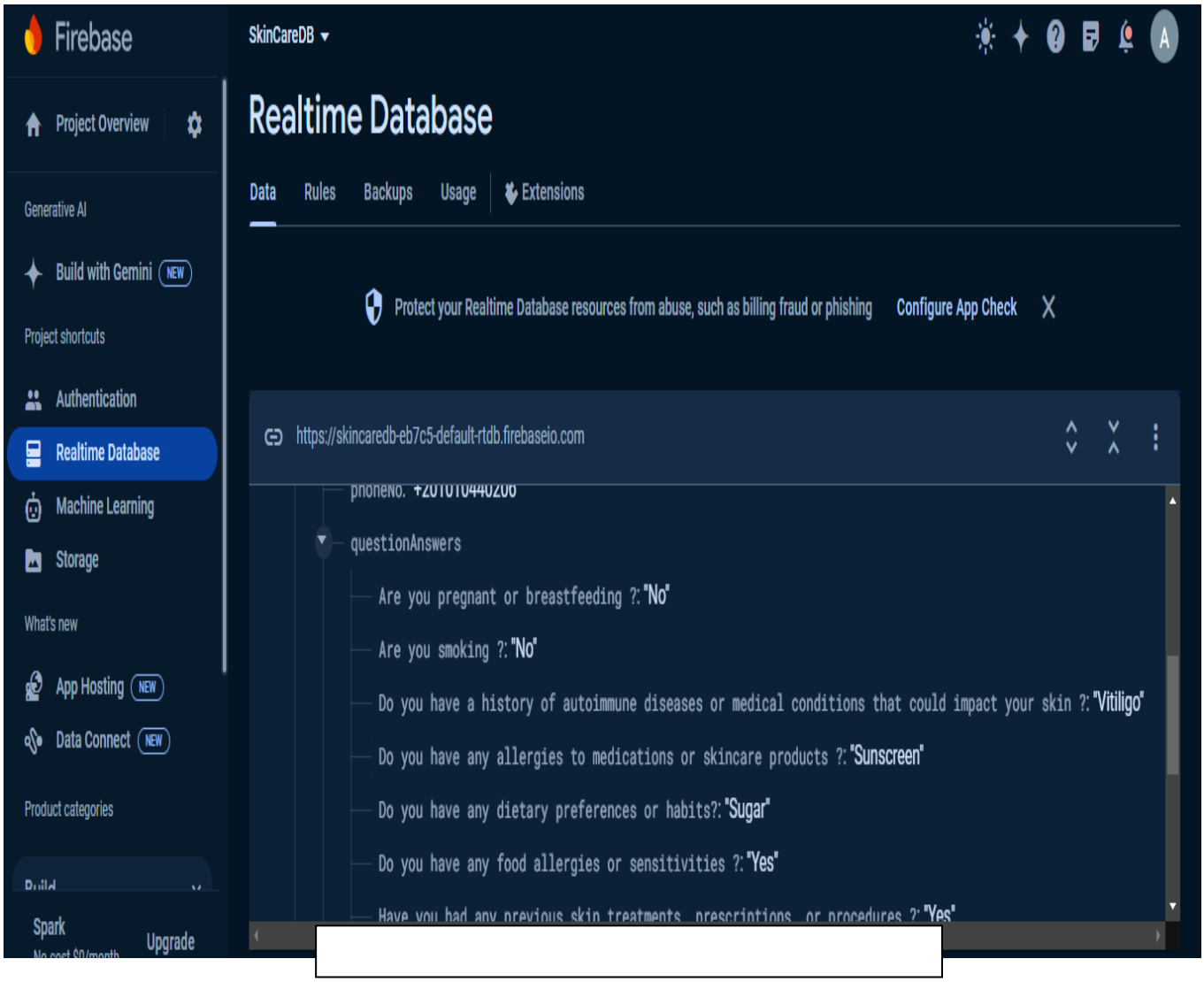
6.Skin analysis result

- Getting the result of skin analysis so user knows his skin concerns .





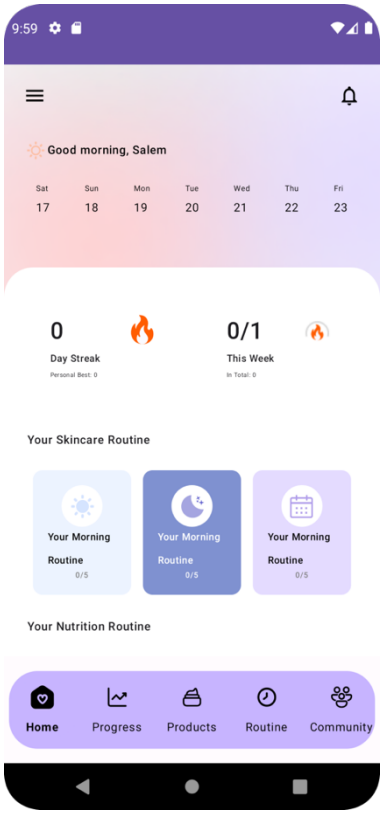


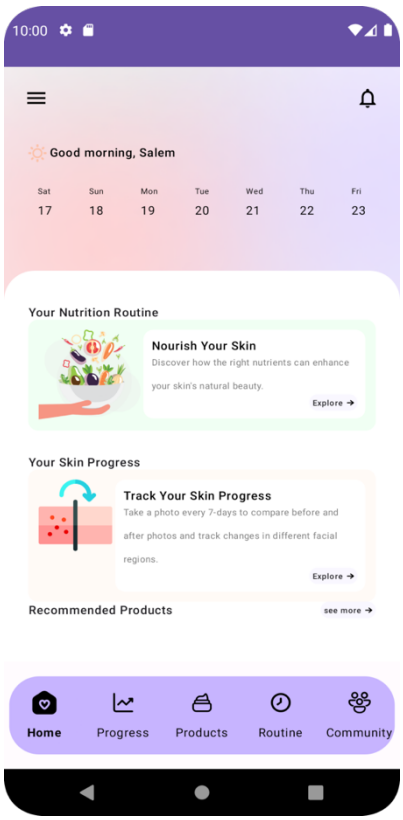

Firebase
SkinCareDB
Project Overview
Realtime Database
Generative AI
Build with Gemini
NEW
Project shortcuts
Authentication
Realtime Database
Machine Learning
Storage
What's new
App Hosting
NEW
Data Connect
NEW
Product categories
Spark
Upgrade
Data
Rules
Backups
Usage
Extensions
Protect your Realtime Database resources from abuse, such as billing fraud or phishing
Configure App Check
https://skincaredb-eb7c5-default-rtdb.firebaseio.com
phoneNo: "+201010440206"
questionAnswers
Are you pregnant or breastfeeding ?: "No"
Are you smoking ?: "No"
Do you have a history of autoimmune diseases or medical conditions that could impact your skin ?: "Vitiligo"
Do you have any allergies to medications or skincare products ?: "Sunscreen"
Do you have any dietary preferences or habits?: "Sugar"
Do you have any food allergies or sensitivities ?: "Yes"
Have you had any previous skin treatments, prescriptions, or procedures ?: "Yes"

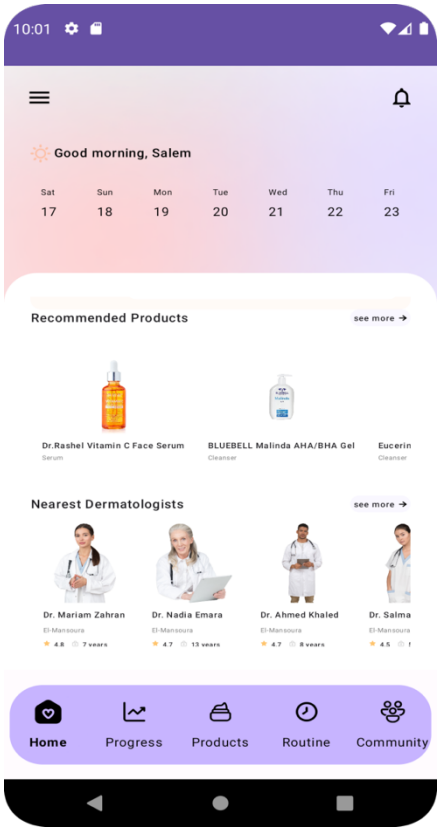
7. Home Screen

- User will get to home screen right after he has finished the survey.

-User can navigate to home screen directly when it’s not its first time he entered the app.

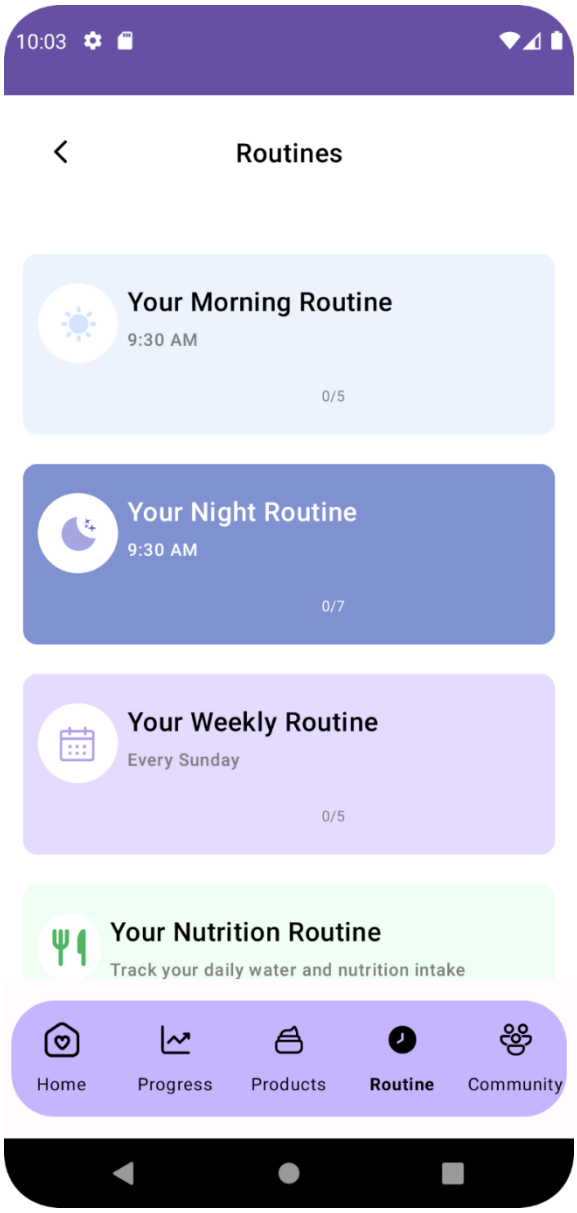




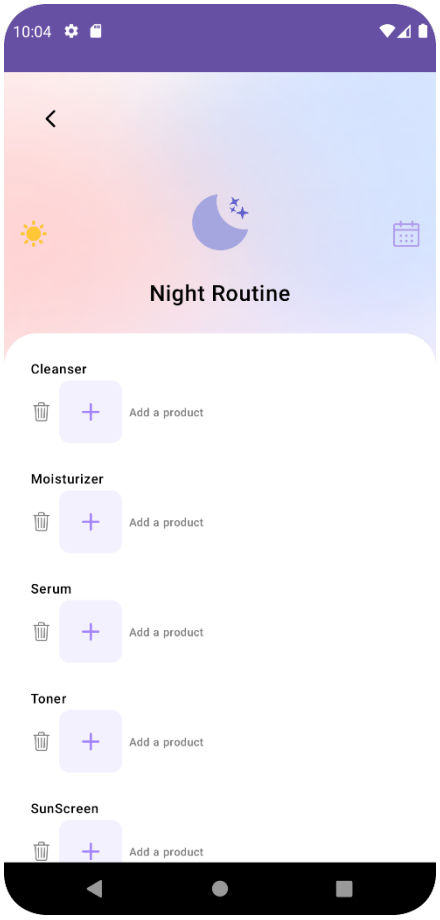


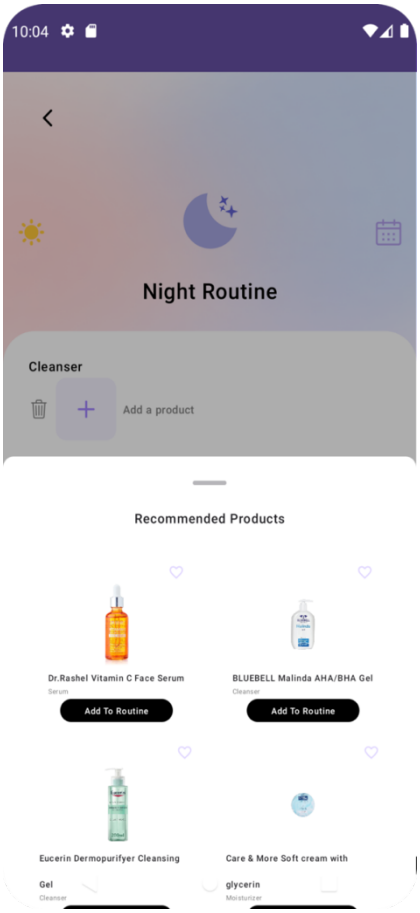
8. Routine Screen

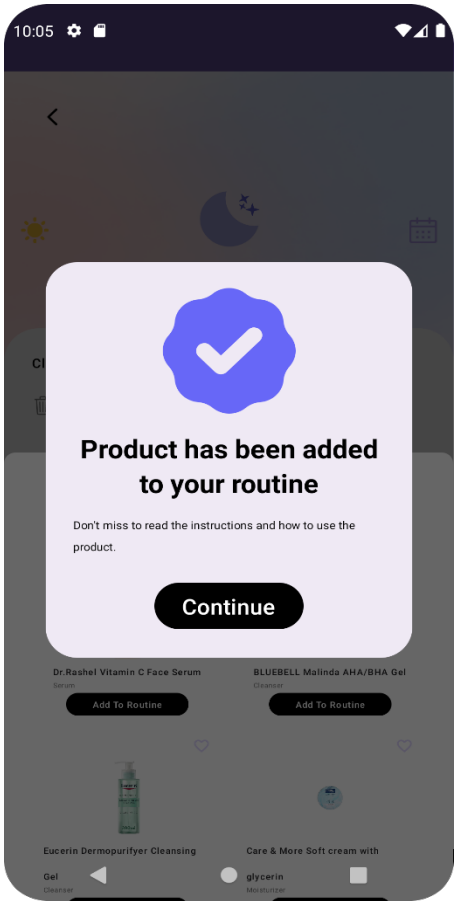
- User can select routine time.

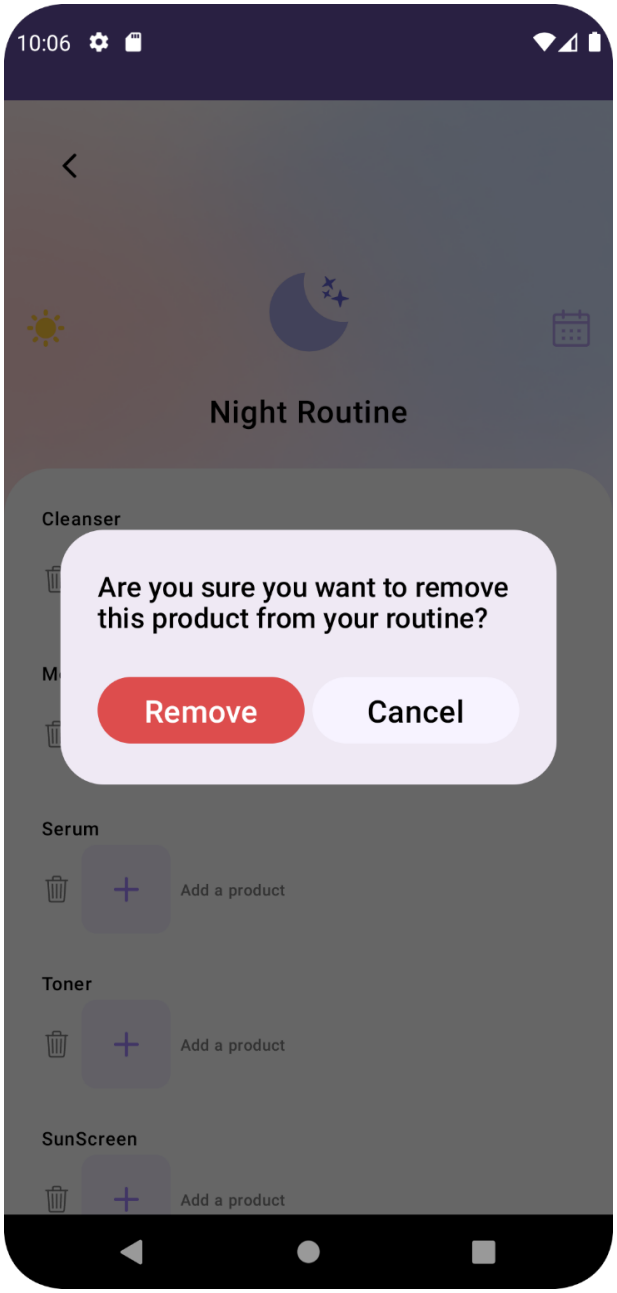


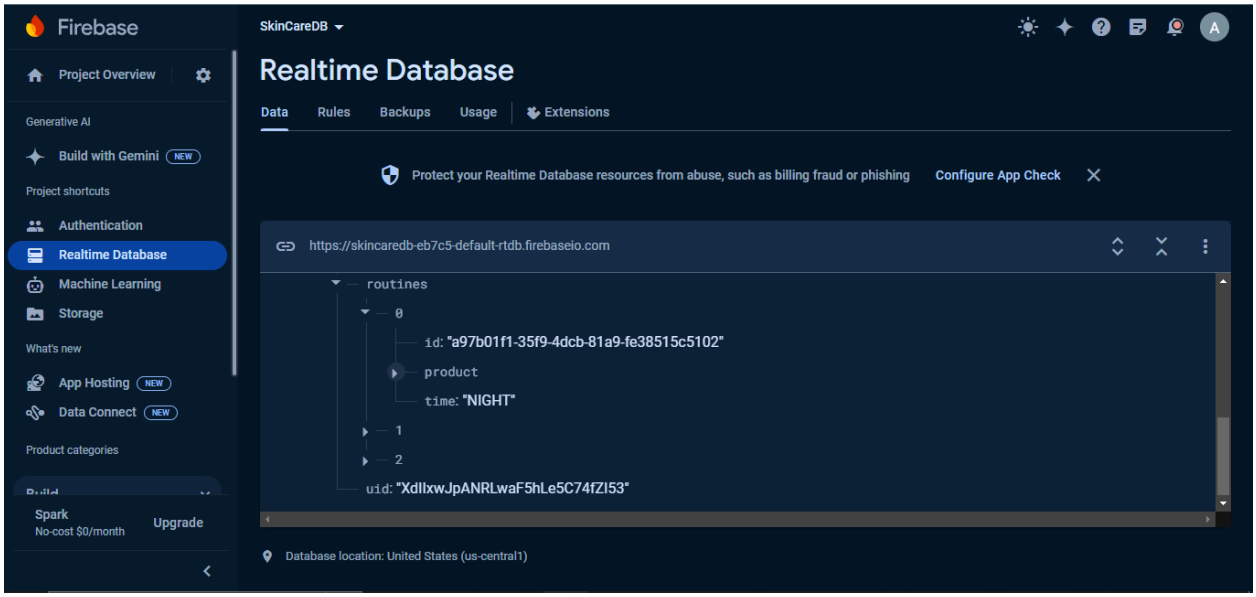
9.Adding routine Screen

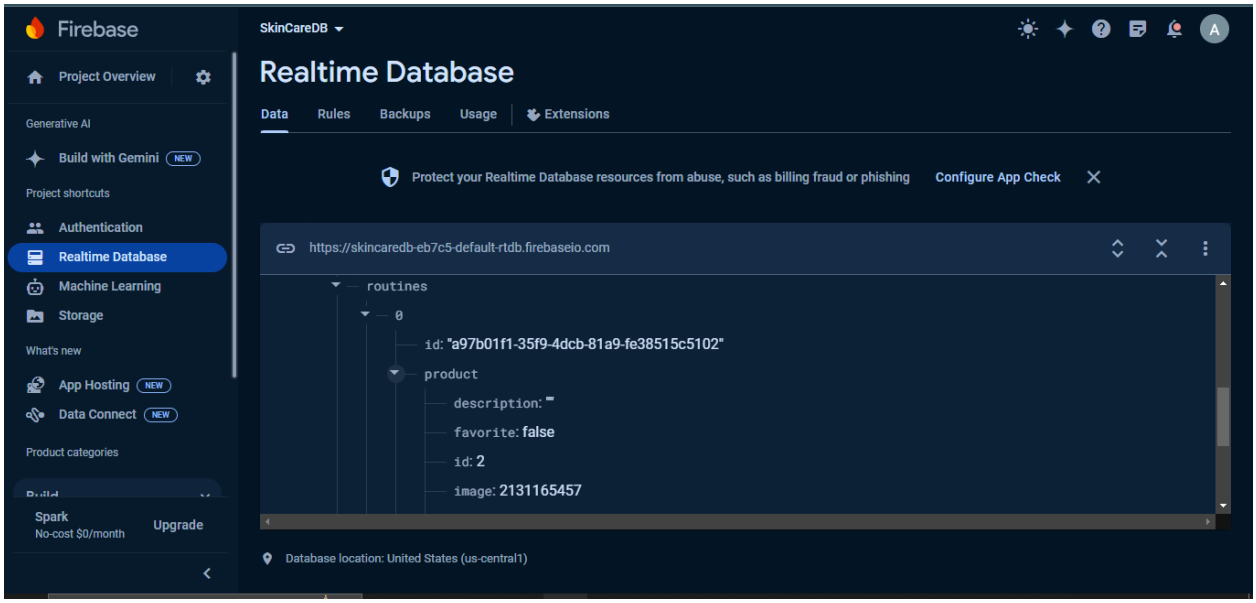







10:06
Night Routine
Cleanser
Are you sure you want to remove this product from your routine?
Remove
Cancel
Serum
Add a product
Toner
Add a product
SunScreen
Add a product

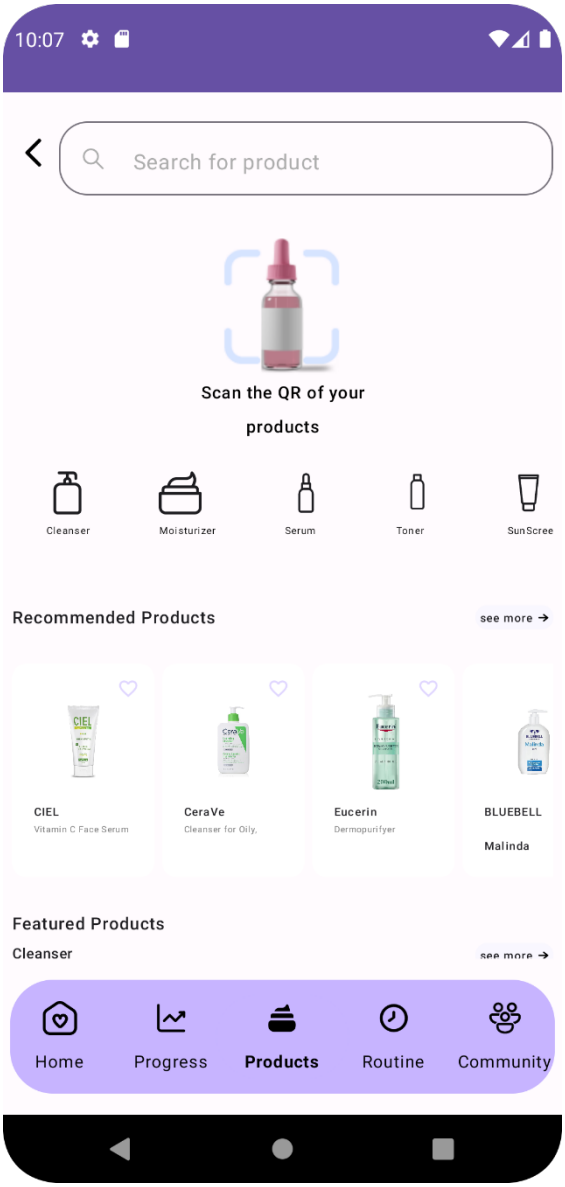




Storing routine in real time database

10.Product Screen

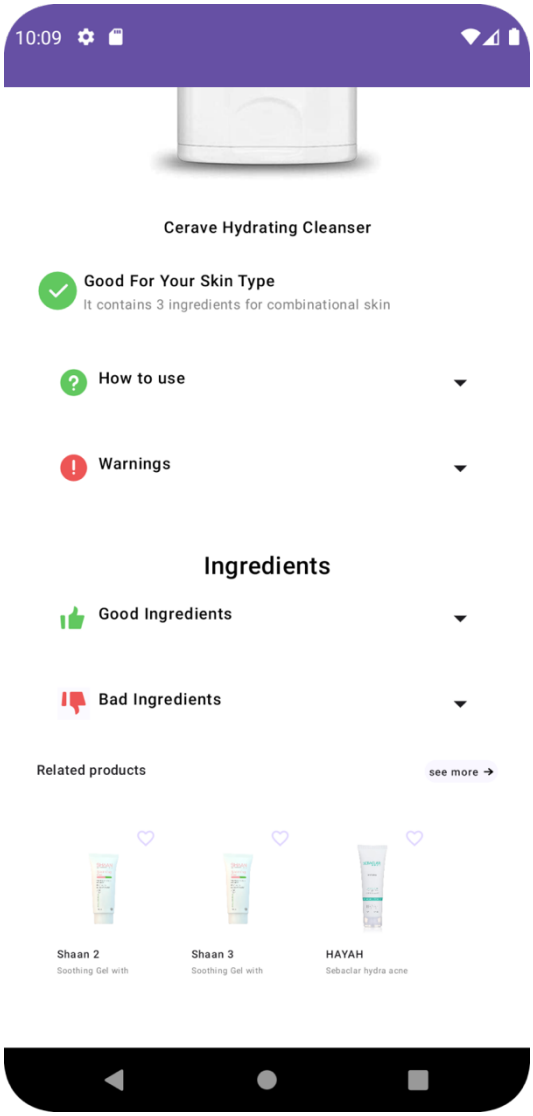
- Display recommended products for users.

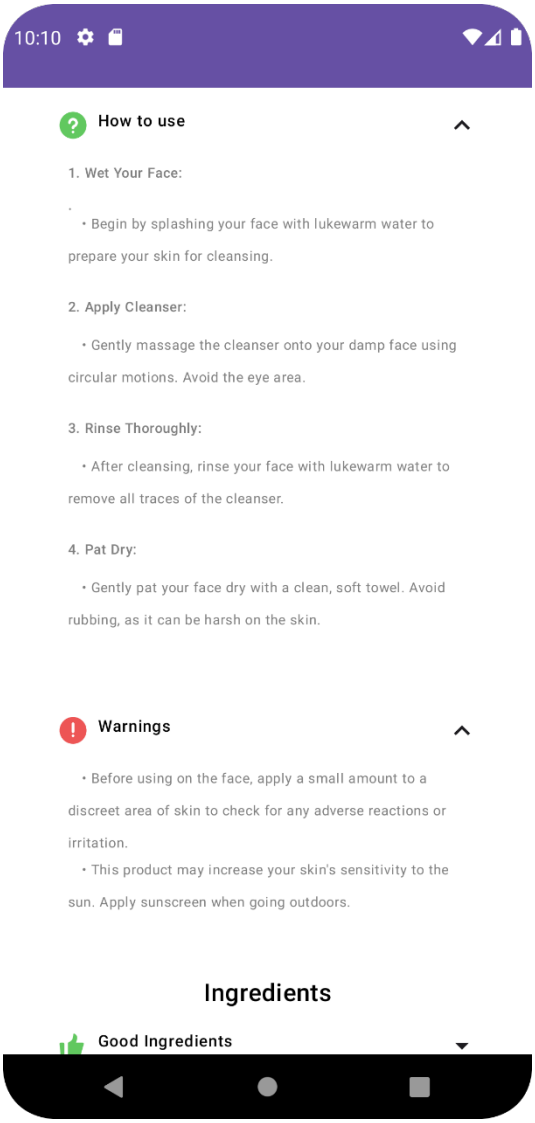


11.Product details screen

- Product details screen user can see how to use the product and read its warnings.



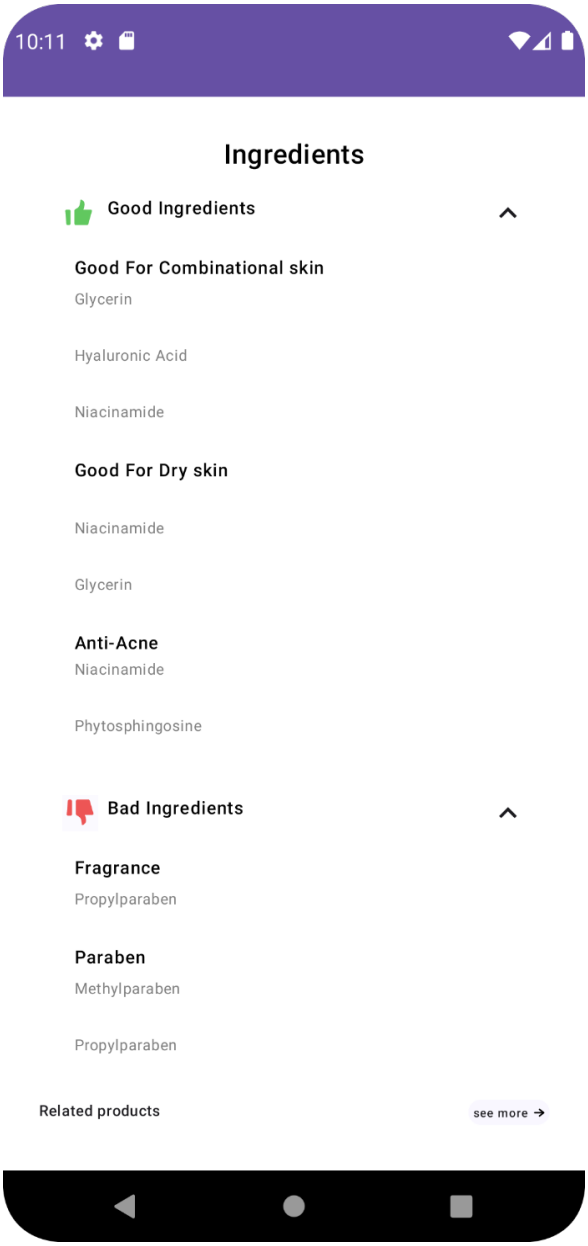

10:09
Cerave Hydrating Cleanser
Good For Your Skin Type
It contains 3 ingredients for combinational skin
How to use
Warnings
Ingredients
Good Ingredients
Bad Ingredients
Related products
see more
Shaan 2
Soothing Gel with
Shaan 3
Soothing Gel with
HAYAH
Sebaclar hydra acne


10:10
How to use
1. Wet Your Face:
• Begin by splashing your face with lukewarm water to prepare your skin for cleansing.
2. Apply Cleanser:
• Gently massage the cleanser onto your damp face using circular motions. Avoid the eye area.
3. Rinse Thoroughly:
• After cleansing, rinse your face with lukewarm water to remove all traces of the cleanser.
4. Pat Dry:
• Gently pat your face dry with a clean, soft towel. Avoid rubbing, as it can be harsh on the skin.
Warnings
• Before using on the face, apply a small amount to a discreet area of skin to check for any adverse reactions or irritation.
• This product may increase your skin's sensitivity to the sun. Apply sunscreen when going outdoors.
Ingredients
Good Ingredients

- **We have products data from API response:**

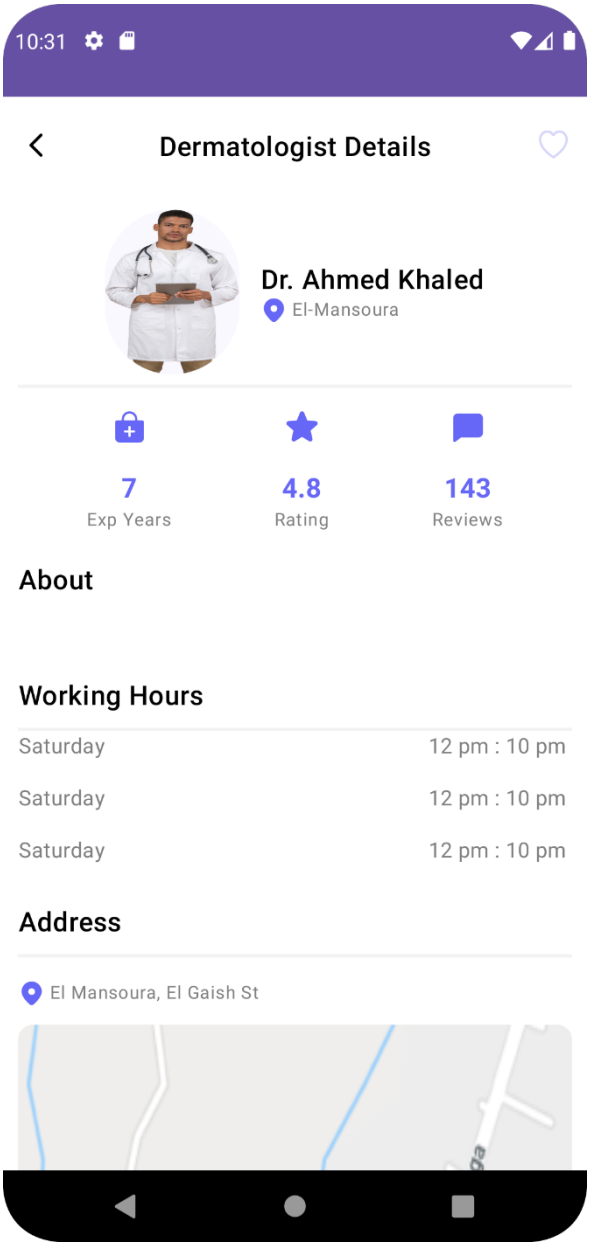

```
{
    "data": [
        {
            "id": "1",
            "name": "Neutrogena Oil-Free Acne Wash",
            "brand": "Neutrogena",
            "product_type": "Cleansear",
            "skin_type": [
                "Oily"
            ],
            "ingredients": [
                "Salicylic Acid",
                " Water",
                " Sodium C14-16 Olefin Sulfonate",
                " Cocamidopropyl Betaine",
                " Sodium Chloride",
                " PEG-80 Sorbitan Laurate",
                " C12-15 Alkyl Lactate",
                " Benzalkonium Chloride",
                " Disodium EDTA",
                " Fragrance",
                " Cocamidopropyl Pg-Dimonium Chloride Phosphate",
                " Propylene Glycol",
                " Aloe Barbadensis Leaf Extract",
                " Anthemis Nobilis Flower Extract",
                " Chamomilla Recutita (Matricaria) Flower Extract",
                " Sodium Hydroxide",
                " Citric Acid",
                " Yellow 5",
                " Red 40"
            ],
            "image": "https://imgur.com/N4F2qSJ"
```

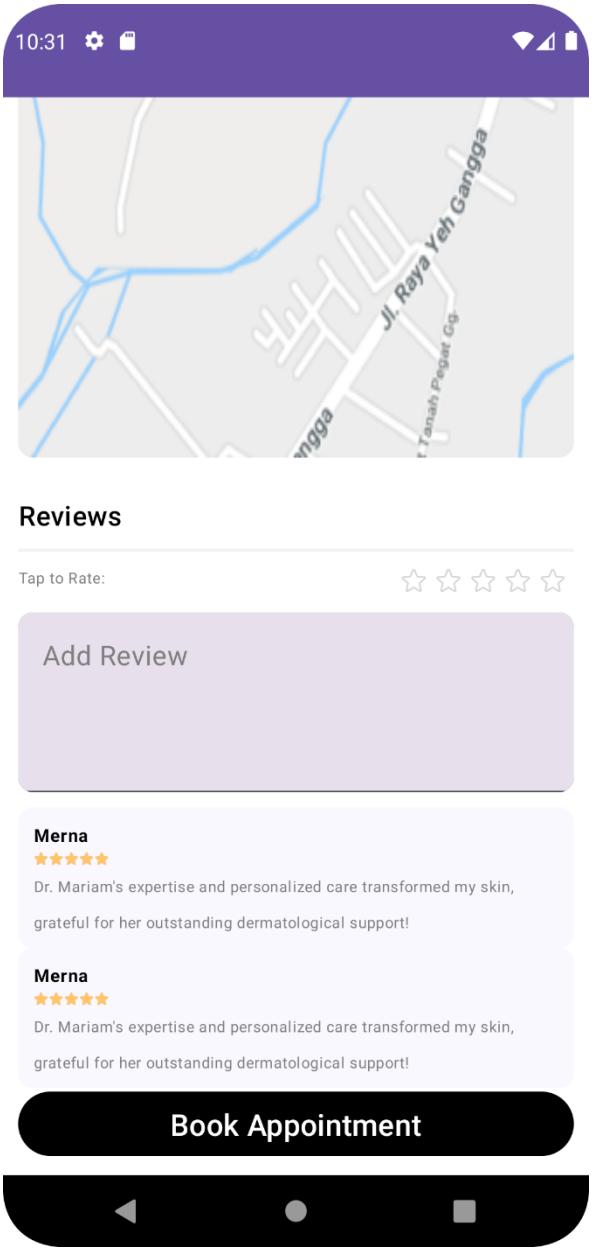
- **Show the good ingredients and bad ingredients for the product**



12.Dermatologist details screen

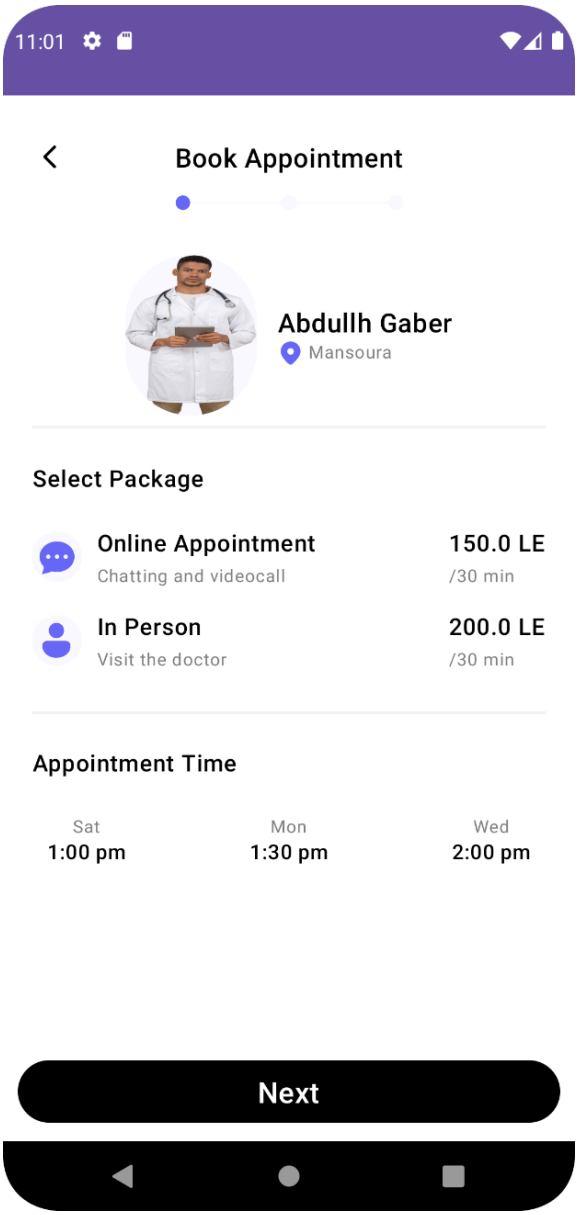
-Details about dermatologist and you can book appointment from here



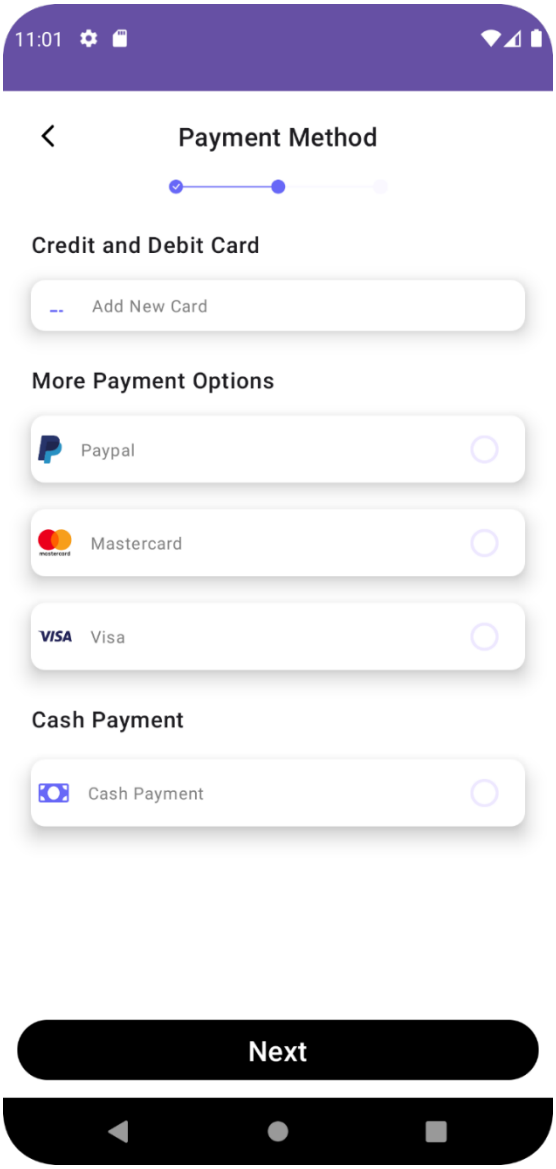


13.Book appointment screen

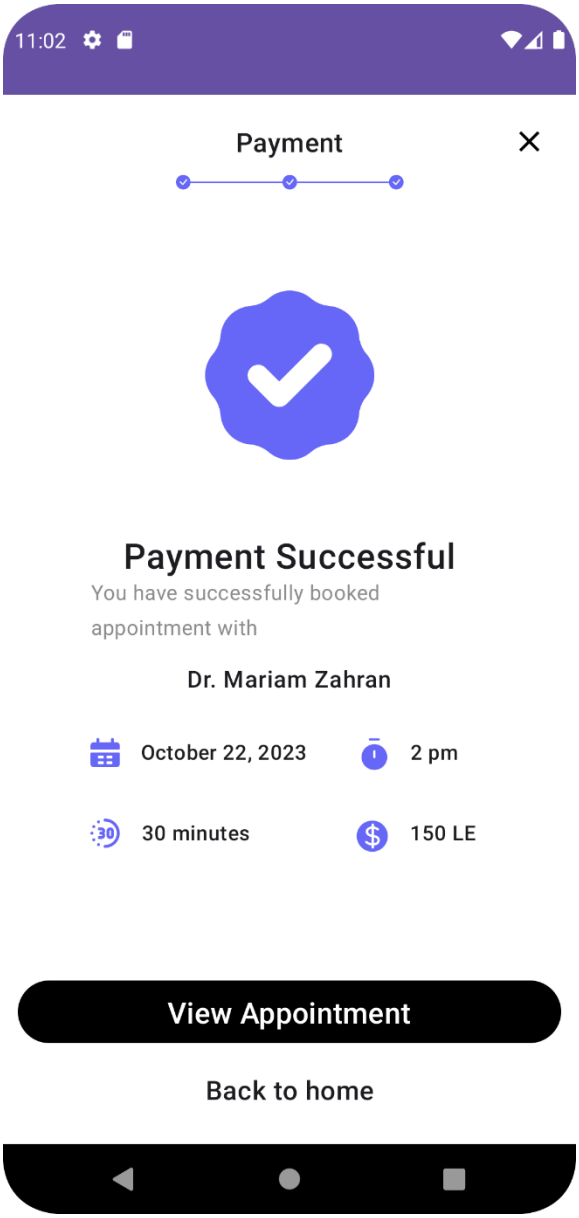
= User can select his best package and know the appointment time.

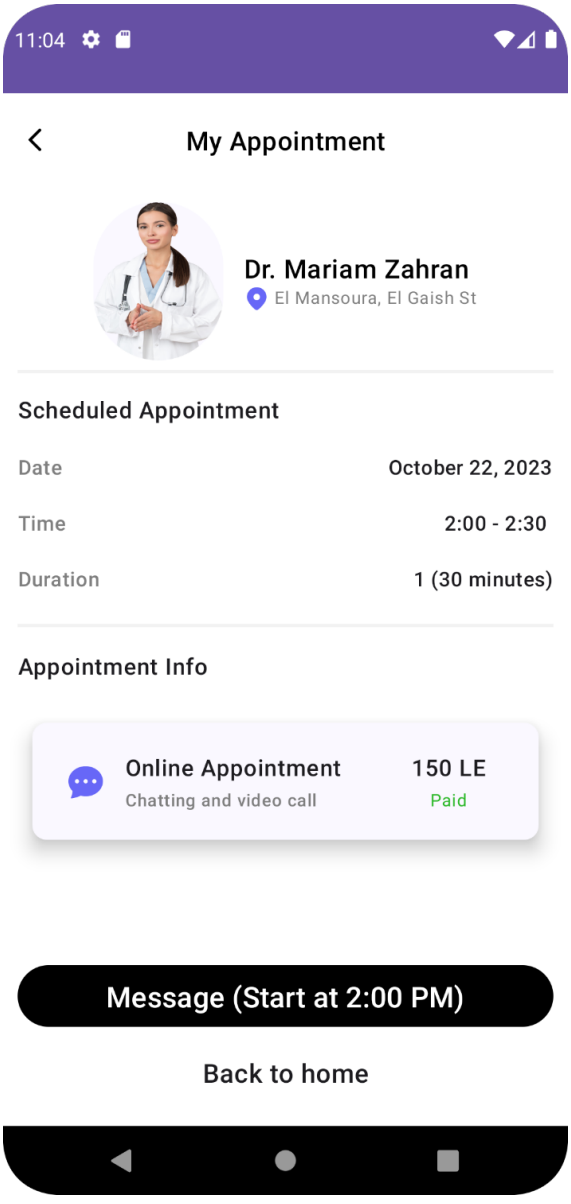


- User can choose his payment method when booking appointment



- Payment Successful Screen which displays data about appointment

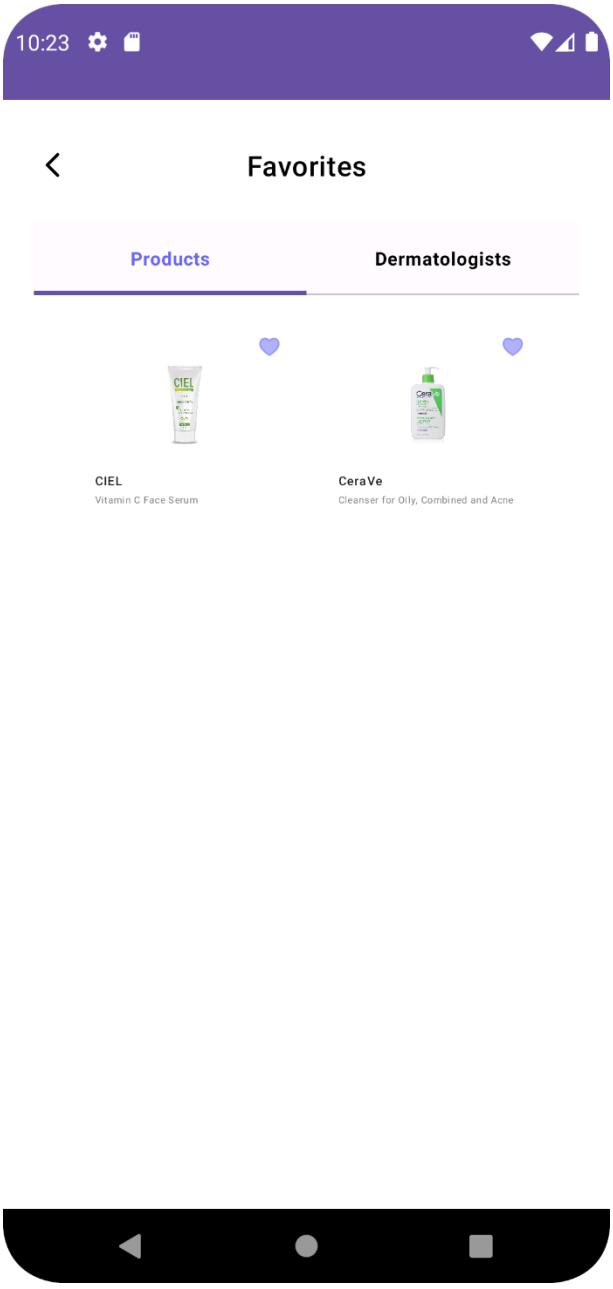


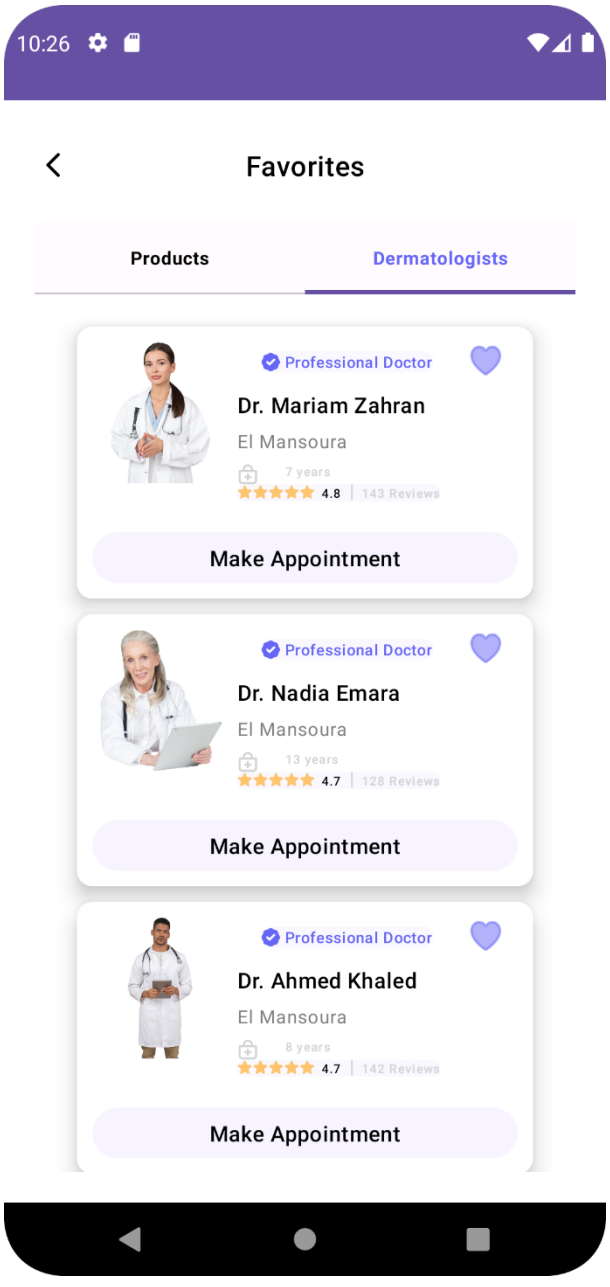


Summary data for the appointment

Favorite screen

- Mark dermatologist and products as favorite





Chapter 6

Chapter 6: Conclusion and Future Work

6.1 Conclusion

Pearl goes beyond just being an app; it's a comprehensive skin care companion designed to empower users on their journey to healthy, radiant skin. We understand the frustration of navigating the world of dermatology and product recommendations. With Pearl, users can take control of their skin health with these key features:

- **Self-Assessment:** A user-friendly skin analysis tool within the app helps identify your unique skin type and potential concerns. This empowers you to make informed decisions about your skincare routine.
- **Personalized Regimen Builder:** Forget generic one-size-fits-all approaches. Pearl creates a customized skincare routine based on your individual needs and goals. Whether it's tackling dryness, acne, or signs of aging, Pearl recommends products tailored to your specific concerns.
- **Direct Dermatologist Connection:** For more complex issues or personalized advice, Pearl seamlessly connects you with board-certified dermatologists directly through the app. This eliminates the hassle of appointment scheduling and provides you with expert guidance at your fingertips.
- **Dermatologist Dashboard:** Pearl also offers dermatologists a powerful, user-friendly dashboard. This comprehensive tool allows them to efficiently review patient information, track progress, and recommend personalized treatment plans – all within a secure and convenient platform.

By offering a holistic approach that combines self-care tools, personalized recommendations, and direct access to dermatologists, Pearl empowers users to confidently navigate their skin care journey. This results in not only improved skin health but also a boost in confidence and overall well-being.

6.2 Future Work

Future work will focus on optimizing our current processes, integrating innovative solutions, and conducting thorough evaluations to improve efficiency and effectiveness.

1- Foster a Collaborative Skincare Community:

- Introduce a dedicated Community section where users can share skincare routines, challenges, and product recommendations, fostering a supportive environment for shared experiences.
  - Users can create posts detailing their skincare journeys, challenges, and successes.
  - Enable interactive discussions with comments and likes on community posts.

2-Empower Users in Skincare Product Selection:

- Implement a Skincare Product Scanner.
  - Users can scan product barcodes for safety and ingredient information.
  - The app processes scanned data to assess the product's impact on the user app.

3- Facilitate Secure Data Sharing with QR Codes:

- Introduce Patient QR Code Functionality.
  - Users can generate and access a unique QR code containing encrypted skincare data.
  - Healthcare professionals can scan the QR code for instant access to the patient's comprehensive skincare history.

4- Enhance Doctor-Patient Collaboration:

- Include Patient QR Code Functionality.
  - Doctors can scan the patient's QR code for efficient access to their skincare history during consultations.

5- The Streaks feature in the app:

- helps patients maintain consistency in their health routines by tracking their daily activities and adherence to treatment plans:

- Patients can earn streaks by completing tasks such as making their routine.
- This gamified approach motivates patients to stay engaged with their health management, making it easier for both patients and healthcare providers to monitor progress and identify areas needing attention.

6- chatting and online communication feature:

- enables seamless interaction between doctors and patients.

7- Detection of combination skin type:

- helping users identify areas of both oily and dry skin for personalized skincare recommendations.

8- Highlight and address Acne types to guide users in their skincare journey.

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